

Project Data Page:





Table of Contents

Cover Page	1
Vessel Design Summary:	2
Nozzle Summary:	5
Nozzle Schedule:	6
Bill of Material:	7
MDMT Summary:	8
Warnings and Errors:	10
Input Echo:	11
XY Coordinate Calculations:	19
Flg Calc [Int P]: LHS Ch Flg	20
Flg Calc [Int P]: RHS Shell Fl	29
Internal Pressure Calculations:	38
Element: LHS Dish End	39
Element: Left Ch shell	41
Element: Main Shell	43
Element: RHS Dish End	45
Shop Pressure Test Results:	48
Field Pressure Test Results:	50
ASME TS Calc: Pressure Test	52
External Pressure Calculations:	60
Element and Detail Weights:	61
Nozzle Flange MAWP:	63
Wind Load Calculation:	65
Earthquake Load Calculation:	67
Center of Gravity Calculation:	69
Saddle Calcs: Bundle Pull New:	70
Saddle Calcs: Bundle Pull Cor:	71
Saddle Calcs: Road Transport:	72
Saddle Calcs: Operating Case:	86
Saddle Calcs: Test Case:	102
Saddle Calcs: Ext. Pressure:	118
Nozzle Calcs.: N1-Tube inlet	131
Nozzle Calcs.: N1-Tube Outlet	140
Nozzle Calcs.: N3-Shell inlet	151
Nozzle Calcs.: N4-Shell outlet	166
ASME TS Calc:	181
Problems/Failures Summary:	190



DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\E-0102.pvdb

Date of Analysis : Sep 9, 2026 10:30am

PV Elite 28, January 2026



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 2 of 190

Vessel Design Summary:

Step: 32 10:31am Sep 9,2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 800.000 mm ID
Vessel Design Length, Tangent to Tangent 3720.00 mm
Specified Datum Line Distance 50.00 mm
Shell Side Design Temperature 82 °C
Channel Side Design Temperature 230 °C
Shell Side Design Pressure 1200.000 KPa.
Channel Side Design Pressure 1050.000 KPa.
Shop Shell Side Test Pressure 2194.171 KPa.
Field Shell Side Test Pressure 1560.000 KPa.
Shop Channel Side Test Pressure 2194.171 KPa.
Field Channel Side Test Pressure 1365.000 KPa.
Wind Design Code ASCE/SEI 7-16
Earthquake Design Code ASCE/SEI 7-16

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	
Head	SA-516 70	...	...	K02700	
Flange	SA-765 II	...	...	K03047	
Nozzle	SA-333 6	...	...	K03006	+IT
Re-Pad	SA-516 70	...	...	K02700	
Nozzle Flg	SA-350 LF2	1	...	K03011	+IT
Tubes	SA-334 1	...	...	K03008	+IT
Tubesheet	SA-765 II	...	...	K03047	
Flg Bolting	SA-193 B7	...	<= 64	G41400	
Hrz Bolting	SA-307 A	...	6 <= t <= 100		
Saddles	SA-516 70	...	...	K02700	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized

+Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (KPa. & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
LHS Dish End	1057.873	0.00	2994.000	3.2000	No	No
Left Ch shell	1057.873	0.00	3630.000	3.2000	N/A	No
LHS Ch Flange	1056.861	0.00	2813.000	3.2000	N/A	No
LHS Shell Flg	1206.862	0.00	2813.000	3.2000	N/A	No
Main Shell	1207.873	101.35	4948.000	3.2000	N/A	No
RHS Dish End	1207.873	101.35	2994.000	3.2000	No	No

Liquid Level: 800.00 mm Dens.: 0.001 kg./cm³ Sp. Gr.: 1.000

Element Types and Properties:

Element	"To" Elev	Element Length	Nominal Thickness	Finished Thickness	Reqd Thk Internal	Reqd Thk External	Long Eff	Circ Eff
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FileName : E-0102

Page 3 of 190

Vessel Design Summary:

Step: 32 10:31am

Sep 9, 2026

Type	mm	mm	mm	mm	mm	mm		
Ellipse	0.0	50.0	14.0	11.9	6.3	...	1.00	1.00
Cylinder	700.0	700.0	14.0	14.0	6.3	...	1.00	1.00
Body Flg	829.0	126.0	103.0	95.0	92.3	92.3	1.00	1.00
Body Flg	1020.0	126.0	103.0	95.0	92.3	92.3	1.00	1.00
Cylinder	3620.0	2600.0	18.0	18.0	6.7	7.5	1.00	1.00
Ellipse	3670.0	50.0	14.0	11.9	6.7	5.3	1.00	1.00

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
N1-Tube inlet	WRC-297	0.974	Shell	Passed
N1-Tube Outlet	WRC-297	0.974	Shell	Passed
N3-Shell inlet	WRC-297	0.974	Shell	Passed
N4-Shell outle	WRC-297	0.974	Shell	Passed

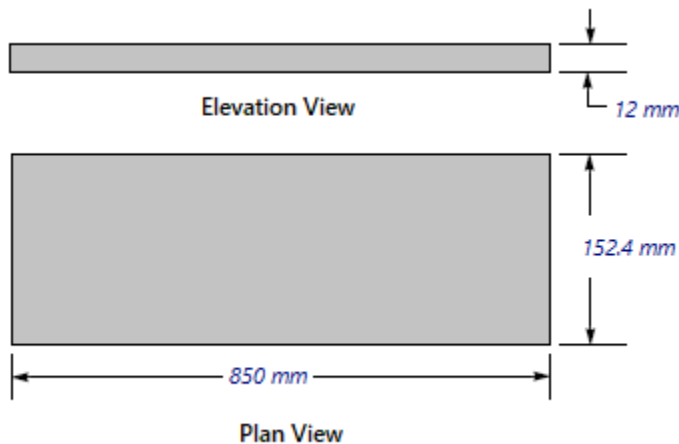
Foundation/Support Summary

Saddle Parameters:

Saddle Width	127.000 mm
Saddle Bearing Angle	120.000 deg.
Centerline Dimension	800.000 mm
Wear Pad Width	152.400 mm
Wear Pad Thickness	10.000 mm
Wear Pad Bearing Angle	132.000 deg.
Distance from Saddle to Tangent	520.000 mm

Baseplate Length	850.000 mm
Baseplate Thickness	12.000 mm
Baseplate Width	152.400 mm
Number of Ribs (including outside ribs)	3
Rib Thickness	10.000 mm
Web Thickness	10.000 mm
Height of Center Web	300.000 mm
Number of Bolts in Baseplate	4

Baseplate Sketch



Maximum Tensile Bolt Load

130. Kgf

Summary of Maximum Saddle Loads, Operating Case, Un-Factored:



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FileName : E-0102

Page 4 of 190

Vessel Design Summary:

Step: 32 10:31am Sep 9,2026

Maximum Vertical Saddle Load	7867.37	Kgf
Maximum Transverse Saddle Shear Load	678.12	Kgf
Maximum Longitudinal Saddle Shear Load	379.81	Kgf

Summary of Maximum Saddle Loads, Pressure Test Case:

Maximum Vertical Saddle Load	6300.80	Kgf
Maximum Transverse Saddle Shear Load	339.06	Kgf
Maximum Longitudinal Saddle Shear Load	334.91	Kgf

Weights:

Fabricated - Bare w/o removable internals	3854.9	kg.
Shop Test - Fabricated + water (full)	5763.6	kg.
Shipping - Fab. + rem. intls.+ shipping app.	3854.9	kg.
Erected - Fab. + rem. intls.+ insul. (etc)	4154.9	kg.
Empty - Fab. + intls. + details + wghts.	4711.4	kg.
Operating - Empty + operating liquid (no ca)	6611.0	kg.
Field Test - Empty weight + test medium (full)	6055.4	kg.

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FileName : E-0102

Page 5 of 190

Nozzle Summary:

Step: 30 10:31am Sep 9, 2026

Nozzle Calculation Summary (KPa. & mm):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
N1-Tube inlet	3630	...	...	OK	6.31	OK	No Calc[*]
N1-Tube Outlet	2850	...	...	OK	6.28	OK	Passed
N3-Shell inlet	4168	OK	...	OK	6.72	OK	Passed
N4-Shell outlet	4197	OK	...	OK	6.75	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 4168 Nozzle : N3-Shell inlet [Shellside]
 Minimum MAWP Nozzles : 2850 Nozzle : N1-Tube Outlet [Tubeside]

[*] - This was a small opening and the areas were not computed.

Note: MAWPs (Internal Case) shown above are at the High Point.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm	Layout Angle deg	Dia. Limit mm
20	N1-Tube inlet	350.000	180.000	160.120
20	N1-Tube Outlet	350.000	0.000	305.459
50	N3-Shell inlet	1570.000	0.000	832.001
50	N4-Shell outlet	2570.000	180.000	832.001

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.

lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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FileName : E-0102

Page 6 of 190

Nozzle Schedule:

Step: 29 10:31am Sep 9,2026

Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia mm	Wall Thk mm	Reinforcing Diameter	Pad Thk mm	Cut Length mm	Flg Class
N1-Tube inlet	80 mm	80	WNF	88.9	7.620	...	...	218	300
N1-Tube Outlet	150 mm	80	WNF	168.3	10.973	...	...	275	300
N3-Shell inlet	450 mm	80	WNF	457.2	23.800	657.20	16.00	339	150
N4-Shell outlet	450 mm	80	WNF	457.2	23.800	657.20	16.00	289	150

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
N1-Tube inl	SA-333 6	12.000	10.000	...	...	...
N1-Tube Out	SA-333 6	12.000	12.000	...	...	...
N3-Shell in	SA-333 6	12.000	12.000	10.000	16.000	...
N4-Shell ou	SA-333 6	18.000	12.000	14.000	16.000	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
N1-Tube inlet	300.000	180.0	200.00	0.00	Left Ch shell
N1-Tube Outlet	300.000	0.0	250.00	0.00	Left Ch shell
N3-Shell inlet	1520.000	0.0	250.00	0.00	Main Shell
N4-Shell outlet	2520.000	180.0	200.00	0.00	Main Shell

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FileName : E-0102

Page 7 of 190

Bill of Material:

Step: 28 10:31am Sep 9,2026

Bill of Materials:

QTY	DESCRIPTION	MATERIAL
2	ELLIPTICAL HEAD: 2.0 X 1, 14.0mm THK X 800.0mm ID X 50.0mm	SA-516 70
1	CYLINDER: 14.0mm THK X 800.0mm ID X 700.0mm	SA-516 70
2	WELD NECK FLANGE: 103.0mm THK X 980.0mm OD	SA-765 II
1	CYLINDER: 18.0mm THK X 800.0mm ID X 2600.0mm	SA-516 70
1	INSULATION: 250mm X 100mm THK	
1	INSULATION: 700mm X 100mm THK	
4	BAFFLES: 780.0mm DIA X 8.0mm THK	SA-516 70
6	TIE RODS: 25mm DIA X 1500mm	SA-193 B7
1	TUBESHEET: 980.0mm X 65.0mm THK	SA-765 II
170	TUBES: 2438.0mm X 25.4mm DIA X 2.3mm THK	SA-334 1
2	SADDLE: 127.0mm X 120 DEG	SA-516 70
1	INSULATION: 2600mm X 140mm THK	Cold Insulation
1	INSULATION: 250mm X 140mm THK	Cold Insulation
1	GASKET: 832.0mm OD X 806.0mm ID	...
36	BODY FLANGE BOLTS: 24.0mm DIA	SA-193 B7
72	NUTS FOR BODY FLANGE BOLTS: 24.0mm DIA	...
1	NAMEPLATE	...

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FileName : E-0102

Page 8 of 190

MDMT Summary:

Step: 31 10:31am Sep 9,2026

Minimum Design Metal Temperature Results Summary (°C):

Description	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
Notes								
LHS Shell Flg[11]	B	-23	-104	-29	0.304	12.000	1.00	No
Main Shell [8]	B	-11	-104	-29	0.299	18.000	1.00	No
RHS Dish End [10]	B	-23	-48	-29	0.501	11.900	1.00	No
RHS Dish End [7]	B	-18	-48	-29	0.410	14.000	1.00	No
LH saddle [24]	B	-29	-104	-29	0.299	10.000	1.00	No
Right saddle [24]	B	-29	-104	-29	0.299	10.000	1.00	No
N3-Shell inlet[1]	B	-46	-104		0.121	20.825	1.00	No
Nozzle Flg [5]	!	-46	-67					
N4-Shell outle[1]	B	-46	-104		0.122	20.825	1.00	No
Nozzle Flg [5]	!	-46	-67					
Tubesheet: SS[13]	B	-14	-30	-29	0.708	16.250	1.00	No
Warmest MDMT:		-11	-30					
LHS Ch Flange[11]	B	-23	-104	-29	0.299	12.000	1.00	No
LHS Dish End [10]	B	-23	-48	-29	0.362	11.900	1.00	No
LHS Dish End [7]	B	-18	-104	-29	0.296	14.000	1.00	No
Left Ch shell [8]	B	-18	-104	-29	0.296	14.000	1.00	No
N1-Tube inlet [1]	B	-46	-104		0.104	6.668	1.00	No
Nozzle Flg [5]	!	-46	-104					
N1-Tube Outlet[1]	B	-46	-104		0.107	9.601	1.00	No
Nozzle Flg [5]	!	-46	-104					
Tubesheet: CS[14]	B	-14	-30	-29	0.708	16.250	1.00	No
Bolting [21]		-48						
Warmest MDMT:		-14	-30					

Exchanger Side	Computed MDMT	Required MDMT	Pass/Fail
Shell	-30.0	-30.0	Pass
Channel/Tube	-30.0	-45.0	Fail

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.



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FileName : E-0102

Page 9 of 190

MDMT Summary:

Step: 31 10:31am Sep 9, 2026

UCS-66(g) was not considered.

UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.

UCS-66(i) applies to impact tested materials not by specification and

UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).

The Basic MDMT includes the (30F) PWHT credit if applicable.

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FileName : E-0102

Page 10 of 190

Warnings and Errors:

Step: 0 10:30am Sep 9, 2026

Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

Warning: LHS Ch Flange and LHS Shell Flg

There are mating flanges in this model that have different design pressures. This will generate different bolt loads. If a tubesheet is in between these flanges, PV Elite will compute the mating flange loads for you. If there is not a tubesheet physically separating these flanges, you must enter in the mating flange loads into the flange dialog.

Warning:

There is a flange in this model connected to an element whose temperature is different. The bolt allowable stress is based on it's parent design temperature. You may need to manually adjust the operating bolt allowable stress to a value based upon a representative maximum temperature the bolts will be subject to.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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FileName : E-0102

Page 11 of 190

Input Echo:

Step: 1 10:30am Sep 9, 2026

Units used in this Analysis (MM):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm
Force	Pounds	0.4536	Kgf
Mass	Pounds	0.4536	kg.
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	Kg-m.
Stress	lbs./sq.in.	0.0069	N./mm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	6.8946	KPa.
Elast. Modulus	lbs./sq.in.	6.8946	KPa.
Pipe Density	lbs./cu.in.	0.0277	kg./cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Wind Speed	miles/hr	1.6093	Km/hr
Tray Weight	lbs./sq.ft.	0.0005	Kgs/cm ²
Inertia	in.4	41.6231	cm4
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8825	Kgs/m ²
Elevation	Feet	304.8000	mm
Volume	in.3	16.3900	cm ³
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Exchanger Design Pressures and Temperatures

Shell Side Design Pressure	1200	KPa.
Channel Side Design Pressure	1050	KPa.
Shell Side Design Temperature	82.0	°C
Channel Side Design Temperature	230.0	°C
Radiography, Shell Side	RT-1	
Radiography, Channel Side	RT-1	
Service Type, Shell Side	None	
Service Type, Channel Side	Sour	
MDMT (CET), Shell Side	-30.0	°C
MDMT (CET), Tube Side	-45.0	°C
User defined MAWP, Shell Side	0	KPa.
User defined MAWP, Channel Side	0	KPa.
User defined MAPnc, Shell Side	0	KPa.
User defined MAPnc, Channel Side	0	KPa.
User defined Test Pres., Shell Side	0	KPa.
User defined Test Pres., Channel Side	0	KPa.

Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Type of Construction	Welded	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	UG-99(c)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	Yes



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FileName : E-0102

Page 12 of 190

Input Echo:

Step: 1 10:30am Sep 9,2026

Field Pressure Test:

Type of Pressure Test UG-99(b) Note [35]
 Pressure Test Position Horizontal
 Test Performed in Corroded Condition Yes

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	ASCE/SEI 7-16	
Wind Load Reduction Scale Factor	1.100	
Basic Wind Speed	349.92	Km/hr
Surface Roughness Category	C: Open Terrain	
Type of Surface	Moderately Smooth	
Base Elevation	300	mm
Percent Wind for Hydrotest	50.0	
Using User defined Wind Press. Vs Elev.	N	
Height of Hill or Escarpment	0	mm
Distance Upwind of Crest	0	mm
Distance from Crest to the Vessel	0	mm
Type of Terrain (Hill, Escarpment)	Flat	
Damping Factor (Beta) for Wind (Ope)	0.0100	
Damping Factor (Beta) for Wind (Empty)	0.0000	
Damping Factor (Beta) for Wind (Filled)	0.0000	
Site Elevation above Sea Level	0	mm

Seismic Design Code	ASCE/SEI 7-16	
Seismic Load Reduction Scale Factor	1.000	
Importance Factor	1.250	
Table Value Fa	1.600	
Table Value Fv	2.400	
Max. Mapped Res. Acceleration	0.147	[Ss]
Max. Eff. Ground Acceleration	0.065	[S]
Force Modification Factor R	3.000	
Site Class	D	
Component Elevation Ratio	0.000	z/h
Amplification Factor	0.000	Ap
Force Factor	0.000	
Consider Vertical Acceleration	Yes	
Minimum Acceleration Multiplier	0.000	
User Value of Sds (used if > 0)	0.000	
User Value of Sd1 (used if > 0)	0.000	
Moment Reduction Factor Tau	1.000	

Design Pressure + Static Head	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 13 of 190

Input Echo:

Step: 1 10:30am Sep 9, 2026

Use ASME VIII-1 Appendix 1-9 N

Perform Blast Load Analysis No

Material Database Year Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk) Yes
 Do not use Nozzle MDMT Interpretation VIII-1 01-37 No
 Use Table G instead of exact equation for "A" Yes
 Shell Head Joints are Tapered Yes
 Compute "K" in corroded condition Yes
 Use Code Case 2286 No
 Use the MAWP to compute the MDMT Yes
 For thickness ratios <= 0.35, MDMT will be -155F (-104C) Yes
 For PWHT & P1 Materials the MDMT can be < -55F (-48C) No

Using Metric Material Databases, ASME II D Yes
 Calculate B31.3 type stress for Nozzles with Loads Yes
 Reduce the MDMT due to lower membrane stress Yes
 Consider Longitudinal Stress in MDMT Calculations Yes

Complete Listing of Vessel Elements and Details:

Element#	1/6	
Element From Node	10	
Element To Node	20	
Element Type	Elliptical	
Description	LHS Dish End	
Distance "FROM" to "TO"	50	mm
Inside Diameter	800	mm
Element Thickness	11.9	mm
Internal Corrosion Allowance	3.2	mm
Nominal Thickness	14	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	1050	KPa.
Design Temperature Internal Pressure	230	°C
Design External Pressure	0	KPa.
Design Temperature External Pressure	230	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Allowable Stress, Ambient	138	N./mm ²
Allowable Stress, Operating	138	N./mm ²
Allowable Stress, Hydrotest	234	N./mm ²
Material Density	0.00775	kg./cm ³
P Number Thickness	32.004	mm
Yield Stress, Operating	220	N./mm ²
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K02700	
Product Form	Plate	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Elliptical Head Factor	2.0	
Weld is pre-Heated	No	
Element From Node	10	
Detail Type	Liquid	
Detail ID	Liquid: 10	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	800	mm
Liquid Density	0.0009984	kg./cm ³
Element From Node	10	
Detail Type	Insulation	



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FileName : E-0102

Page 14 of 190

Input Echo:

Step: 1 10:30am Sep 9,2026

Detail ID	Hot insulation
Dist. from "FROM" Node / Offset dist	-200 mm
Height/Length of Insulation	250 mm
Thickness of Insulation	100 mm
Density	0.000352 kg./cm ³

Element#	2/6
Element From Node	20
Element To Node	30
Element Type	Cylinder
Description	Left Ch shell
Distance "FROM" to "TO"	700 mm
Inside Diameter	800 mm
Element Thickness	14 mm
Internal Corrosion Allowance	3.2 mm
Nominal Thickness	14 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1050 KPa.
Design Temperature Internal Pressure	230 °C
Design External Pressure	0 KPa.
Design Temperature External Pressure	230 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	20
Detail Type	Liquid
Detail ID	Liquid: 20
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	800 mm
Liquid Density	0.0009984 kg./cm ³

Element From Node	20
Detail Type	Insulation
Detail ID	Hot insulation
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	700 mm
Thickness of Insulation	100 mm
Density	0.000352 kg./cm ³

Element From Node	20
Detail Type	Nozzle
Detail ID	N1-Tube inlet
Dist. from "FROM" Node / Offset dist	300 mm
Nozzle Diameter	80 mm
Nozzle Schedule	80
Nozzle Class	300
Layout Angle	180.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	14.058 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-333 6 +IT

Element From Node	20
Detail Type	Nozzle
Detail ID	N1-Tube Outlet
Dist. from "FROM" Node / Offset dist	300 mm
Nozzle Diameter	150 mm
Nozzle Schedule	80
Nozzle Class	300
Layout Angle	0.0



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 15 of 190

Input Echo:

Step: 1 10:30am Sep 9, 2026

Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	35.022 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-333 6 +IT

Element From Node	20
Detail Type	Weight
Detail ID	Partition plate
Dist. from "FROM" Node / Offset dist	350 mm
Miscellaneous Weight	100 Kg
Offset from Element Centerline	0 mm

Element#	3/6
Element From Node	30
Element To Node	40
Element Type	Flange
Description	LHS Ch Flange
Distance "FROM" to "TO"	126 mm
Flange Inside Diameter	800 mm
Element Thickness	95 mm
Internal Corrosion Allowance	3.2 mm
Nominal Thickness	103 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1050 KPa.
Design Temperature Internal Pressure	230 °C
Design External Pressure	0 KPa.
Design Temperature External Pressure	230 °C
Effective Diameter Multiplier	1.2
Material Name	SA-765 II
Allowable Stress, Ambient	138 N./mm ²
Allowable Stress, Operating	136.8 N./mm ²
Allowable Stress, Hydrotest	225 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	32.004 mm
Yield Stress, Operating	208 N./mm ²
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K03047
Product Form	Forgings
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No

Element From Node	30
Detail Type	Liquid
Detail ID	Liquid: 30
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	700 mm
Liquid Density	0.0009984 kg./cm ³

Element#	4/6
Element From Node	40
Element To Node	50
Element Type	Flange
Description	LHS Shell Flg
Distance "FROM" to "TO"	126 mm
Flange Inside Diameter	800 mm
Element Thickness	95 mm
Internal Corrosion Allowance	3.2 mm
Nominal Thickness	103 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1200 KPa.



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FileName : E-0102

Page 16 of 190

Input Echo:

Step: 1 10:30am Sep 9, 2026

Design Temperature Internal Pressure	82	°C
Design External Pressure	0	KPa.
Design Temperature External Pressure	82	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-765 II	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	
Element From Node	40	
Detail Type	Liquid	
Detail ID	Liquid: 40	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	700	mm
Liquid Density	0.0009984	kg./cm ³

Element#	5/6	
Element From Node	50	
Element To Node	60	
Element Type	Cylinder	
Description	Main Shell	
Distance "FROM" to "TO"	2600	mm
Inside Diameter	800	mm
Element Thickness	18	mm
Internal Corrosion Allowance	3.2	mm
Nominal Thickness	18	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	1200	KPa.
Design Temperature Internal Pressure	82	°C
Design External Pressure	101.35	KPa.
Design Temperature External Pressure	82	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Allowable Stress, Ambient	138	N./mm ²
Allowable Stress, Operating	138	N./mm ²
Allowable Stress, Hydrotest	234	N./mm ²
Material Density	0.00775	kg./cm ³
P Number Thickness	32.004	mm
Yield Stress, Operating	242.6	N./mm ²
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K02700	
Product Form	Plate	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	50	
Detail Type	Saddle	
Detail ID	LH saddle	
Dist. from "FROM" Node / Offset dist	520	mm
Width of Saddle	127	mm
Height of Saddle at Bottom	800	mm
Saddle Contact Angle	120.0	
Height of Composite Ring Stiffener	0	mm
Width of Wear Plate	152.4	mm
Thickness of Wear Plate	10	mm
Contact Angle, Wear Plate (degrees)	132.0	
Friction coefficient	0.30000001	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	mm



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FileName : E-0102

Page 17 of 190

Input Echo:

Step: 1 10:30am Sep 9,2026

Element From Node	50
Detail Type	Saddle
Detail ID	Right saddle
Dist. from "FROM" Node / Offset dist	2080 mm
Width of Saddle	127 mm
Height of Saddle at Bottom	800 mm
Saddle Contact Angle	120.0
Height of Composite Ring Stiffener	0 mm
Width of Wear Plate	152.4 mm
Thickness of Wear Plate	10 mm
Contact Angle, Wear Plate (degrees)	132.0
Friction coefficient	0.30000001
Moment Factor	3.0
Dimension E at base (optional)	0 mm
Circumferential Eff. over Saddle	1.0
Circumferential Eff. at Midspan	1.0
Tangent to Tangent dist. (optional)	0 mm
Element From Node	50
Detail Type	Liquid
Detail ID	Liquid: 50
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	800 mm
Liquid Density	0.0009984 kg./cm ³
Element From Node	50
Detail Type	Insulation
Detail ID	Cold insulation
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	2600 mm
Thickness of Insulation	140 mm
Density	0.00034 kg./cm ³
Element From Node	50
Detail Type	Nozzle
Detail ID	N3-Shell inlet
Dist. from "FROM" Node / Offset dist	500 mm
Nozzle Diameter	450 mm
Nozzle Schedule	80
Nozzle Class	150
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	146.13 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-333 6 +IT
Element From Node	50
Detail Type	Nozzle
Detail ID	N4-Shell outlet
Dist. from "FROM" Node / Offset dist	1500 mm
Nozzle Diameter	450 mm
Nozzle Schedule	80
Nozzle Class	150
Layout Angle	180.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	153.61 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-333 6 +IT
Element From Node	50
Detail Type	Weight
Detail ID	Fireproofing we
Dist. from "FROM" Node / Offset dist	500 mm
Miscellaneous Weight	150 Kg



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 18 of 190

Input Echo:

Step: 1 10:30am Sep 9,2026

Offset from Element Centerline	585	mm
Element From Node	50	
Detail Type	Weight	
Detail ID	Fireproofing	
Dist. from "FROM" Node / Offset dist	2025	mm
Miscellaneous Weight	150	Kgf
Offset from Element Centerline	585	mm

Element#	6/6	
Element From Node	60	
Element To Node	80	
Element Type	Elliptical	
Description	RHS Dish End	
Distance "FROM" to "TO"	50	mm
Inside Diameter	800	mm
Element Thickness	11.9	mm
Internal Corrosion Allowance	3.2	mm
Nominal Thickness	14	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	1200	KPa.
Design Temperature Internal Pressure	82	°C
Design External Pressure	101.35	KPa.
Design Temperature External Pressure	82	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Elliptical Head Factor	2.0	
Weld is pre-Heated	No	

Element From Node	60	
Detail Type	Liquid	
Detail ID	Liquid: 60	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	800	mm
Liquid Density	0.0009984	kg./cm ³

Element From Node	60	
Detail Type	Insulation	
Detail ID	Cold insulation	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	250	mm
Thickness of Insulation	140	mm
Density	0.00034	kg./cm ³



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 19 of 190

XY Coordinate Calculations:

Step: 2 10:30am

Sep 9, 2026

XY Coordinate Calculations:

From	To	X (Horiz.) mm	Y (Vert.) mm	DX (Horiz.) mm	DY (Vert.) mm
LHS Dish End		50	...	50	...
Left Ch shell		750	...	700	...
LHS Ch Flange		879	...	126	...
LHS Shell Flg		1070	...	126	...
Main Shell		3670	...	2600	...
RHS Dish End		3720	...	50	...

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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FileName : E-0102

Page 20 of 190

Flg Calc [Int P]: LHS Ch Flg

Flng: 11 10:30am Sep 9,2026

Flange Input Data Values

Description: LHS Ch Flg :

LHS Ch Flange

Flange Type	Integral Weld Neck		
Design Pressure	P	1056.86	KPa.
Design Temperature		230	°C
Internal Corrosion Allowance	ci	3.2000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	800.000	mm
Flange Outside Diameter	A	980.000	mm
Flange Thickness	t	95.0000	mm
Thickness of Hub at Small End	go	12.0000	mm
Thickness of Hub at Large End	gl	18.0000	mm
Length of Hub	h	31.0000	mm
Flange Material		SA-765 II	
Flange Material UNS number		K03047	
Flange Allowable Stress At Temperature	Sfo	136.80	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Diameter of Bolt Circle	C	920.000	mm
Nominal Bolt Diameter	a	24.0000	mm
Type of Threads		TEMA Metric	
Number of Bolts		36	
Flange Face Outside Diameter	Fod	835.000	mm
Flange Face Inside Diameter	Fid	800.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	832.000	mm
Gasket Inside Diameter	Gi	806.000	mm
Gasket Factor	m	4.2500	
Gasket Design Seating Stress	y	69.64	N./mm ²
Gasket Thickness	tg	3.0000	mm
Raised face depth	Fd	0.0000	mm
Length of Partition Gasket	lp	732.0000	mm
Width of Partition Gasket	tp	10.0000	mm
Partition Gasket Factor	mPart	2.7500	
Partition Gasket Design Seating Stress	yPart	25.51	N./mm ²



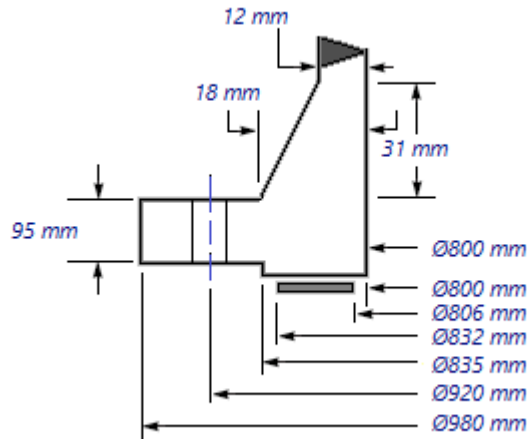
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FileName : E-0102

Page 21 of 190

Flg Calc [Int P]: LHS Ch Flg

Flng: 11 10:30am Sep 9, 2026



ASME Code, Section VIII Division 1, 2025

Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + C_a)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (1056.86 \cdot (800.0/2 + 3.2)) / (138.0 \cdot 1.0 - 0.6 \cdot 1056.86) + C_a$$

$$= 6.3024 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 8.8) / (403.2 + 0.6 \cdot 8.8)$$

$$= 2972.747 \text{ KPa.}$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 12.0) / (400.0 + 0.6 \cdot 12.0)$$

$$= 4066.488 \text{ KPa.}$$

Corroded Flange Thickness, $t_c = T - c_i$	91.800 mm
Corroded Flange ID, $B_{cor} = B + 2 \cdot c_i$	806.400 mm
Corroded Large Hub, $g1_{cor} = g1 - c_i$	14.800 mm
Corroded Small Hub, $g0_{cor} = g0 - c_i$	8.800 mm
Code R Dimension, $R = ((C - B_{cor})/2) - g1_{cor}$	42.000 mm

Gasket Contact Width, $N = (G_o - G_i)/2$	12.800 mm
Basic Gasket Width, $b_o = N/2$	6.400 mm
Effective Gasket Width, $b = C_b \cdot \sqrt{b_o}$	6.375 mm
Gasket Reaction Diameter, $G = G_o - 2 \cdot b$	819.250 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 819.2501^2 \cdot 1056.861$$

$$= 56811.664 \text{ Kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P + 2 \cdot l_p \cdot b_{Part} \cdot m_{Part} \cdot P$$

$$= 2 \cdot 6.3749 \cdot \pi \cdot 819.2501 \cdot 4.25 \cdot 1056.86$$

$$+ 2 \cdot 732.0 \cdot 5.0 \cdot 2.75 \cdot 1056.8615$$

$$= 17200.105 \text{ Kgf}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot B_{cor}^2 \cdot P$$

$$= \pi/4 \cdot 806.4^2 \cdot 1056.8615$$

$$= 55043.441 \text{ Kgf}$$



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FileName : E-0102

Page 22 of 190

Flg Calc [Int P]: LHS Ch Flg

Flng: 11 10:30am

Sep 9, 2026

Pressure Force on Flange Face [Ht]:

$$\begin{aligned}
 &= H - H_d \\
 &= 56812 - 55043 \\
 &= 1768.221 \text{ Kgf}
 \end{aligned}$$

Design Bolt Load for the Operating Condition [Wo]:

$$\begin{aligned}
 &= \max(W_{o,1} + W_{o,2} + H'p, 0) \\
 &= \max(56812 + 17200 + 0, 0) \\
 &= 74011.773 \text{ Kgf} \\
 &= 84516.242 \text{ Kgf, Mating Flange Load Governs}
 \end{aligned}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$\begin{aligned}
 &= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p \\
 &= 69.64 \cdot 6.3749 \cdot \pi \cdot 819.25 + 25.51 \cdot 5.0 \cdot 732.0 \\
 &= 126032.969 \text{ Kgf}
 \end{aligned}$$

Required Bolt Area [Am]:

$$\begin{aligned}
 &= \max((W_o + F_a + 4Me/G) / S_b, W_g / S_a) \\
 &= \max((84516 + 0 + 4 \cdot 0 / 819.25) / 172, 126033 / 172) \\
 &= 71.861 \text{ cm}^2
 \end{aligned}$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$\begin{aligned}
 &= 2a + 6t / (m + 0.5) \\
 &= 2 \cdot 24.0 + 6 \cdot 91.8 / (4.25 + 0.5) \\
 &= 163.958 \text{ mm}
 \end{aligned}$$

Actual Circumferential Bolt Spacing [Bs]:

$$\begin{aligned}
 &= C \cdot \sin(\pi / n) \\
 &= 920.0 \cdot \sin(\pi / 36) \\
 &= 80.183 \text{ mm}
 \end{aligned}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$\begin{aligned}
 &= \max(\sqrt{ Bs / (2a + t) }, 1) \\
 &= \max(\sqrt{ 80.183 / (2 \cdot 24.0 + 91.8) }, 1) \\
 &= 1.0000
 \end{aligned}$$

Bolting Information for TEMA Metric Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	71.861	112.589	
Radial Distance between Hub and Bolts:	33.500	42.000	
Radial Distance between Bolts and Edge:	29.000	30.000	
Circ. Spacing between the Bolts:	57.500	80.183	163.958

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$\begin{aligned}
 &= A_b \cdot S_a / (y \cdot \pi (G_o + G_i)) \\
 &= 112.589 \cdot 172.0 / (69.64 \cdot \pi (832.0 + 806.0)) \\
 &= 5.404 \text{ mm}
 \end{aligned}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 16 mm
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$\begin{aligned}
 &= S_a \cdot A_b \\
 &= 172.0 \cdot 112.5893 \\
 &= 197464.48 \text{ Kgf}
 \end{aligned}$$

Gasket Load for the Operating Condition [HG]:

$$\begin{aligned}
 &= W_o - H \\
 &= 84516 - 56812 \\
 &= 27704.58 \text{ Kgf}
 \end{aligned}$$



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FileName : E-0102

Page 23 of 190

Flg Calc [Int P]: LHS Ch Flg

Flng: 11 10:30am Sep 9, 2026

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (920.0 - 819.2501) / 2$$

$$= 50.3750 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5 ((C - B) / 2 + hg)$$

$$= 0.5 ((920.0 - 806.4) / 2 + 50.375)$$

$$= 53.5875 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (920.0 - 806.4 - 14.8) / 2$$

$$= 49.4000 \text{ mm}$$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	55043.	49.4000	1.0000	2719.
Face Pressure, Mt	1768.	53.5875	1.0000	95.
Gasket Load, Mg	27705.	50.3750	1.0000	1396.
Gasket Seating, Matm	197464.	50.3750	1.0000	9947.

Total Moment for Operation, Mo 4210. Kg-m.

Total Moment for Gasket seating, Mg 9947. Kg-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 84.240 mm

Hub Ratio, h/h0 = HL / H0 0.368

Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.682

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.862

Factor V per 4.16.5 0.314

Factor f per 4.16.5 1.230

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 980.0 / 806.4$$

$$= 1.215$$

Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169 (K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$$

$$= (0.66845 + (5.7169 (1.215^2 \cdot \log_{10}(1.215))) / (1.215^2 - 1)) / (1.215 - 1)$$

$$= 10.069$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (1.215^2(1 + 8.55246 \cdot \log_{10}(1.215)) - 1) / ((1.0472 + 1.9448 \cdot 1.215^2) \cdot (1.215 - 1))$$

$$= 1.833$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136 (K^2 - 1) \cdot (K - 1))$$

$$= (1.215^2(1 + 8.55246 \cdot \log_{10}(1.215)) - 1) / (1.36136 (1.215^2 - 1) \cdot (1.215 - 1))$$

$$= 11.065$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$

$$= (1.215^2 + 1) / (1.215^2 - 1)$$

$$= 5.194$$

Stress factor from Table 4.16.4 for integral flanges [d]:



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FileName : E-0102

Page 24 of 190

Flg Calc [Int P]: LHS Ch Flg

Flng: 11 10:30am

Sep 9, 2026

$$\begin{aligned}
 &= U \cdot g_0^2 \cdot h_0 / V \\
 &= 11.065 \cdot 8.8^2 \cdot 84.24 / 0.314 \\
 &= 229828.969 \text{ mm}^3
 \end{aligned}$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$\begin{aligned}
 &= F / h_0 \\
 &= 0.862 / 84.24 \\
 &= 0.010 \text{ 1/mm}
 \end{aligned}$$

Stress factor from Table 4.16.4 [L]:

$$\begin{aligned}
 &= (t \cdot e + 1) / T + t^3 / d \\
 &= (91.8 \cdot 0.01 + 1) / 1.833 + 91.8^3 / 229828.969 \\
 &= 4.424
 \end{aligned}$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$\begin{aligned}
 &= (f \cdot Mop / Bcor) / (L \cdot g_1^2) \\
 &= (1.2302 \cdot 4210 / 806.4) / (4.424 \cdot 14.8^2) \\
 &= 64.99 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Hub Stress, Seating [SHa]:

$$\begin{aligned}
 &= (f \cdot Matm / Bcor) / (L \cdot g_1^2) \\
 &= (1.2302 \cdot 9947 / 806.4) / (4.424 \cdot 14.8^2) \\
 &= 153.57 \text{ N./mm}^2
 \end{aligned}$$

Radial Flange Stress, Operating [SRo]:

$$\begin{aligned}
 &= (\beta \cdot Mop / Bcor) / (L \cdot t^2) \\
 &= (2.2518 \cdot 4210 / 806.4) / (4.424 \cdot 91.8^2) \\
 &= 3.09 \text{ N./mm}^2
 \end{aligned}$$

Radial Flange Stress, Seating [SRa]:

$$\begin{aligned}
 &= (\beta \cdot Matm / Bcor) / (L \cdot t^2) \\
 &= (2.2518 \cdot 9947 / 806.4) / (4.424 \cdot 91.8^2) \\
 &= 7.31 \text{ N./mm}^2
 \end{aligned}$$

Tangential Flange Stress, Operating [STo]:

$$\begin{aligned}
 &= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo \\
 &= (10.0688 \cdot 4210 / (91.8^2 \cdot 806.4)) - 5.1938 \cdot 3 \\
 &= 45.11 \text{ N./mm}^2
 \end{aligned}$$

Tangential Flange Stress, Seating [STa]:

$$\begin{aligned}
 &= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa \\
 &= (10.0688 \cdot 9947 / (91.8^2 \cdot 806.4)) - 5.1938 \cdot 7 \\
 &= 106.59 \text{ N./mm}^2
 \end{aligned}$$

Average Flange Stress, Operating [SAo]:

$$\begin{aligned}
 &= (SHo + \max(SRo, STo)) / 2 \\
 &= (65 + \max(3, 45)) / 2 \\
 &= 55.05 \text{ N./mm}^2
 \end{aligned}$$

Average Flange Stress, Seating [SAa]:

$$\begin{aligned}
 &= (SHa + \max(SRa, STa)) / 2 \\
 &= (154 + \max(7, 107)) / 2 \\
 &= 130.08 \text{ N./mm}^2
 \end{aligned}$$

Bolt Stress, Operating [BSo]:

$$\begin{aligned}
 &= Wm1 / Ab \\
 &= 84516 / 112.5893 \\
 &= 73.62 \text{ N./mm}^2
 \end{aligned}$$

Bolt Stress, Seating [BSa]:

$$\begin{aligned}
 &= (Wm2 / Ab) \\
 &= (126033 / 112.5893) \\
 &= 109.78 \text{ N./mm}^2
 \end{aligned}$$



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FileName : E-0102

Page 25 of 190

Flg Calc [Int P]: LHS Ch Flg

Flng: 11 10:30am Sep 9,2026

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	64.99	205.20	153.57	207.00
Radial Flange	3.09	136.80	7.31	138.00
Tangential Flange	45.11	136.80	106.59	138.00
Maximum Average	55.05	136.80	130.08	138.00
Bolting	73.62	172.00	109.78	172.00

Minimum Required Flange Thickness 92.278 mm

Estimated M.A.W.P. (Operating) 2820 KPa.

Estimated Finished Weight of Flange at given Thk. 194.5 kg.

Estimated Unfinished Weight of Forging at given Thk 245.7 kg.

Flange Stress results per WRC-538 at computed Sfmax (N./mm²):

		Actual	Operating Allowed	stress ratio	Gasket Actual	Seating Allowed	stress ratio
Longitudinal Hub	Sh	474.84	475.43	0.999	474.84	499.99	0.950
Tangential Flange	St	328.45	356.57	0.921	328.45	374.99	0.876
Tangential Flange	Sto	160.20	237.71	0.674	160.20	250.00	0.641
Combined	Sh/f + St	713.14	713.14	1.000	713.14	749.99	0.951
Combined	Sr + St	350.97	713.14	0.492	350.97	749.99	0.468
Combined	Sh + Sto	635.04	713.14	0.890	635.04	749.99	0.847
Radial Flange	Sr	22.52	475.43	0.047	22.52	499.99	0.045
Maximum Stress Ratio				1.000			0.951



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 26 of 190

Flg Calc [Int P]: LHS Ch Flg

Flng: 11 10:30am Sep 9, 2026

Coefficients for f and $f2$

	f	$f2$	$f3$
a	-0.71375912	-0.01447638	-0.06456312
b	-0.12846279	-0.27814745	-0.08232239
c	1.08037907	0.01035395	0.13691677
d	0.99766848	1.37984158	-0.77879888
e	1.21823466	-0.34346764	0.14909019
f	0.01024612	0.02536192	0.00660062
g	0.4262313	-0.0014508	0.0616252
h	1.4204976	2.25543162	0.74639834
i	-0.70278181	-0.60889597	-0.079595
j	-0.02483937	-0.15735462	0.09851673
k	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g1}{g0} \right) + e \left(\frac{h}{h0} \right) + g \left(\frac{g1}{g0} \right)^2 + i \left(\frac{h}{h0} \right)^2 + k \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}{\left[1 + b \left(\frac{g1}{g0} \right) + d \left(\frac{h}{h0} \right) + f \left(\frac{g1}{g0} \right)^2 + h \left(\frac{h}{h0} \right)^2 + j \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}$$

Hub Stress Correction Factor [f]:

= max(1; F1num / F1den)

= max(1; 1.6751/1.3571)

= 1.2343

Hub Stress Correction Factor [f2]:

= max(0.25; fn2) if $h/h0 < 0.35$, else max (0.25; fn3)

= max(0.25; 0.5482/1.3198) if $h/h0 < 0.35$, else max (0.25; 0.3147/0.7557)

= 0.4164

In this case, the hub ratio $h/h0$, 0.3680 was ≥ 0.35 .

PV Elite iterates on the assembly bolt load stress, Sfmax, until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	Sbmax	507.5 N./mm ²
Minimum permissible bolt stress	Sbmin	290.0 N./mm ²
Max. permissible bolt stress prior to flange damage	Sfmax	530.0 N./mm ²
Target assembly gasket stress	SgT	207.0 N./mm ²
Maximum permissible gasket stress	Sgmax	276.0 N./mm ²
Minimum gasket operating stress	Sgmin-O	97.0 N./mm ²
Minimum gasket seating stress	Sgmin-S	140.0 N./mm ²
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	Phig	0.700



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 27 of 190

Flg Calc [Int P]: LHS Ch Flg Flng: 11 10:30am Sep 9, 2026

Root Area of a Single Bolt	Abs	3.1275 cm ²
Max. permissible single flange rotation for gasket	Thetagmax	1.000 deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	250.0 N./mm ²
Flange Material Operating Yield Stress	Syo	208.0 N./mm ²
Single flange rotation at Sfmax	Thetafmax	0.392 deg

Gasket Contact Area [Ag]:

$$= \pi/4 (GO.D.^2 - GI.D.^2) + (PartFactor \cdot Ip \cdot bPart)$$

$$= \pi/4 (832.0^2 - 806.0^2) + (0.5 \cdot 732.0 \cdot 5.0)$$

$$= 352.8 \text{ cm}^2$$

Target Bolt Stress [Sbsel]: (Eq. O-1)

$$= SgT \cdot Ag / (nb \cdot Ab)$$

$$= 207.0 \cdot 352.785 / (36 \cdot 3.127)$$

$$= 648.6 \text{ N./mm}^2$$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)

$$= \min(Sbsel; Sbmax)$$

$$= \min(648.61; 507.5)$$

$$= 507.5 \text{ N./mm}^2$$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)

$$= \max(Sbsel; Sbmin)$$

$$= \max(507.5; 290.0)$$

$$= 507.5 \text{ N./mm}^2$$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)

$$= \min(Sbsel; Sfmax)$$

$$= \min(507.5; 530.016)$$

$$= 507.5 \text{ N./mm}^2$$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)

$$:: Sbsel \geq Sgmin-S(Ag / (nb \cdot Abs))$$

$$:: 507.5 \geq 140.0(352.785 / (36 \cdot 3.127))$$

$$:: 507.5 \geq 438.7 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Operating Stress is maintained. (Eq. O-8)

$$:: Sbsel \geq (Sgmin-O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$$

$$:: 507.5 \geq (97.0 \cdot 352.785 + \pi/4 \cdot 1056.861 \cdot 806.0^2) / (0.7 \cdot 3.127 \cdot 36)$$

$$:: 507.5 \geq 502.6 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Maximum Stress is achieved. (Eq. O-9)

$$:: Sbsel \leq Sgmax(Ag / (nb \cdot Abs))$$

$$:: 507.5 \leq 276.0(352.785 / (36 \cdot 3.127))$$

$$:: 507.5 \leq 864.8 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Flange Rotation Limit is exceeded. (Eq. O-10)

$$:: Sbsel \leq Sfmax(\text{thetagmax} / \text{thetafmax})$$

$$:: 507.5 \leq 530.016(1.0/0.392)$$

$$:: 507.5 \leq 1353.1 \text{ N./mm}^2 \text{ Check Passes}$$

Assembly Bolt Torque [Tb]: (Eq. O-2)

$$= Sbsel \cdot K \cdot Abs \cdot \phi b$$

$$= 507.5 \cdot 0.2 \cdot 3.127 \cdot 24.0$$

$$= 77.7 \text{ Kg-m.}$$

Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 138 N./mm²

Bolt Allowable Stress for: SA-193 B7 Sb 171 N./mm²

Note: Since Sb >= SbB8, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:



Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 9947.5 / 1.0 \cdot 9806.637 \cdot 0.314 / (4.117 \cdot 201334 \cdot 8.8^2 \cdot 84.24 \cdot 0.3)$
 $= 0.984$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 4209.6 / 1.0 \cdot 9806.637 \cdot 0.314 / (4.117 \cdot 189090 \cdot 8.8^2 \cdot 84.24 \cdot 0.3)$
 $= 0.443$ (should be ≤ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 9947.5 / 1.0 \cdot 9806.637 \cdot 0.314 / (4.424 \cdot 201334 \cdot 8.8^2 \cdot 84.24 \cdot 0.3)$
 $= 0.916$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 4209.6 / 1.0 \cdot 9806.637 \cdot 0.314 / (4.424 \cdot 189090 \cdot 8.8^2 \cdot 84.24 \cdot 0.3)$
 $= 0.412$ (should be ≤ 1)

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.299, Temperature Reduction per Fig. UCS 66.1 = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	59.04	207.00	136.55	207.00
Radial Flange	4.04	138.00	9.33	138.00
Tangential Flange	36.47	138.00	84.35	138.00
Maximum Average	47.76	138.00	110.45	138.00
Bolting	73.62	172.00	110.56	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 83.160 mm

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FileName : E-0102

Page 29 of 190

Flg Calc [Int P]: RHS Shell Fl

Flng: 12 10:30am Sep 9,2026

Flange Input Data Values

Description: RHS Shell Fl :

LHS Shell Flg

Flange Type	Integral Weld Neck		
Design Pressure	P	1206.86	KPa.
Design Temperature		82	°C
Internal Corrosion Allowance	ci	3.2000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	800.000	mm
Flange Outside Diameter	A	980.000	mm
Flange Thickness	t	95.0000	mm
Thickness of Hub at Small End	go	12.0000	mm
Thickness of Hub at Large End	gl	18.0000	mm
Length of Hub	h	31.0000	mm
Flange Material		SA-765 II	
Flange Material UNS number		K03047	
Flange Allowable Stress At Temperature	Sfo	138.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Diameter of Bolt Circle	C	920.000	mm
Nominal Bolt Diameter	a	24.0000	mm
Type of Threads		TEMA Metric	
Number of Bolts		36	
Flange Face Outside Diameter	Fod	835.000	mm
Flange Face Inside Diameter	Fid	800.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	832.000	mm
Gasket Inside Diameter	Gi	806.000	mm
Gasket Factor	m	4.2500	
Gasket Design Seating Stress	y	69.64	N./mm ²
Gasket Thickness	tg	3.0000	mm
Raised face depth	Fd	0.0000	mm
Length of Partition Gasket	lp	732.0000	mm
Width of Partition Gasket	tp	10.0000	mm
Partition Gasket Factor	mPart	2.7500	
Partition Gasket Design Seating Stress	yPart	25.51	N./mm ²



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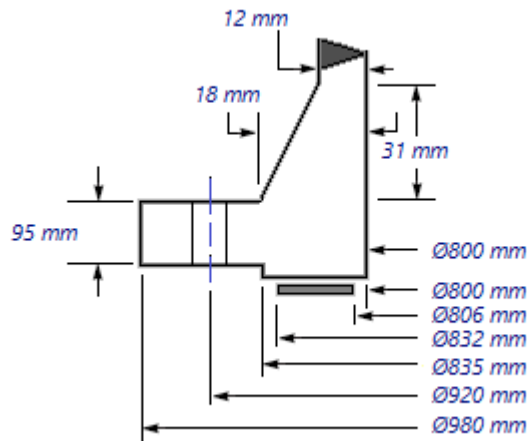
FileName : E-0102

Page 30 of 190

Flg Calc [Int P]: RHS Shell F1

Flng: 12 10:30am

Sep 9, 2026



ASME Code, Section VIII Division 1, 2025

Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + C_a)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (1206.86 \cdot (800.0/2 + 3.2)) / (138.0 \cdot 1.0 - 0.6 \cdot 1206.86) + C_a$$

$$= 6.7450 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 8.8) / (403.2 + 0.6 \cdot 8.8)$$

$$= 2972.747 \text{ KPa.}$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 12.0) / (400.0 + 0.6 \cdot 12.0)$$

$$= 4066.488 \text{ KPa.}$$

Corroded Flange Thickness, $t_c = T - c_i$	91.800 mm
Corroded Flange ID, $B_{cor} = B + 2 \cdot c_i$	806.400 mm
Corroded Large Hub, $g1_{cor} = g1 - c_i$	14.800 mm
Corroded Small Hub, $g0_{cor} = g0 - c_i$	8.800 mm
Code R Dimension, $R = ((C - B_{cor})/2) - g1_{cor}$	42.000 mm

Gasket Contact Width, $N = (G_o - G_i)/2$	12.800 mm
Basic Gasket Width, $b_o = N/2$	6.400 mm
Effective Gasket Width, $b = C_b \cdot \sqrt{b_o}$	6.375 mm
Gasket Reaction Diameter, $G = G_o - 2 \cdot b$	819.250 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 819.2501^2 \cdot 1206.862$$

$$= 64874.926 \text{ Kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P + 2 \cdot l_p \cdot b_{Part} \cdot m_{Part} \cdot P$$

$$= 2 \cdot 6.3749 \cdot \pi \cdot 819.2501 \cdot 4.25 \cdot 1206.86$$

$$+ 2 \cdot 732.0 \cdot 5.0 \cdot 2.75 \cdot 1206.8616$$

$$= 19641.311 \text{ Kgf}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot B_{cor}^2 \cdot P$$

$$= \pi/4 \cdot 806.4^2 \cdot 1206.8616$$

$$= 62855.742 \text{ Kgf}$$



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FileName : E-0102

Page 31 of 190

Flg Calc [Int P]: RHS Shell Fl Flng: 12 10:30am Sep 9, 2026

Pressure Force on Flange Face [Ht]:

$$= H - H_d$$

$$= 64875 - 62856$$

$$= 2019.186 \text{ Kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(W_{o,1} + W_{o,2} + H'p, 0)$$

$$= \max(64875 + 19641 + 0, 0)$$

$$= 84516.242 \text{ Kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 69.64 \cdot 6.3749 \cdot \pi \cdot 819.25 + 25.51 \cdot 5.0 \cdot 732.0$$

$$= 126032.969 \text{ Kgf}$$

Required Bolt Area [Am]:

$$= \max((W_o + F_a + 4Me/G) / S_b, W_g / S_a)$$

$$= \max((84516 + 0 + 4 \cdot 0 / 819.25) / 172, 126033 / 172)$$

$$= 71.861 \text{ cm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmx]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 24.0 + 6 \cdot 91.8 / (4.25 + 0.5)$$

$$= 163.958 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 920.0 \cdot \sin(\pi / 36)$$

$$= 80.183 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{ (Bs / (2a + t)) }, 1)$$

$$= \max(\sqrt{ (80.183 / (2 \cdot 24.0 + 91.8)) }, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Metric Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	71.861	112.589	
Radial Distance between Hub and Bolts:	33.500	42.000	
Radial Distance between Bolts and Edge:	29.000	30.000	
Circ. Spacing between the Bolts:	57.500	80.183	163.958

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= A_b \cdot S_a / (y \cdot \pi (G_o + G_i))$$

$$= 112.589 \cdot 172.0 / (69.64 \cdot \pi (832.0 + 806.0))$$

$$= 5.404 \text{ mm}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 16 mm
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= S_a \cdot A_b$$

$$= 172.0 \cdot 112.5893$$

$$= 197464.48 \text{ Kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= W_o - H$$

$$= 84516 - 64875$$

$$= 19641.31 \text{ Kgf}$$



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FileName : E-0102

Page 32 of 190

Flg Calc [Int P]: RHS Shell Fl Flng: 12 10:30am Sep 9, 2026

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (920.0 - 819.2501) / 2$$

$$= 50.3750 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5 ((C - B) / 2 + hg)$$

$$= 0.5 ((920.0 - 806.4) / 2 + 50.375)$$

$$= 53.5875 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (920.0 - 806.4 - 14.8) / 2$$

$$= 49.4000 \text{ mm}$$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	62856.	49.4000	1.0000	3105.
Face Pressure, Mt	2019.	53.5875	1.0000	108.
Gasket Load, Mg	19641.	50.3750	1.0000	989.
Gasket Seating, Matm	197464.	50.3750	1.0000	9947.

Total Moment for Operation, Mo 4203. Kg-m.
Total Moment for Gasket seating, Mg 9947. Kg-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 84.240 mm
Hub Ratio, h/h0 = HL / H0 0.368
Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.682

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.862
Factor V per 4.16.5 0.314
Factor f per 4.16.5 1.230

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 980.0 / 806.4$$

$$= 1.215$$

Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169 (K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$$

$$= (0.66845 + (5.7169 (1.215^2 \cdot \log_{10}(1.215))) / (1.215^2 - 1)) / (1.215 - 1)$$

$$= 10.069$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (1.215^2 (1 + 8.55246 \cdot \log_{10}(1.215)) - 1) / ((1.0472 + 1.9448 \cdot 1.215^2) \cdot (1.215 - 1))$$

$$= 1.833$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136 (K^2 - 1) \cdot (K - 1))$$

$$= (1.215^2 (1 + 8.55246 \cdot \log_{10}(1.215)) - 1) / (1.36136 (1.215^2 - 1) \cdot (1.215 - 1))$$

$$= 11.065$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$

$$= (1.215^2 + 1) / (1.215^2 - 1)$$

$$= 5.194$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$= U \cdot g0^2 \cdot h0 / V$$



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 33 of 190

Flg Calc [Int P]: RHS Shell Fl Flng: 12 10:30am Sep 9, 2026

$$= 11.065 \cdot 8.8^2 \cdot 84.24 / 0.314$$

$$= 229828.969 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$= F / h_0$$

$$= 0.862 / 84.24$$

$$= 0.010 \text{ 1/mm}$$

Stress factor from Table 4.16.4 [L]:

$$= (t \cdot e + 1) / T + t^3/d$$

$$= (91.8 \cdot 0.01 + 1) / 1.833 + 91.8^3 / 229828.969$$

$$= 4.424$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot Mop / Bcor) / (L \cdot g1^2)$$

$$= (1.2302 \cdot 4203 / 806.4) / (4.424 \cdot 14.8^2)$$

$$= 64.88 \text{ N./mm}^2$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot Matm / Bcor) / (L \cdot g1^2)$$

$$= (1.2302 \cdot 9947 / 806.4) / (4.424 \cdot 14.8^2)$$

$$= 153.57 \text{ N./mm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$

$$= (2.2518 \cdot 4203 / 806.4) / (4.424 \cdot 91.8^2)$$

$$= 3.09 \text{ N./mm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$$

$$= (2.2518 \cdot 9947 / 806.4) / (4.424 \cdot 91.8^2)$$

$$= 7.31 \text{ N./mm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$$

$$= (10.0688 \cdot 4203 / (91.8^2 \cdot 806.4)) - 5.1938 \cdot 3$$

$$= 45.03 \text{ N./mm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$$

$$= (10.0688 \cdot 9947 / (91.8^2 \cdot 806.4)) - 5.1938 \cdot 7$$

$$= 106.59 \text{ N./mm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (SHo + \max(SRo, STo)) / 2$$

$$= (65 + \max(3, 45)) / 2$$

$$= 54.96 \text{ N./mm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (SHa + \max(SRa, STa)) / 2$$

$$= (154 + \max(7, 107)) / 2$$

$$= 130.08 \text{ N./mm}^2$$

Bolt Stress, Operating [BSO]:

$$= Wm1 / Ab$$

$$= 84516 / 112.5893$$

$$= 73.62 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (Wm2 / Ab)$$

$$= (126033 / 112.5893)$$

$$= 109.78 \text{ N./mm}^2$$



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FileName : E-0102

Page 34 of 190

Flg Calc [Int P]: RHS Shell Fl

Flng: 12 10:30am

Sep 9,2026

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	64.88	207.00	153.57	207.00
Radial Flange	3.09	138.00	7.31	138.00
Tangential Flange	45.03	138.00	106.59	138.00
Maximum Average	54.96	138.00	130.08	138.00
Bolting	73.62	172.00	109.78	172.00

Minimum Required Flange Thickness 92.278 mm

Estimated M.A.W.P. (Operating) 2820 KPa.

Estimated Finished Weight of Flange at given Thk. 194.5 kg.

Estimated Unfinished Weight of Forging at given Thk 245.7 kg.

Flange Stress results per WRC-538 at computed S_{fmax} (N./mm²):

		Actual	Operating Allowed	stress ratio	Gasket Actual	Seating Allowed	stress ratio
Longitudinal Hub	Sh	499.37	499.99	0.999	499.37	499.99	0.999
Tangential Flange	St	345.42	374.99	0.921	345.42	374.99	0.921
Tangential Flange	Sto	168.48	250.00	0.674	168.48	250.00	0.674
Combined	Sh/f + St	749.99	749.99	1.000	749.99	749.99	1.000
Combined	Sr + St	369.10	749.99	0.492	369.10	749.99	0.492
Combined	Sh + Sto	667.85	749.99	0.890	667.85	749.99	0.890
Radial Flange	Sr	23.68	499.99	0.047	23.68	499.99	0.047
Maximum Stress Ratio				1.000			1.000



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 35 of 190

Flg Calc [Int P]: RHS Shell F1

Flng: 12 10:30am Sep 9, 2026

Coefficients for f and $f2$

	f	$f2$	$f3$
a	-0.71375912	-0.01447638	-0.06456312
b	-0.12846279	-0.27814745	-0.08232239
c	1.08037907	0.01035395	0.13691677
d	0.99766848	1.37984158	-0.77879888
e	1.21823466	-0.34346764	0.14909019
f	0.01024612	0.02536192	0.00660062
g	0.4262313	-0.0014508	0.0616252
h	1.4204976	2.25543162	0.74639834
i	-0.70278181	-0.60889597	-0.079595
j	-0.02483937	-0.15735462	0.09851673
k	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g1}{g0} \right) + e \left(\frac{h}{h0} \right) + g \left(\frac{g1}{g0} \right)^2 + i \left(\frac{h}{h0} \right)^2 + k \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}{\left[1 + b \left(\frac{g1}{g0} \right) + d \left(\frac{h}{h0} \right) + f \left(\frac{g1}{g0} \right)^2 + h \left(\frac{h}{h0} \right)^2 + j \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}$$

Hub Stress Correction Factor [f]:

= max(1; F1num / F1den)

= max(1; 1.6751/1.3571)

= 1.2343

Hub Stress Correction Factor [f2]:

= max(0.25; fn2) if $h/h0 < 0.35$, else max (0.25; fn3)

= max(0.25; 0.5482/1.3198) if $h/h0 < 0.35$, else max (0.25; 0.3147/0.7557)

= 0.4164

In this case, the hub ratio $h/h0$, 0.3680 was ≥ 0.35 .

PV Elite iterates on the assembly bolt load stress, Sfmax, until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	Sbmax	507.5 N./mm ²
Minimum permissible bolt stress	Sbmin	290.0 N./mm ²
Max. permissible bolt stress prior to flange damage	Sfmax	557.4 N./mm ²
Target assembly gasket stress	SgT	207.0 N./mm ²
Maximum permissible gasket stress	Sgmax	276.0 N./mm ²
Minimum gasket operating stress	Sgmin-O	97.0 N./mm ²
Minimum gasket seating stress	Sgmin-S	140.0 N./mm ²
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	Phig	0.700



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 36 of 190

Flg Calc [Int P]: RHS Shell Fl Flng: 12 10:30am Sep 9, 2026

Root Area of a Single Bolt	Abs	3.1275 cm ²
Max. permissible single flange rotation for gasket	Thetagmax	1.000 deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	250.0 N./mm ²
Flange Material Operating Yield Stress	Syo	230.1 N./mm ²
Single flange rotation at Sfmax	Thetaamax	0.409 deg

Gasket Contact Area [Ag]:

$$= \pi/4 (GO.D.^2 - GI.D.^2) + (PartFactor \cdot Ip \cdot bPart)$$

$$= \pi/4 (832.0^2 - 806.0^2) + (0.5 \cdot 732.0 \cdot 5.0)$$

$$= 352.8 \text{ cm}^2$$

Target Bolt Stress [Sbsel]: (Eq. O-1)

$$= SgT \cdot Ag / (nb \cdot Ab)$$

$$= 207.0 \cdot 352.785 / (36 \cdot 3.127)$$

$$= 648.6 \text{ N./mm}^2$$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)

$$= \min(Sbsel; Sbmax)$$

$$= \min(648.61; 507.5)$$

$$= 507.5 \text{ N./mm}^2$$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)

$$= \max(Sbsel; Sbmin)$$

$$= \max(507.5; 290.0)$$

$$= 507.5 \text{ N./mm}^2$$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)

$$= \min(Sbsel; Sfmax)$$

$$= \min(507.5; 557.397)$$

$$= 507.5 \text{ N./mm}^2$$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)

$$:: Sbsel \geq Sgmin-S(Ag / (nb \cdot Abs))$$

$$:: 507.5 \geq 140.0(352.785 / (36 \cdot 3.127))$$

$$:: 507.5 \geq 438.7 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Operating Stress is maintained. (Eq. O-8)

$$:: Sbsel \geq (Sgmin-O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$$

$$:: 507.5 \geq (97.0 \cdot 352.785 + \pi/4 \cdot 1206.862 \cdot 806.0^2) / (0.7 \cdot 3.127 \cdot 36)$$

$$:: 507.5 \geq 512.3 \text{ N./mm}^2 \text{ Check Fails}$$

Check if Gasket Maximum Stress is achieved. (Eq. O-9)

$$:: Sbsel \leq Sgmax(Ag / (nb \cdot Abs))$$

$$:: 507.5 \leq 276.0(352.785 / (36 \cdot 3.127))$$

$$:: 507.5 \leq 864.8 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Flange Rotation Limit is exceeded. (Eq. O-10)

$$:: Sbsel \leq Sfmax(\text{thetagmax} / \text{thetaamax})$$

$$:: 507.5 \leq 557.397(1.0/0.409)$$

$$:: 507.5 \leq 1362.4 \text{ N./mm}^2 \text{ Check Passes}$$

Assembly Bolt Torque [Tb]: (Eq. O-2)

$$= Sbsel \cdot K \cdot Abs \cdot \phi b$$

$$= 507.5 \cdot 0.2 \cdot 3.127 \cdot 24.0$$

$$= 77.7 \text{ Kg-m.}$$

Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 138 N./mm²

Bolt Allowable Stress for: SA-193 B7 Sb 171 N./mm²

Note: Since Sb >= SbB8, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 37 of 190

Flg Calc [Int P]: RHS Shell F1 Flng: 12 10:30am Sep 9, 2026

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 9947.5 / 1.0 \cdot 9806.637 \cdot 0.314 / (4.117 \cdot 202713 \cdot 8.8^2 \cdot 84.24 \cdot 0.3)$
 $= 0.978$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 4202.8 / 1.0 \cdot 9806.637 \cdot 0.314 / (4.117 \cdot 199226 \cdot 8.8^2 \cdot 84.24 \cdot 0.3)$
 $= 0.420$ (should be ≤ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 9947.5 / 1.0 \cdot 9806.637 \cdot 0.314 / (4.424 \cdot 202713 \cdot 8.8^2 \cdot 84.24 \cdot 0.3)$
 $= 0.910$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 4202.8 / 1.0 \cdot 9806.637 \cdot 0.314 / (4.424 \cdot 199226 \cdot 8.8^2 \cdot 84.24 \cdot 0.3)$
 $= 0.391$ (should be ≤ 1)

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.304, Temperature Reduction per Fig. UCS 66.1 = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	59.21	207.00	136.55	207.00
Radial Flange	4.05	138.00	9.33	138.00
Tangential Flange	36.58	138.00	84.35	138.00
Maximum Average	47.89	138.00	110.45	138.00
Bolting	73.72	172.00	110.56	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 83.160 mm

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FileName : E-0102

Page 38 of 190

Internal Pressure Calculations: Step: 5 10:30am Sep 9,2026

Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd KPa.	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) N./mm ²	Element#
LHS Dish End		1057.9	14	3.2	800	138	1
Left Ch shell		1057.9	14	3.2	800	138	2
LHS Ch Flange		1056.9	103	3.2	800	136.8	3
LHS Shell Flg		1206.9	103	3.2	800	138	4
Main Shell		1207.9	18	3.2	800	138	5
RHS Dish End		1207.9	14	3.2	800	138	6

Element Required Thickness and MAWP :

From	To	Design Pressure KPa.	M.A.W.P. Corroded KPa.	M.A.P. New & Cold KPa.	Minimum Thickness mm	Required Thickness mm	Element#
LHS Dish End		1050	2994.5	4093.01	11.9	6.26109	1
Left Ch shell		1050	3629.81	4730.3	14	6.30535	2
LHS Ch Flange		1050	2812.86	2815.86	95	92.2782	3
LHS Shell Flg		1200	2812.86	2815.87	95	92.2782	4
Main Shell		1200	4948.07	6046.28	18	6.74799	5
RHS Dish End		1200	2994.5	4093.01	11.9	6.69551	6

Warning: Channel Side MAWP governed by a standard Flange!

Summary of Heat Exchanger Maximum Allowable Working Pressures :

Note:

For Exchanger designs, the following values include MAWPs that consider the tubesheet, tubes, tube/tubesheet joint etc. These values were determined by iteration. Review the tubesheet analysis report for more information.

Shell Side MAWP = 1506 KPa.
 Shell Side MAPnc = 1694 KPa.
 Channel Side MAWP = 1088 KPa.
 Channel Side MAPnc = 1694 KPa.

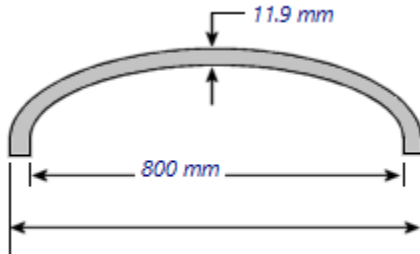
Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: LHS Dish End

Elliptical Head Design Information:			
Design Pressure P	1057.873 KPa.	Int. Design Temperature	230 °C
Material	SA-516 70	Ext. Design Temperature	230 °C
Length	50.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.2000 mm	Allowable Stress (ope) S	138.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	800.000 mm	Actual Stress	48.6 N./mm ²
Spec. Min. Thk t,e, or sa	11.9000 mm	Spec. Nominal Thk.	14.000 mm
Surface Area	8732.3 cm ²	SG of Contents	1.000
Element Volume	92169.9 cm ³	Weight of Contents	92.1 Kg
Empty Weight	99.9 Kg	Operating Weight	192.0 Kg
		Comp. All. Ope. Stress	108.3 N./mm ²
		Aspect Ratio	2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 10.000 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 5.000 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (1057.873 \cdot 806.4 \cdot 0.99) / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 1057.873)$$

$$= 3.0611 + 3.2000 = 6.2611 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 7.873 KPa.

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 8.7) / (0.99 \cdot 806.4 + 0.2 \cdot 8.7)$$

$$= 3002 - 8 = 2994 \text{ KPa.}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 11.9) / (1.0 \cdot 800.0 + 0.2 \cdot 11.9)$$

$$= 4093 \text{ KPa.}$$



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FileName : E-0102

Page 40 of 190

Element: LHS Dish End

Step: 6 10:30am Sep 9,2026

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (1057.873 \cdot (0.99 \cdot 806.4 + 0.2 \cdot 8.7)) / (2 \cdot 1.0 \cdot 8.7)$$

$$= 48.623 \text{ N./mm}^2$$

Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \text{ per UG-27 (c)(1)}$$

$$= (1057.873 \cdot 403.2) / (138.0 \cdot 1.0 - 0.6 \cdot 1057.873) + 3.2$$

$$= 6.305 \text{ mm}$$

Straight Flange Maximum Allowable Working Pressure:

[Less Operating Hydrostatic Head Pressure of 7.873 KPa.](#)

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 10.8) / (403.2 + 0.6 \cdot 10.8)$$

$$= 3638 - 8 = 3630 \text{ KPa.}$$

Factor K, corroded condition [Kcor]:

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth})))^2 / 6$$

$$= (2 + (806.4 / (2 \cdot 203.2)))^2 / 6$$

$$= 0.989543$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 7.343 %

[Note: Please Check Requirements of UCS-79 as Elongation is > 5%.](#)

MDMT Calculations in the Knuckle Portion:

Govrn. thk, tg = 11.9, tr = 3.149, c = 3.2 mm, E* = 1.0

Thickness Ratio = $tr \cdot E* / (tg - c) = 0.362$, Temp. Reduction = 69 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, tg = 14.0, tr = 3.195, c = 3.2 mm, E* = 1.0

Thickness Ratio = $tr \cdot E* / (tg - c) = 0.296$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-18 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1088.365 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-45 °C
Governing Thk.	11.900 mm	Basic MDMT	-23 °C
Stress/Thk. Ratio	0.362	Temperature Difference	25 °C
UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

[Note: Temperature values displayed above are for the location with the highest basic MDMT](#)

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 10.000 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 5.000 mm



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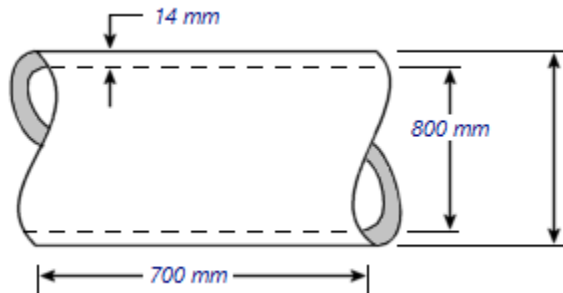
Page 41 of 190

Element: Left Ch shell

Step: 7 10:30am Sep 9,2026

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Left Ch shell

Cylindrical Shell Design Information:			
Design Pressure P	1057.873 KPa.	Int. Design Temperature	230 °C
Material	SA-516 70	Ext. Design Temperature	230 °C
Length	700.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.2000 mm	Allowable Stress (ope) S	138.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	800.000 mm	Actual Stress	40.1 N./mm ²
Spec. Min. Thk t,e, or sa	14.0000 mm	Spec. Nominal Thk.	14.000 mm
Surface Area	18208.7 cm ²	SG of Contents	1.000
Element Volume	351921.4 cm ³	Weight of Contents	351.7 Kg
Empty Weight	194.2 Kg	Operating Weight	545.9 Kg
		Comp. All. Ope. Stress	112.2 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (1057.873 \cdot 403.2) / (138.0 \cdot 1.0 - 0.6 \cdot 1057.873) \\
 &= 3.1053 + 3.2000 = 6.3053 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 7.873 KPa.

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 1.0 \cdot 10.8) / (403.2 + 0.6 \cdot 10.8) \\
 &= 3638 - 8 = 3630 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 1.0 \cdot 14.0) / (400.0 + 0.6 \cdot 14.0) \\
 &= 4730 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (1057.873 \cdot (403.2 + 0.6 \cdot 10.8)) / (1.0 \cdot 10.8) \\
 &= 40.131 \text{ N./mm}^2
 \end{aligned}$$



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FileName : E-0102

Page 42 of 190

Element: Left Ch shell

Step: 7 10:30am

Sep 9,2026

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 1.720 %

Minimum Design Metal Temperature Results:

Govrn. thk, $t_g = 14.0$, $t_r = 3.195$, $c = 3.2$ mm, $E^* = 1.0$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.296$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1088.365 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-45 °C
Governing Thk.	14.000 mm	Basic MDMT	-18 °C
Stress/Thk. Ratio	0.296	Temperature Difference	86 °C
UCS-66(c) red.	No	Computed Min. Temp.	-104 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	100 %

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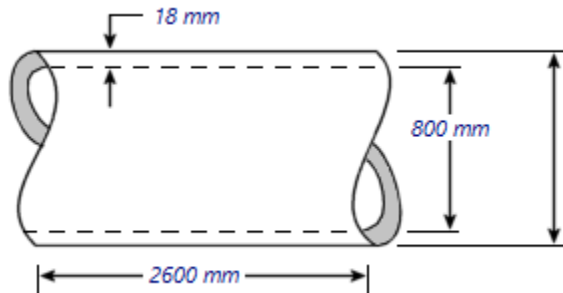
Page 43 of 190

Element: Main Shell

Step: 8 10:30am Sep 9,2026

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Main Shell

Cylindrical Shell Design Information:			
Design Pressure P	1207.873 KPa.	Int. Design Temperature	82 °C
Material	SA-516 70	Ext. Design Temperature	82 °C
Length	2600.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.2000 mm	Allowable Stress (ope) S	138.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	800.000 mm	Actual Stress	33.6 N./mm ²
Spec. Min. Thk t,e, or sa	18.0000 mm	Spec. Nominal Thk.	18.000 mm
Surface Area	68285.7 cm ²	SG of Contents	1.000
Element Volume	1063378.6 cm ³	Weight of Contents	1062.7 Kg
Empty Weight	932.1 Kg	Operating Weight	1994.9 Kg
		Comp. All. Ope. Stress	141.5 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (1207.873 \cdot 403.2) / (138.0 \cdot 1.0 - 0.6 \cdot 1207.873) \\
 &= 3.5480 + 3.2000 = 6.7480 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 7.873 KPa.

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 1.0 \cdot 14.8) / (403.2 + 0.6 \cdot 14.8) \\
 &= 4956 - 8 = 4948 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 1.0 \cdot 18.0) / (400.0 + 0.6 \cdot 18.0) \\
 &= 6046 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (1207.873 \cdot (403.2 + 0.6 \cdot 14.8)) / (1.0 \cdot 14.8) \\
 &= 33.633 \text{ N./mm}^2
 \end{aligned}$$



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 44 of 190

Element: Main Shell

Step: 8 10:30am Sep 9,2026

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 2.200 %

Minimum Design Metal Temperature Results:

Govrn. thk, $t_g = 18.0$, $t_r = 4.429$, $c = 3.2$ mm, $E^* = 1.0$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.299$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -11 °C

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1505.975 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-30 °C
Governing Thk.	18.000 mm	Basic MDMT	-11 °C
Stress/Thk. Ratio	0.299	Temperature Difference	93 °C
UCS-66(c) red.	No	Computed Min. Temp.	-104 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	100 %

EXTERNAL PRESSURE RESULTS for: Main Shell

Cylindrical Shell From 50 to 60 Ext. Chart: CS-2 at 82 °C

Elastic Modulus from Chart: CS-2 at 82 °C: 199943392 KPa.

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
14.800	836.00	2716.67	56.49	3.2496	0.0008922	81.71

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 81.7149) / (3 \cdot 56.4865) = 1928.7239 \text{ KPa.}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
4.346	836.00	2716.67	192.36	3.2496	0.0001463	14.62

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 14.6235) / (3 \cdot 192.3602) = 101.3562 \text{ KPa.}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
14.800	836.00	96957.34	56.49	50.0000	0.0003491	34.90

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 34.9004) / (3 \cdot 56.4865) = 823.7584 \text{ KPa.}$$



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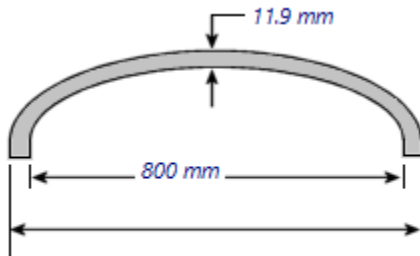
Page 45 of 190

Element: RHS Dish End

Step: 9 10:30am Sep 9,2026

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: RHS Dish End

Elliptical Head Design Information:				
Design Pressure	P	1207.873 KPa.	Int. Design Temperature	82 °C
Material		SA-516 70	Ext. Design Temperature	82 °C
Length		50.000 mm	External Pressure Chart	CS-2
			UNS Number	K02700
Int. Corr. All.	c	3.2000 mm	Allowable Stress (ope)	S 138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa 138.0 N./mm ²
Inside Diameter	Di	800.000 mm	Actual Stress	55.5 N./mm ²
Spec. Min. Thk t,e, or sa		11.9000 mm	Spec. Nominal Thk.	14.000 mm
Surface Area		8732.3 cm ²	SG of Contents	1.000
Element Volume		92169.9 cm ³	Weight of Contents	92.1 Kg
Empty Weight		99.9 Kg	Operating Weight	192.0 Kg
			Comp. All. Ope. Stress	141.5 N./mm ²
			Aspect Ratio	2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 10.000 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 5.000 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (1207.873 \cdot 806.4 \cdot 0.99) / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 1207.873)$$

$$= 3.4955 + 3.2000 = 6.6955 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 7.873 KPa.

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 8.7) / (0.99 \cdot 806.4 + 0.2 \cdot 8.7)$$

$$= 3002 - 8 = 2994 \text{ KPa.}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 11.9) / (1.0 \cdot 800.0 + 0.2 \cdot 11.9)$$

$$= 4093 \text{ KPa.}$$



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 46 of 190

Element: RHS Dish End

Step: 9 10:30am Sep 9,2026

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (1207.873 \cdot (0.99 \cdot 806.4 + 0.2 \cdot 8.7)) / (2 \cdot 1.0 \cdot 8.7)$$

$$= 55.517 \text{ N./mm}^2$$

Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \text{ per UG-27 (c)(1)}$$

$$= (1207.873 \cdot 403.2) / (138.0 \cdot 1.0 - 0.6 \cdot 1207.873) + 3.2$$

$$= 6.748 \text{ mm}$$

Straight Flange Maximum Allowable Working Pressure:

[Less Operating Hydrostatic Head Pressure of 7.873 KPa.](#)

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 10.8) / (403.2 + 0.6 \cdot 10.8)$$

$$= 3638 - 8 = 3630 \text{ KPa.}$$

Factor K, corroded condition [Kcor]:

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth})))^2 / 6$$

$$= (2 + (806.4 / (2 \cdot 203.2)))^2 / 6$$

$$= 0.989543$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 7.343 %

[Note: Please Check Requirements of UCS-79 as Elongation is > 5%.](#)

MDMT Calculations in the Knuckle Portion:

Govrn. thk, tg = 11.9, tr = 4.359, c = 3.2 mm, E* = 1.0

Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.501$, Temp. Reduction = 32 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, tg = 14.0, tr = 4.429, c = 3.2 mm, E* = 1.0

Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.41$, Temp. Reduction = 48 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-18 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1505.975 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-30 °C
Governing Thk.	11.900 mm	Basic MDMT	-23 °C
Stress/Thk. Ratio	0.501	Temperature Difference	25 °C
UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

[Note: Temperature values displayed above are for the location with the highest basic MDMT](#)

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 10.000 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 5.000 mm

EXTERNAL PRESSURE RESULTS for: RHS Dish End

Elliptical Head From 60 to 80 Ext. Chart: CS-2 at 82 °C



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FileName : E-0102

Page 47 of 190

Element: RHS Dish End

Step: 9 10:30am Sep 9,2026

Elastic Modulus from Chart: CS-2 at 82 °C: 199943392 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Outer Dia	Do/t	Factor A	Factor B
8.700	823.80	94.69	0.0014668	96.21

$$MAEP = B / (K0 * Do / t) = 96.2056 / (0.9 * 94.6897) = 1128.8339 \text{ KPa.}$$

Results for Required Thickness				Tca
Tca	Outer Dia	Do/t	Factor A	Factor B
2.112	823.80	390.13	0.0003560	35.59

$$MAEP = B / (K0 * Do / t) = 35.5925 / (0.9 * 390.1323) = 101.3627 \text{ KPa.}$$

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PV Elite 28 Licensee: SL User

FileName : E-0102

Page 48 of 190

Shop Pressure Test Results: Step: 10 10:30am Sep 9,2026

Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	1957.767 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	1560.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	2194.171 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	1656.572 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	1716.000 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	1716.000 KPa.
App 27-4	= M.A.W.P.	1505.975 KPa.

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	1414.874 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	1365.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	2194.171 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	1197.201 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	1501.500 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	1501.500 KPa.
App 27-4	= M.A.W.P.	1088.365 KPa.

Horizontal Test performed per: UG-99c

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & KPa.):

From To	Stress	Allowable	Ratio	Pressure
LHS Dish End	101.2	234.0	0.433	2202.01
Left Ch shell	83.5	234.0	0.357	2202.01
Main Shell	61.3	234.0	0.262	2202.01
RHS Dish End	101.2	234.0	0.433	2202.01

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N1-Tube inlet	Nozzle	118.00	118.00	1.000
N1-Tube Outlet	Nozzle	118.00	118.00	1.000
N3-Shell inlet	Nozzle	118.00	118.00	1.000
N3-Shell inlet	Pad	138.00	138.00	1.000
N4-Shell outlet	Nozzle	118.00	118.00	1.000
N4-Shell outlet	Pad	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
LHS Dish End	138.00	138.00	1.000
Left Ch shell	138.00	138.00	1.000
LHS Ch Flange	138.00	136.80	1.009
LHS Shell Flg	138.00	138.00	1.000
Main Shell	138.00	138.00	1.000
RHS Dish End	138.00	138.00	1.000

Minimum 1.000



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 49 of 190

Shop Pressure Test Results:

Step: 10 10:30am

Sep 9, 2026

Stress ratios for Exchanger Materials (N./mm²):

Description	Ambient	Operating	Ratio
Tube Material	108.00	108.00	1.000
Tubesheet Material	138.00	136.80	1.009

Minimum

1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
N1-Tube inlet	21.27	216.00	0.098
N1-Tube Outlet	22.96	216.00	0.106
N3-Shell inlet	23.56	216.00	0.109
N4-Shell outlet	23.56	216.00	0.109

Elements Suitable for Test Pressure.

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PV Elite 28 Licensee: SL User

FileName : E-0102

Page 50 of 190

Field Pressure Test Results: Step: 11 10:30am Sep 9, 2026

Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	1957.767 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	1560.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	2194.171 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	1656.572 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	1716.000 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	1716.000 KPa.
App 27-4	= M.A.W.P.	1505.975 KPa.

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	1414.874 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	1365.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	2194.171 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	1197.201 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	1501.500 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	1501.500 KPa.
App 27-4	= M.A.W.P.	1088.365 KPa.

UG-99(b)(35), Test Pressure Calculation [Shell Side]:

= Test Factor • Design Pressure • Stress Ratio

= 1.3 • 1200.0 • 1.0

= 1560.000 KPa.

UG-99(b) Note 35, Test Pressure Calculation [Channel Side]:

= Test Factor • Design Pressure • Stress Ratio

= 1.3 • 1050.0 • 1.0

= 1365.000 KPa.

Horizontal Test performed per: UG-99b (Note 35)

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & KPa.):

From To	Stress	Allowable	Ratio	Pressure
LHS Dish End	63.1	234.0	0.270	1372.84
Left Ch shell	52.1	234.0	0.223	1372.84
Main Shell	43.7	234.0	0.187	1567.84
RHS Dish End	72.1	234.0	0.308	1567.84

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N1-Tube inlet	Nozzle	118.00	118.00	1.000
N1-Tube Outlet	Nozzle	118.00	118.00	1.000
N3-Shell inlet	Nozzle	118.00	118.00	1.000
N3-Shell inlet	Pad	138.00	138.00	1.000
N4-Shell outlet	Nozzle	118.00	118.00	1.000
N4-Shell outlet	Pad	138.00	138.00	1.000

Minimum

1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
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PV Elite 28 Licensee: SL User

FileName : E-0102

Page 51 of 190

Field Pressure Test Results: Step: 11 10:30am Sep 9, 2026

LHS Dish End	138.00	138.00	1.000
Left Ch shell	138.00	138.00	1.000
LHS Ch Flange	138.00	136.80	1.009
LHS Shell Flg	138.00	138.00	1.000
Main Shell	138.00	138.00	1.000
RHS Dish End	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Exchanger Materials (N./mm²):

Description	Ambient	Operating	Ratio
Tube Material	108.00	108.00	1.000
Tubesheet Material	138.00	136.80	1.009

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
N1-Tube inlet	13.26	216.00	0.061
N1-Tube Outlet	14.31	216.00	0.066
N3-Shell inlet	16.77	216.00	0.078
N4-Shell outlet	16.77	216.00	0.078

Elements Suitable for Test Pressure.

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PV Elite 28 Licensee: SL User

FileName : E-0102

Page 52 of 190

ASME TS Calc: Pressure Test

Case: 1 10:30a Sep 9, 2026

Input Echo, Tubesheet Number 1, Description: U-Tube

Shell Data:

Main Shell Description: Main Shell

Shell Test Pressure		1560.00 KPa.
Shell Thickness	ts	18.0000 mm
Shell Internal Corrosion Allowance	cas	0.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	800.000 mm
Shell Temperature for Internal Pressure	Ts	21.11 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	138.00 N./mm ²
Shell Allowable Stress at Ambient		138.00 N./mm ²

Channel Data:

Channel Description: Left Ch shell

Channel Type:		Cylinder
Channel Test Pressure		1365.00 KPa.
Channel Thickness	tc	14.0000 mm
Channel Internal Corrosion Allowance	cac	0.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	800.000 mm
Channel Design Temperature	TEMPC	21.11 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	138.00 N./mm ²
Channel Allowable Stress at Ambient		138.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	170
Tube Wall Thickness	et	2.3200 mm
Tube Outside Diameter	D	25.4000 mm
Total Straight Tube Length	Lt	2438.00 mm
Straight Tube Length (bet. inner tubsht faces) L		2373.00 mm
Design Temperature of the Tubes		21.11 °C
Tube Material		SA-334 1
Tube Material UNS Number		K03008
Is this a Welded Tube		No
Tube Material Specification used	Smls. & wld. tube	
Tube Allowable Stress at Temperature		108.00 N./mm ²
Tube Allowable Stress At Ambient		108.00 N./mm ²
Tube Yield Stress At design Temperature	Syt	205.00 N./mm ²
Tube Pitch (Center to Center Spacing)	P	38.1000 mm
Tube Layout Pattern		Triangular

Fillet Weld Leg	af	2.3200 mm
Groove Weld Leg	ag	2.3200 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No

Radius to Outermost Tube Hole Center	ro	370.687 mm
Largest Center-to-Center Tube Distance	U1	0.0000 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	5.0000 mm
Area of Tube Layout	Ap	2517.1 cm ²
Perimeter of Tube Layout	Cp	2062.1 mm



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FileName : E-0102

Page 53 of 190

ASME TS Calc: Pressure Test

Case: 1 10:30a Sep 9,2026

Tubesheet Data:

Tubesheet TYPE: U-tube, Gasketed both Sides, Conf. d

Tubesheet Design Metal Temperature T 21.11 °C
Tubesheet Material SA-765 II

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number K03047
Tubesheet Allowable Stress at Temperature S 138.00 N./mm²
Tubesheet Allowable Stress at Ambient Tt 138.00 N./mm²
Thickness of Tubesheet h 65.0000 mm
Tubesheet Corr. Allowance (Shell side) Cats 0.0000 mm
Tubesheet Corr. Allowance (Channel side) Catc 0.0000 mm
Tubesheet Outside Diameter A 980.000 mm

Additional Data for Stepped Tubesheets:

Is the Tubesheet Stepped? Yes
Is the Tubesheet Flat on Tubeside? No
Step 1 Diameter on the Tubeside dt1 832.00 mm
Step 1 Depth on the Tubeside ht1 8.00 mm
Step 2 Diameter on the Tubeside dt2 797.00 mm
Step 2 Depth on the Tubeside ht2 8.00 mm
Is the Tubesheet Flat on Shellside? No
Calculated Tubesheet Diameter per Div 2, 4.18.4 832.00 mm

Note: Tubesheet diameter is now: 832.000 mm Div 2, 4.18.4.

Dimension G for the Channel Side Gc 819.250 mm
Area of the Untubed Lanes AL 0.0 cm²

Junction Stress Reduction option Perform Plastic Calculation
Perform Differential Pressure Design No
Run Multiple Load Cases Yes
Channel Side Min. Design Pressure Ptd,min 103.4195 KPa.

Additional Data for Gasketed Tubesheets:

Tubesheet Gasket on which Side Both
Flange Outside Diameter A 980.000 mm
Flange Inside Diameter B 800.000 mm
Flange Face Outside Diameter Fod 835.000 mm
Flange Face Inside Diameter Fid 800.000 mm
Gasket Outside Diameter Go 832.000 mm
Gasket Inside Diameter Gi 806.000 mm
Small end Hub thk. g0 12.0000 mm
Large end Hub thk. g1 18.0000 mm
Gasket Factor, m 4.25
Gasket Design Seating Stress y 69.64 N./mm²
Flange Facing Sketch Code Sketch 1a
Gasket Thickness tg 3.0000 mm
Full face Gasket Flange Option Program Selects
Length of Partition Gasket lp 732.000 mm
Width of Partition Gasket wp 10.0000 mm
Partition Gasket Factor, mPart 2.7500
Partition Gasket Design Seating Stress yPart 25.51 N./mm²
Partition Gasket Facing Sketch Code Sketch 1a
Partition Gasket Column for Gasket Seating Code Column II

Bolting Information:

Diameter of Bolt Circle C 920.000 mm
Nominal Bolt Diameter dB 24.0000 mm
Thread Series TEMA Metric Thread
Number of Bolts n 36



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 54 of 190

ASME TS Calc: Pressure Test

Case: 1 10:30a Sep 9,2026

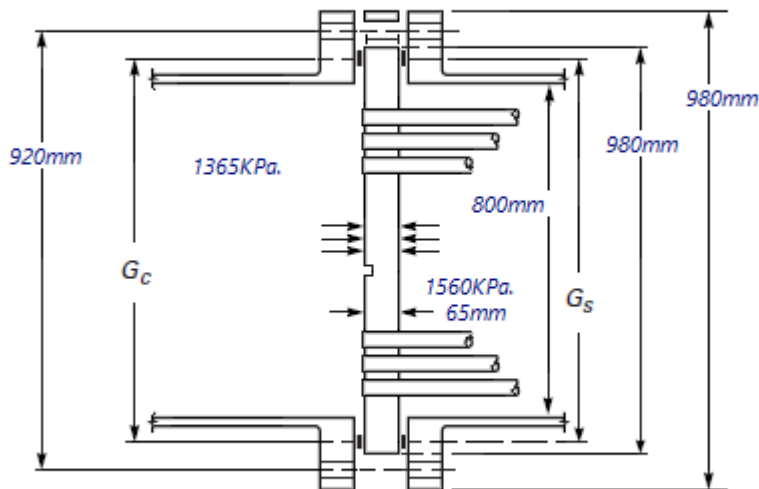
Bolt Material	SA-193 B7
Bolt Allowable Stress At Temperature	Sb 172.00 N./mm ²
Bolt Allowable Stress At Ambient	Sa 172.00 N./mm ²
Weld between Flange and Shell/Channel	0.0000 mm

Alternate Flange Operating Bolt Load, Wm1	109388.98 Kg
Alternate Flange Seating Bolt Load, Wm2	126883.20 Kg
Alternate Flange Design Bolt Load, W	197464.48 Kg

Note: The full bolt load Sa*Ab, was used for all load cases.

Tubesheet Integral with	None
Tubesheet Extended as Flange	Yes
Thickness of Extended Portion of Tubesheet	Tf 49.0000 mm
Is Bolt Load Transferred to the Tubesheet	Yes

Tubesheet Gasketed w/Shell and Channel



Configuration d:

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties :

Shell - TM-1 Carbon Steels with C> 0.3%

Elastic Mod. at Design Temperature	21.1 °C	0.20132E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20132E+09 KPa.

Channel - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tubes - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Tubesheet Design Temp.	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

TubeSheet - TM-1 Carbon Steels with C<= 0.3%



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 55 of 190

ASME TS Calc: Pressure Test

Case: 1 10:30a Sep 9, 2026

Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:

$$= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P) \text{ per Appendix 1-1 (a)(1)}$$

$$= (1365.0 \cdot (25.4/2 - 0)) / (108.0 \cdot 1.0 + 0.4 \cdot 1365.0)$$

$$= 0.1597 + 0.0000 = 0.1597 \text{ mm}$$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart	CS-2	at	21.11 °C
Elastic Modulus for Material			199943392.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLen	D/T	L/D	Factor A	B
2.3200	25.40	2743.69	10.95	50.0000	0.0091770	122.73

$$EMAWP = (4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 122.731) / (3 \cdot 10.9483) = 14945.9004 \text{ KPa.}$$

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLen	D/T	L/D	Factor A	B
0.5707	25.40	2743.69	44.50	50.0000	0.0005554	55.53

$$EMAWP = (4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 55.5268) / (3 \cdot 44.5037) = 1663.4885 \text{ KPa.}$$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.5707 mm
Allowable Internal Pressure at Corroded thickness	21282.78 KPa.
Required Internal Design Pressure	1365.00 KPa.
Allowable External Pressure at Corroded thickness	14945.90 KPa.
Required External Design Pressure	1663.42 KPa.
Required Thickness due to Shell Side pressure	0.5707 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Intermediate Calculations For Tubesheets Extended As Flanges:

ASME Code, Section VIII Division 1, 2025

Gasket Contact Width,	$N = (G_{oc} - G_{ic}) / 2$	13.000 mm
Basic Gasket Width,	$b_0 = N / 2$	6.500 mm
Effective Gasket Width,	$b = \sqrt{b_0} \cdot 2.5$	6.422 mm
Gasket Reaction Diameter,	$G = G_o - 2.0 \cdot b$	819.156 mm

Results for ASME U-tube Tubesheet Calculations for Configuration d:

Per Edition 2025, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:

$$= 2 \cdot r_o + dt$$

$$= 2 \cdot 370.687 + 25.4$$

$$= 766.774 \text{ mm}$$

Basic ligament efficiency for shear [μ]:

$$= (p - dt) / p$$

$$= (38.1 - 25.4) / 38.1$$

$$= 0.333$$

ASME Section VIII-2 Part 4.18.7.4 Step 2:



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 56 of 190

ASME TS Calc: Pressure Test

Case: 1 10:30a Sep 9, 2026

Ratio [ps]:

$$= G_s / D_o \text{ (Configurations d, e, f)}$$

$$= 819.1555 / 766.774$$

$$= 1.0683$$

Ratio [pc]:

$$= G_c / D_o \text{ (Configurations d)}$$

$$= 819.2501 / 766.774$$

$$= 1.0684$$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [p]:

$$= l_{tx} / h$$

$$= 0 / 65.0$$

$$= 0.000 \text{ (must be } 0 \leq p \leq 1 \text{)}$$

Effective tube hole Diameter [d*]:

$$= \max(d_t - 2t_t(E_t T / E) \cdot (S_t T / S) \cdot (p); d_t - 2t_t)$$

$$= \max(25.4 - 2 \cdot 2.32 (0.20270E+09 / 0.20270E+09) \cdot (107 / 137) \cdot (0); 25.4 - 2 \cdot 2.32)$$

$$= 25.4000 \text{ mm}$$

Effective tube pitch [p*]:

$$= p / \sqrt{ (1 - 4 \cdot \min(A_l; 4 \cdot D_o \cdot p) / (\pi \cdot D_o^2)) }$$

$$= 38.1 / \sqrt{ (1 - 4 \cdot \min(0; 4 \cdot 766.774 \cdot 38.1) / (\pi \cdot 766.774^2)) }$$

$$= 38.1000 \text{ mm}$$

Effective ligament efficiency for bending [μ*]:

$$= (p^* - d^*) / p^*$$

$$= (38.1 - 25.4) / 38.1$$

$$= 0.3333$$

$h/p = 1.706037$; $\mu^* = 0.333333$; $E^* = 68162000 \text{ KPa}$.

$E^*/E = 0.336268$; $v^* = 0.324996$

Skip Step 4 for Configuration d :

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$= A / D_o$$

$$= 832.0 / 766.774$$

$$= 1.0851$$

Coefficient [F]:

$$= (1 - v^*) / E^* \cdot (E \cdot \ln(K))$$

$$= (1 - 0.32) / 68162000 \cdot (0.20270E+09 \cdot \ln(1.09))$$

$$= 0.1639$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$= D_o^2 / 16 \cdot [(p_s - 1) \cdot (p_s^2 + 1) \cdot P_s - (p_c - 1) \cdot (p_c^2 + 1) \cdot P_t]$$

$$= 766.774^2 / 16 \cdot [(1.068 - 1) \cdot (1.068^2 + 1) \cdot 1560.0 - (1.068 - 1) \cdot (1.068^2 + 1) \cdot 1365.0]$$

$$= 1034023.9375 \text{ KPa} \cdot \text{mm}^2$$

Moment acting on unperforated tubesheet rim [M*]

$$= M_{ts} + W^* \cdot (G_c - G_s) / (2 \cdot \pi \cdot D_o)$$

$$= 1034023 + 197464.5 \cdot (819.25 - 819.25) / (2 \cdot \pi \cdot 766.774)$$

$$= 1034023.9375 \text{ KPa} \cdot \text{mm}^2$$

Note: W is the maximum of the bolt loads between the shell and channel sides.*



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 57 of 190

ASME TS Calc: Pressure Test

Case: 1 10:30a Sep 9, 2026

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$= ((M \cdot) - D_o^2/32 \cdot F (P_s - P_t)) / (1 + F)$$

$$= ((1034023) - 766.774^2/32 \cdot 0.164 (1560.0 - 1365.0)) / (1 + 0.16)$$

$$= 383956.7812 \text{ KPa} \cdot \text{mm}^2$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$= M_p + D_o^2/64 \cdot (3 + \nu) \cdot (P_s - P_t)$$

$$= 383956.78 + 766.774^2/64 \cdot (3 + 0.325) \cdot (1560.0 - 1365.0)$$

$$= 6340310.5000 \text{ KPa} \cdot \text{mm}^2$$

Maximum bending moment acting on tubesheet [M]:

$$= \max(| M_p | ; | M_o |)$$

$$= \max(| 383956.781 | ; | 6340310 |)$$

$$= 6340310.5000 \text{ KPa} \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$= 6 \cdot M / (\mu \cdot (h - h_g')^2)$$

$$= 6 \cdot 6340310 / (0.3333 (65.0 - 5.0)^2)$$

$$= 31.7034 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$= 2 \cdot \beta_T \cdot S_y$$

$$= 2 \cdot 0.95 \cdot 250.0$$

$$= 474.99 \text{ N./mm}^2$$

Tubesheet bending stress at final thickness [σ]:

$$= 6 \cdot M / (\mu \cdot (h - h_g')^2)$$

$$= 6 \cdot 6215223 / (0.3333 (20.3607 - 5.0)^2)$$

$$= 474.1669 \text{ N./mm}^2$$

Required tubesheet thickness for Bending Stress [HreqB]:

$$= h + C_{ats} + C_{atc}$$

$$= 20.3607 + 0 + 0$$

$$= 20.3607 \text{ mm}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$= \max(H_{reqB}; H_{reqS})$$

$$= \max(20.3607 ; 0.3759)$$

$$= 20.3607 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [| P_s - P_t |]:

$$= | P_s - P_t |$$

$$= | 1560.0 - 1365.0 |$$

$$= 195.000 \text{ KPa.}$$

Shear Stress check [τ_{limit}]:

$$= 4 \cdot \mu \cdot h / D_o \cdot 0.8 \cdot \beta_T \cdot S_y$$

$$= 4 \cdot 0.333 \cdot 65.0 / 766.77 \cdot 0.8 \cdot 0.95 \cdot 249.995$$

$$= 21.475 \text{ N./mm}^2$$

| P_s - P_t | is ≤ the Shear Stress limit (τ_{limit}). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [HreqS]:

$$= 1 / (4 \cdot \mu \cdot (0.8 \cdot \beta_T \cdot S_y)) \cdot (4 \cdot A_p / C_p) \cdot | P_s - P_t | + C_{ats} + C_{atc}$$

$$= 1 / (4 \cdot 0.333 \cdot 189.997) \cdot (4 \cdot 2517.1 / 2062.09) \cdot | 1560.0 - 1365.0 | + 0$$

$$= 0.3759 \text{ mm}$$

Note: Analysis Completed for Tubesheet Configuration d.



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FileName : E-0102

Page 58 of 190

ASME TS Calc: Pressure Test

Case: 1 10:30a Sep 9, 2026

Tube Metal Area [At]:

$$= \pi (d_o - t) t$$

$$= \pi (25.4 - 2.32) 2.32$$

$$= 1.682 \text{ cm}^2$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$= \pi \cdot t (d_o - t) S$$

$$= \pi \cdot 2.32 (25.4 - 2.32) 194.746$$

$$= 3340.524 \text{ Kg}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t)$$

$$= 0.85 \cdot \pi \cdot 2.32 (25.4 + 0.67 \cdot 2.32) 194.75$$

$$= 3316.0984 \text{ Kg}$$

Weld Strength Factor [fw]:

$$= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$$

$$= 194.746 / 194.746$$

$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t)$$

$$= 0.55 \cdot \pi \cdot 2.32 (25.4 + 0.67 \cdot 2.32) 194.75$$

$$= 2145.7107 \text{ Kg}$$

Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{((0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d)} - 0.75 \cdot d_o \right)$$

$$= 2 \left(\sqrt{((0.75 \cdot 25.4)^2 + 1.07 \cdot 2.32 \cdot (25.4 - 2.32) 1.0 \cdot 1.0)} - 0.75 \cdot 25.4 \right)$$

$$= 2.897 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.4487 mm

Minimum Required Groove Weld Leg [agr]: 1.4487 mm

Tube-Tubesheet Jt allowable, 3340.52 is $\geq$ tube strength 3340.52 Kg

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	31.7 <=	475.0 N./mm ²	Ok
Tubesheet shear stress	1.1 <=	190.0 N./mm ²	Ok

Thickness results for loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	20.361	65.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.449	2.320	Ok
Tube-Tubesheet Groove Weld Leg :	1.449	2.320	Ok

U-Tube Tubesheet results per ASME Div. 2 Part 4.18.7 2025

Results for 3 Load Cases:

--Reqd. Thk. + CA ---- Tubesheet Stresses Case Pass/



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FileName : E-0102

Page 59 of 190

ASME TS Calc: Pressure Test

Case: 1 10:30a Sep 9, 2026

Case#	Tbsht	Extnsn	Bend	Allwd	Shear	Allwd	Type	Fail
D1uc	46.027	...	222	475	8	190	Ps+Pt D1	Ok
D2uc	50.292	30.879	271	475	9	190	Ps+Pt D2	Ok
D3uc	20.361	...	32	475	1	190	Ps+Pt D3	Ok
Max:	50.2916	30.879	mm	0.570		0.049	(Str. Ratio)	

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the Load Case Tables

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

Note:

Because there was no net pressure differential across the tubesheet due to the specified pressures psd,min and ptd,min; load case D4 was not processed as the resulting tubesheet bending stress would be zero.

Summary of Thickness Comparisons for 3 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	50.292	65.000	Ok
Tubesheet thickness flanged extension :	30.879	49.000	Ok
Tube thickness :	0.571	2.320	Ok
Tube-tubesheet fillet weld leg :	1.449	2.320	Ok
Tube-tubesheet groove weld leg :	1.449	2.320	Ok

Note: This is a full strength Tube to Tubesheet Joint.

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PV Elite 28 Licensee: SL User
 FileName : E-0102 Page 60 of 190
 External Pressure Calculations: Step: 12 10:30am Sep 9,2026

External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

External Pressure Calculations:

From	To	Section Length mm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B N./mm ²
10	20	No Calc	823.8	8.7	0.0014668	80.3141
20	30	816.667	828	10.8	0.002059	86.2021
30	40	No Calc	...	91.8	No Calc	No Calc
40	50	No Calc	...	91.8	No Calc	No Calc
50	60	2716.67	836	14.8	0.00089218	81.7149
60	80	No Calc	823.8	8.7	0.0014668	96.2056

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure KPa.	External M.A.W.P. KPa.
10	20	11.9	No Calc	...	942.37
20	30	14	No Calc	...	1499.08
30	40	95	92.2782	...	No Calc
40	50	95	92.2782	...	No Calc
50	60	18	7.54601	101.353	1928.72
60	80	11.9	5.31159	101.353	1128.83

Minimum

942

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	30	816.667	No Calc	No Calc	No Calc
30	40	No Calc	No Calc	No Calc	No Calc
40	50	No Calc	No Calc	No Calc	No Calc
50	60	2716.67	96957.3	No Calc	No Calc
60	80	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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FileName : E-0102

Page 61 of 190

Element and Detail Weights:

Step: 13 10:30am Sep 9, 2026

Element and Detail Weights:

From	To	Element Metal Wgt. kg.	Element ID Volume cm ³	Corroded Metal Wgt. kg.	Corroded ID Volume cm ³	Extra due Misc % kg.
10	20	99.9233	92169.9	77.0837	94195.4	9.99233
20	30	194.237	351921	150.429	357575	19.4237
30	40	210.114	68199.6	201.727	68450	21.0114
40	50	210.114	68199.6	201.727	68450	21.0114
50	60	932.135	1063378	769.421	1084376	93.2135
60	80	99.9233	92169.9	77.0837	94195.4	9.99233
Total		1746	1736039.25	1477	1767242.00	174



For elements specified as shell side elements, the volume(s) shown above for those elements, reflects the displacement of the tubes.

Weight of Details:

From	Type	Weight of Detail kg.	X Offset, Dtl. Cent. mm	Y Offset, Dtl. Cent. mm	Z Offset Dtl. Cent. mm	Description
10	Liqd	92.1137	-66.6667	...	...	Liquid: 10
10	Insl	43.4426	-75	...	...	Hot insulation
20	Liqd	351.707	350	...	...	Liquid: 20
20	Insl	71.9195	350	...	...	Hot insulation
20	Nozl	15.4641	300	-436.83	...	N1-Tube inlet
20	Nozl	38.5239	300	473.165	...	N1-Tube Outlet
20	Wght	100	350	...	...	Partition plate
30	Liqd	63.2407	63	50	...	Liquid: 30
40	Liqd	63.2407	63	50	...	Liquid: 40
50	Sadl	75.8404	520	586	...	LH saddle
50	Sadl	75.8404	2080	586	...	Right saddle
50	Liqd	1062.73	1300	...	...	Liquid: 50
50	Insl	379.916	1300	...	...	Cold insulation
50	Nozl	160.748	500	604.8	...	N3-Shell inlet
50	Nozl	168.972	1500	-604.8	...	N4-Shell outlet
50	Wght	150	500	-585	...	Fireproofing we
50	Wght	150	2025	-585	...	Fireproofing
60	Liqd	92.1137	116.667	...	...	Liquid: 60
60	Insl	61.29	125	...	...	Cold insulation
30	FTsh	370.266	161.5	...	...	U-Tube
30	Tube	681.229	1348	...	...	Tubes
40	Bafl	61.5324	847.73	...	...	Baffles
40	Bafl	61.5324	1205.92	...	...	Baffles
40	Bafl	61.5324	1564.11	...	...	Baffles
40	Bafl	62.2809	1922.3	...	...	Baffles

Total Weight of Each Detail Type:

Saddles	151.7
Liquid	1725.1
Insulation	556.6
Nozzles	383.7
Weights	400.0
Exchanger Components	1298.4
Liquid in Tubes	174.5
Sum of the Detail Weights	4689.9 kg.



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FileName : E-0102

Page 62 of 190

Element and Detail Weights:

Step: 13 10:30am Sep 9,2026

Weight Summation Results: (kg.)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	1921.1	1921.1	1921.1	1921.1	1921.1	1921.1
Saddles	151.7	151.7	151.7	151.7	151.7	151.7
Nozzles	383.7	383.7	383.7	383.7	383.7	383.7
Wld Weights	100.0	100.0	100.0	100.0	100.0	100.0
Exchanger	1298.4	1298.4	1298.4	1298.4	1298.4	1298.4
Insulation	...	...	...	...	556.6	556.6
Empty Weights	...	...	...	300.0	300.0	...
Ope Weights	...	...	...	...	...	300.0
Ope. Liquid	...	...	...	...	...	1725.1
Tube Ope Lqd	...	...	...	...	...	174.5
Test Liquid	...	1735.0	...	...	...	...
Tube Tst Lqd	...	173.7	...	...	...	...
Totals	3854.9	5763.6	3854.9	4154.9	4711.4	6611.0

Field Installation Options:

* Insulation installed after lifting.

Miscellaneous Weight Percent: 10.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	3854.9 kg.
Shop Test Wt.	- Fabricated weight + water (full)	5763.6 kg.
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	3854.9 kg.
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	4154.9 kg.
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	4711.4 kg.
Operating Wt.	- Empty weight + operating liq. uncorroded	6611.0 kg.
Oper. Wt. + CA	- Corr wt. + operating liquid	6315.1 kg.
Field Test Wt.	- Empty weight + test medium (full)	6055.4 kg.

Exchanger Tube Data

Volume of Exchanger tubes :	173827.2 cm ³
Weight of Ope Liq in tubes :	174.5 kg.
Weight of Water in tubes :	173.7 kg.

Field Test weight is computed in the corroded condition.

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	8732.25	8194.14
20	30	18208.7	17592.9
30	40	6501.71	3367.79
40	50	6501.71	3367.79
50	60	68285.7	65345.1
60	80	8732.25	8194.14

Total	116962.3	106061.9 cm ²
Total	125.9	114.2 ft ²



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 63 of 190

Nozzle Flange MAWP:

Step: 14 10:30am Sep 9,2026

Nozzle Flange MAWP Results: (KPa. & °C)

Nozzle Description	Flange Rating		Des. Temp	Class	Grade/Group	Equiv. Press	Max Pressure		
	Ope.	Ambient					UG-44	50%	DNV
N1-Tube inlet	4272.0	5100.0	230	300	GR 1.1	5319.7	1088.4	4272.0	1088.4
N1-Tube Outlet	4272.0	5100.0	230	300	GR 1.1	3380.5	3027.5	4272.0	3027.5
N3-Shell inlet	1820.4	1960.0	82	150	GR 1.1	1795.5	1820.4	1820.4	935.1
N4-Shell outle	1820.4	1960.0	82	150	GR 1.1	1795.5	1820.4	1820.4	935.1

Shellside Flange Rating

Lowest Flange Pressure Rating was (Ope)[ShellSide]: 1820.434 KPa.

Lowest Flange Pressure Rating was (Amb)[ShellSide]: 1960.006 KPa.

Channelside Flange Rating

Lowest Flange Pressure Rating was (Ope)[TubeSide]: 1088.365 KPa.

Lowest Flange Pressure Rating was (Amb)[TubeSide]: 5100.005 KPa.

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm	Fm Value	Resolved Moment ME Kg-m.	Axial Force FE Kgf	MAWP PD KPa.	Maximum Pressure Pmax KPa.	Pass Fail
N1-Tube inlet	111.125	0.50	129	612	1088.4	1088.4	Pass
N1-Tube Outlet	196.088	0.50	454	1147	1088.4	3027.5	Pass
N3-Shell inlet	500.888	1.20	4087	3442	1506.0	1820.4	Pass
N4-Shell outlet	500.888	1.20	4087	3442	1506.0	1820.4	Pass

Evaluating Flange on Nozzle [N1-Tube inlet]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [((PR - PD) + FM \cdot PR)]$$

$$= 16(129) + 4 \cdot \text{abs}(612) 111.125 \leq \pi \cdot 111.125^3 [(4272.041 - 1088.365) + .500 \cdot 4272.041]$$

$$= 0.23387E+04 \leq 0.23387E+04 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N1-Tube inlet [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(4272.041, 0.500 \cdot 4272.041 + 4272.041 - 2338.7 / (\pi \cdot 111.125^3))$$

$$= \min(4272.041, 1088.365)$$

$$= 1088.365 \text{ KPa.}$$

Evaluating Flange on Nozzle [N1-Tube Outlet]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [((PR - PD) + FM \cdot PR)]$$

$$= 16(454) + 4 \cdot \text{abs}(1147) 196.088 \leq \pi \cdot 196.088^3 [(4272.041 - 1088.365) + .500 \cdot 4272.041]$$

$$= 0.81657E+04 \leq 0.12850E+05 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N1-Tube Outlet [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(4272.041, 0.500 \cdot 4272.041 + 4272.041 - 8165.7 / (\pi \cdot 196.088^3))$$

$$= \min(4272.041, 3027.510)$$

$$= 3027.510 \text{ KPa.}$$

Evaluating Flange on Nozzle [N3-Shell inlet]:



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 64 of 190

Nozzle Flange MAWP:

Step: 14 10:30am Sep 9, 2026

$$\begin{aligned}
 &= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR] \\
 &= 16(4087) + 4 \cdot \text{abs}(3442) 500.888 \leq \pi \cdot 500.888^3 [(1820.434 - 1505.975) + 1.200 \cdot 1820.434] \\
 &= 0.72289E+05 \leq 0.10061E+06 \text{ Check Passes}
 \end{aligned}$$

Maximum Pressure for Flange on Nozzle N3-Shell inlet [Pmax]:

$$\begin{aligned}
 &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\
 &= \min(1820.434, 1.200 \cdot 1820.434 + 1820.434 - 72288.7 / (\pi \cdot 500.888^3)) \\
 &= \min(1820.434, 2209.423) \\
 &= 1820.434 \text{ KPa.}
 \end{aligned}$$

Evaluating Flange on Nozzle [N4-Shell outlet]:

$$\begin{aligned}
 &= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR] \\
 &= 16(4087) + 4 \cdot \text{abs}(3442) 500.888 \leq \pi \cdot 500.888^3 [(1820.434 - 1505.975) + 1.200 \cdot 1820.434] \\
 &= 0.72289E+05 \leq 0.10061E+06 \text{ Check Passes}
 \end{aligned}$$

Maximum Pressure for Flange on Nozzle N4-Shell outlet [Pmax]:

$$\begin{aligned}
 &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\
 &= \min(1820.434, 1.200 \cdot 1820.434 + 1820.434 - 72288.7 / (\pi \cdot 500.888^3)) \\
 &= \min(1820.434, 2209.423) \\
 &= 1820.434 \text{ KPa.}
 \end{aligned}$$

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

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PV Elite 28 Licensee: SL User

FileName : E-0102

Page 65 of 190

Wind Load Calculation:

Step: 15 10:30am Sep 9, 2026

Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	349.92 Km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	300 mm
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm
Distance Upwind of Crest Lh	0 mm
Distance from Crest to the Vessel x	0 mm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:

$$\begin{aligned}
 &= \min(0.85, 0.925((1 + 1.7 \cdot gQ \cdot I_{zbar} \cdot Q)/(1 + 1.7 \cdot gV \cdot I_{zbar}))) \\
 &= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.228 \cdot 0.985)/(1 + 1.7 \cdot 3.4 \cdot 0.228))) \\
 &= \min(0.85, 0.917) \\
 &= 0.850
 \end{aligned}$$

Natural Frequency of Vessel (Operating)	33.000 Hz
Natural Frequency of Vessel (Empty)	33.000 Hz
Natural Frequency of Vessel (Test)	33.000 Hz

User Entered Importance Factor is	1.000
Force Coefficient [Cf]	0.549
Structure Height to Diameter ratio	3.916

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of [α] and [Zg]:

Exposure Category: C from Table 26.11-1
 $\alpha = 9.5$: $Z_g = 274320.0$ mm

Effective Height [z]:

$$\begin{aligned}
 &= \text{Centroid Height} + \text{Vessel Base Elevation} \\
 &= 800.0 + 300.0 = 1100.0 \text{ mm} \\
 &= 3.609 \text{ ft. Imperial Units}
 \end{aligned}$$

Velocity Pressure coefficient evaluated at height z [Kz]:

$$\begin{aligned}
 &\text{Because } z (3.609 \text{ ft.}) < 15 \text{ ft.} \\
 &= 2.01(15 / Z_g)^{2/\alpha} \\
 &= 2.01(15/900.0)^{2/9.5} \\
 &= 0.849
 \end{aligned}$$

Type of Hill: No Hill



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 66 of 190

Wind Load Calculation:

Step: 15 10:30am Sep 9, 2026

Wind Directionality Factor [Kd]:

= 1, per Table 26.6-1

As there is No Hill Present: [Kzt]:

K1 = 0, K2 = 0, K3 = 0

Topographical Factor [Kzt]:

= (1 + K1 • K2 • K3)²

= (1 + 0.0 • 0.0 • 0.0)²

= 1.0

Velocity Pressure evaluated at height z, Imperial Units [qz]:

= max(16, 0.00256 • Kz • Kzt • Kd • Ke • V(mph)²)

= max(16, 0.00256 • 0.849 • 1.0 • 1.0 • 1.0 • 217.435²)

= 102.7 psf [501.637] Kgs/m²

Force on the first element [F]:

= qz • G • Cf • WindArea

= 102.742 • 0.85 • 0.549 • 3.361

= 161.0 lbs. [73.0] Kgf

Element	Hgt (z) mm	K1	K2	K3	Kz	Kzt	qz Kgs/m ²
LHS Dish End	1100.0	0.000	0.000	0.000	0.849	1.000	501.637
Left Ch shell	1100.0	0.000	0.000	0.000	0.849	1.000	501.637
LHS Ch Flange	1100.0	0.000	0.000	0.000	0.849	1.000	501.637
LHS Shell Flg	1100.0	0.000	0.000	0.000	0.849	1.000	501.637
Main Shell	1100.0	0.000	0.000	0.000	0.849	1.000	501.637
RHS Dish End	1100.0	0.000	0.000	0.000	0.849	1.000	501.637

Wind Loads on Masses/Equipment/Piping

ID	Wind Area cm ²	Elevation mm	Pressure Kgs/m ²	Force Kgf
Partition pla	0.00	300.00	501.64	0.00
Fireproofing	0.00	300.00	501.64	0.00
Fireproofing	0.00	300.00	501.64	0.00

Wind Load Calculation:

From	To	Wind Height mm	Wind Diameter mm	Wind Area cm ²	Wind Pressure Kgs/m ²	Element Wind Load Kgf
10	20	1100	1228.56	3122.24	501.637	80.3379
20	30	1100	1233.6	8635.2	501.637	222.191
30	40	1100	960	1209.6	501.637	31.1241
40	50	1100	960	1209.6	501.637	31.1241
50	60	1100	1339.2	34819.2	501.637	895.928
60	80	1100	1324.56	3712.98	501.637	95.5382

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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FileName : E-0102

Page 67 of 190

Earthquake Load Calculation:

Step: 16 10:30am Sep 9,2026

Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code		ASCE/SEI 7-16
Seismic Load Reduction Scale Factor		1.000
Importance Factor		1.250
Table Value Fa		1.600
Table Value Fv		2.400
Max. Mapped Res. Acceleration	[Ss]	0.147
Max. Eff. Ground Acceleration	[S]	0.065
Force Modification Factor R		3.000
Site Class		D
Component Elevation Ratio	z/h	0.000
Amplification Factor	Ap	0.000
Force Factor		0.000
Consider Vertical Acceleration		Yes
Minimum Acceleration Multiplier		0.000
User Value of Sds (used if > 0)		0.000
User Value of Sd1 (used if > 0)		0.000
Moment Reduction Factor Tau		1.000

Seismic Analysis Results:

$$\begin{aligned} SMS &= Fa * Ss = 1.6 * 0.147 = 0.235 \\ SM1 &= Fv * S1 = 2.4 * 0.065 = 0.156 \\ SDS &= 2/3 * Sms = 2/3 * 0.235 = 0.157 \\ SD1 &= 2/3 * Sm1 = 2/3 * 0.156 = 0.104 \end{aligned}$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 $= Ct \cdot h_n^x$ where $Ct = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
 $= 0.020(3.937^{0.75})$
 $= 0.056$ seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 $= (1/\text{Natural Frequency}) = (1/33.0)$
 $= 0.030$

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
 $= \text{Minimum Value of } (1.7 \cdot 0.056, 0.03) \text{ per 12.8.2}$
 $= 0.030$
 As the time period is < 0.06 second, use section 15.4.2.

Base Shear per equation 15.4-5, [V]:
 $= 0.3 \cdot Sds \cdot W \cdot I$
 $= 0.3 \cdot 0.157 \cdot 6459 \cdot 1.25$
 $= 379.809$ Kg

Vertical load per 12.4-4, [Ev]:
 $= 0.2 \cdot Sds \cdot W$
 $= 0.2 \cdot 0.157 \cdot 6459$
 $= 202.6$ Kg

Earthquake Load Calculation:

From	To	Earthquake Height mm	Earthquake Weight Kgf	Element Ope Load Kgf
10	20	400	807.417	47.4761



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FileName : E-0102

Page 68 of 190

Earthquake Load Calculation: Step: 16 10:30am Sep 9, 2026

20	30	400	807.417	47.4761
30	40	400	807.417	47.4761
40	50	400	807.417	47.4761
50	Sad1	400	807.417	47.4761
Sad1	60	400	807.417	47.4761
50	60	400	807.417	47.4761
60	80	400	807.417	47.4761

The Earthquake Weight does not include the weight of the saddles.

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FileName : E-0102

Page 69 of 190

Center of Gravity Calculation: Step: 17 10:30am Sep 9,2026

Shop/Field Installation Options :

Insulation is installed in the Field after being lifted.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	2370.000 mm
Center of Gravity of the Liquid	1806.980 mm
Center of Gravity of the Insulation	2081.517 mm
Center of Gravity of the Nozzles	1838.711 mm
Center of Gravity of the Added Weights (Operating)	1849.375 mm
Center of Gravity of the Added Weights (Empty)	1849.375 mm
Center of Gravity of the Tubesheet(s)	911.500 mm
Center of Gravity of the Tubes	2098.000 mm
Center of Gravity of the Baffles	2330.644 mm
Center of Gravity of Bare Shell New and Cold	1741.599 mm
Center of Gravity of Bare Shell Corroded	1717.934 mm
Vessel CG in the Operating Condition	1833.796 mm
Vessel CG in the Fabricated (Shop/Empty) Condition	1803.345 mm
Vessel CG in the Test Condition	1811.374 mm

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FileName : E-0102

Page 70 of 190

Saddle Calcs: Bundle Pull New: Step: 18 10:30am Sep 9, 2026

ASME Horizontal Vessel Analysis: Bundle pullout, Left Saddle

This saddle was not processed as the coefficient of friction was specified and the directive was set to only analyze the fixed saddle for this case.

ASME Horizontal Vessel Analysis: Bundle pullout, Right Saddle

This saddle was not processed as the coefficient of friction was specified and the directive was set to only analyze the fixed saddle for this case.

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FileName : E-0102

Page 71 of 190

Saddle Calcs: Bundle Pull Cor: Step: 19 10:30am Sep 9, 2026

ASME Horizontal Vessel Analysis: Bundle pullout, Left Saddle

This saddle was not processed as the coefficient of friction was specified and the directive was set to only analyze the fixed saddle for this case.

ASME Horizontal Vessel Analysis: Bundle pullout, Right Saddle

This saddle was not processed as the coefficient of friction was specified and the directive was set to only analyze the fixed saddle for this case.

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FileName : E-0102

Page 72 of 190

Saddle Calcs: Road Transport: Step: 20 10:30am Sep 9,2026

ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Road Transport

Note:

Wear Pad Width (152.40) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
and less than $2a$. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

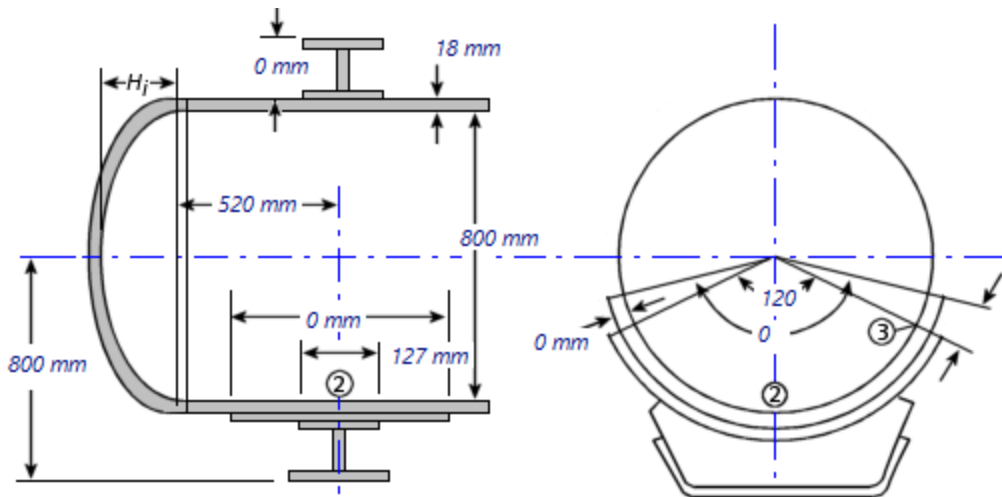
$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(127.0 + 1.56 \cdot \sqrt{409.0 \cdot 18.0}, 2 \cdot 520.0)$$

$$= 260.8513 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	409.00	mm
Shell Thickness used in this Case	t	18.000	mm
Stiffened Vessel Length per 4.15.6	L	2776.00	mm
Distance from Saddle to Vessel tangent	a1 or a	520.00	mm
Saddle Width	b1 or b	127.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	800.00	mm
Coefficient of Friction	mu	0.30	



Results for Horizontal Vessel Analysis:

Saddle Force Q for this Transportation Case		4004.11	Kgf
Shipping Longitudinal Acceleration	gx	0.10	g
Shipping Transverse Acceleration	gz	0.10	g
Shipping Vertical Acceleration	gy	0.05	g
Vertical Acceleration Acting with Transverse	gw	1.00	g
Internal Pressure during Transport		0.00	KPa.
Transportation Wind Load Scalar Factor	wlsf	1.00	
Transportation Wind Velocity	wv	92.7	Km/hr



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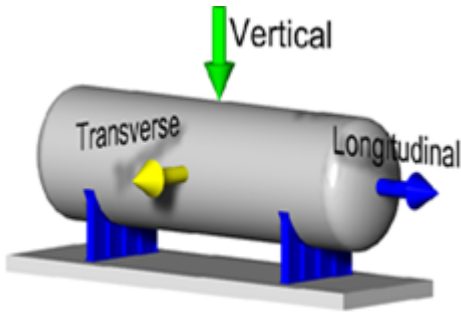
FileName : E-0102

Page 73 of 190

Saddle Calcs: Road Transport:

Step: 20 10:30am

Sep 9, 2026



Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	-0.85	-234.00
Long. Stress at Top of Midspan	0.85	138.00
Long. Stress at Bottom of Midspan	0.85	138.00
Long. Stress at Top of Saddles	2.62	138.00
Long. Stress at Bottom of Saddles	-1.45	-234.00
Long. Stress at Bottom of Saddles	1.45	138.00
Tangential Shear in Shell	3.90	110.40
Circ. Stress at Horn of Saddle	13.42	207.00
Circ. Compressive Stress in Shell	0.64	138.00

Intermediate Results: Saddle Reaction Q due to Road Transport Load:

Transverse Force due to Wind during Transport [F_{tw}]:

$$\begin{aligned}
 &= \text{Transverse Area} \cdot 0.00256 \cdot wv^2 \cdot wlsf \\
 &= 56.74 \cdot 0.00256 \cdot 57.6^2 \cdot 1.0 \\
 &= 481.9 \text{ lbf} \\
 &= 218.601 \text{ Kg}
 \end{aligned}$$

Longitudinal Force due to Wind during Transport [F_{lw}]:

$$\begin{aligned}
 &= \pi \cdot D^2/4 \cdot 0.00256 \cdot wv^2 \cdot wlsf \cdot \text{Wind Dia. Mult} \\
 &= \pi \cdot 3.399^2/4 \cdot 0.00256 \cdot 57.6^2 \cdot 1.0 \cdot 1.2 \\
 &= 92.5 \text{ lbf} \\
 &= 42.0 \text{ Kg}
 \end{aligned}$$

Transverse Force due to Lateral Acceleration [F_t]:

$$\begin{aligned}
 &= g_z \cdot \text{Shipping Weight} \cdot g_w + F_{tw} \\
 &= 0.1 \cdot 3703.17 \cdot 1.0 + 218.6 \\
 &= 588.9 \text{ Kg}
 \end{aligned}$$

Longitudinal Force due to Axial Acceleration [F_l]:

$$\begin{aligned}
 &= g_x \cdot \text{Shipping Weight} \cdot g_w + F_{lw} \\
 &= 0.1 \cdot 3703.17 \cdot 1.0 + 41.95 \\
 &= 412.3 \text{ Kg}
 \end{aligned}$$

Saddle Reaction Force due to Transverse Load [F_{wt}]:

$$\begin{aligned}
 &= F_t / \text{Num of Saddles} \cdot B / E \\
 &= 588.9/2 \cdot 800.0/741.3178 \\
 &= 317.8 \text{ Kg}
 \end{aligned}$$

Saddle Reaction Force due to Longitudinal Load [F_{wl}]:

$$\begin{aligned}
 &= F_l \cdot B / L_s \\
 &= 412 \cdot 800.0/1560.0 \\
 &= 211.4 \text{ Kg}
 \end{aligned}$$

Saddle Reaction Force due to Vertical Transportation Acceleration [F_{rv}]:

$$\begin{aligned}
 &= \text{Saddle Load} \cdot g_y \\
 &= 3510.8 \cdot 0.05
 \end{aligned}$$



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FileName : E-0102

Page 74 of 190

Saddle Calcs: Road Transport: Step: 20 10:30am Sep 9, 2026

= 175.540 Kgf

Load Combination Results due to Transport Loading [Q]:

= Saddle Load + Frv + max(Fwl, Fwt)

= 3511 + 176 + max(211, 318)

= 4004.1 Kgf

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	4079.95	Kgf
Transverse Shear Load	294.46	Kgf
Longitudinal Shear Load	412.27	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

= min(2(ShellID/2 + t + WearPadThickness)sin(θ/2), 2•Rm)

= min(2(800.0/2 + 18.0 + 10.0)sin(120.0/2), 2•409.0)

= 741.318 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

= -Q•a [1 - (1- a/L + (Rm²-hm²)/(2a•L))/(1+(4hm)/3L)]

= -4004•520.0[1-(1-520.0/2776.0+(409.0²-0.0²)/

(2•520.0•2776.0))/(1+(4•0.0)/(3•2776.0))]

= -269.4 Kg-m.

Moment per Equation 4.15.2 [M2]:

= Q•L/4((1+2(Rm²-hm²)/(L²))/(1+(4hm²)/(3L))-4a/L)

= 4004•2776/4((1+2(409²-0²)/(2776²))/(1+(4•0)/

(3•2776))-4•520/2776)

= 817.4 Kg-m.

Longitudinal Stress at Top of Shell (4.15.4) [σ1]:

= P • Rm/(2t) - M2/(π•Rm²t)

= 0.0 • 409.0/(2•18.0) - 817.4/(π•409.0²•18.0)

= -0.85 N./mm²

Longitudinal Stress at Bottom of Shell (4.15.5) [σ2]:

= P • Rm/(2t) + M2/(π • Rm² • t)

= 0.0 • 409.0/(2 • 18.0) + 817.4/(π • 409.0² • 18.0)

= 0.85 N./mm²

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ*3]:

= P • Rm/(2t) - M1/(K1•π•Rm²t)

= 0.0•409.0/(2•18.0)--269.4/(0.1066•π•409.0²•18.0)

= 2.62 N./mm²

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ*4]:

= P • Rm/(2t) + M1/(K1* • π • Rm² • t)

= 0.0•409.0/(2•18.0)+-269.4/(0.1923•π•409.0²•18.0)

= -1.45 N./mm²

Maximum Shear Force in the Saddle (4.15.3) [T]:

= Q(L-2a)/(L+(4•hm/3))

= 4004(2776.0 - 2 • 520.0)/(2776.0 + (4 • 0.0/3))

= 2504.0 Kgf

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ2]:

= K2 • T / (Rm • t)



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 75 of 190

Saddle Calcs: Road Transport: Step: 20 10:30am Sep 9,2026

$$= 1.1707 \cdot 2504.01 / (409.0 \cdot 18.0)$$

$$= 3.90 \text{ N./mm}^2$$

Decay Length (4.15.20) [x1,x2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(409.0 \cdot 18.0)}$$

$$= 66.926 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.7603 \cdot 4004 \cdot 0.1 / (18.0(127.0 + 66.93 + 66.93))$$

$$= -0.64 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, L<8Rm (4.15.23) [σ_7^*]:

$$= -Q / (4 \cdot t(b + X1 + X2)) - 12 \cdot K7 \cdot Q \cdot R_m / (L \cdot t^2)$$

$$= -4004 / (4 \cdot 18.0(127.0 + 66.926 + 66.926)) -$$

$$12 \cdot 0.053 \cdot 4004 \cdot 409.0 / (2776.0 \cdot 18.0^2)$$

$$= -13.42 \text{ N./mm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(127.0 + 1.56 \cdot \sqrt{(409.0 \cdot 18.0)}, 2 \cdot 520.0)$$

$$= 260.85 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	850.0001	mm
Baseplate Thickness	Bpthk	12.0000	mm
Baseplate Width	Bpwid	152.4000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	300.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(400.0 \cdot 18.0)} + 152.4)18.0^3) / 12$$

$$= 13.840 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 13.84 + 51.259(128.369 - 9.0)^2$$

$$= 7317.716 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	284.8	18.0	9.0	51.3	46132.8	0.732E+04
Wearplate	152.4	10.0	23.0	15.2	35052.0	0.169E+04
Web	10.0	360.0	208.0	36.0	748799.8	0.617E+04
BasePlate	152.4	12.0	394.0	18.3	720547.1	0.129E+05
Totals	...	...	...	120.8	1550531.8	0.281E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 1550532 / 120.787$$

$$= 128.369 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120.0 / 2$$



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 76 of 190

Saddle Calcs: Road Transport: Step: 20 10:30am Sep 9, 2026

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 4004.105$$

$$= 814.9222 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 1.1494 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 800.0 - 400.0 \cdot \sin(120.0/2) / 1.0472$$

$$= 469.203 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 382.3712 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 1.7137 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(4004 + 76) \cdot 152.4) / (4 \cdot 850.0 \cdot 173.33)}$$

$$= 5.571 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (400.0 + 18.0 + 10.0)$$

$$= 741.318 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (N_{\text{ribs}} - 1)$$

$$= 741.3177 / (3 - 1)$$

$$= 370.659 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 370.6589 \cdot 152.4 / 2$$

$$= 282.442 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 4079.945 / (850.0 \cdot 152.4)$$

$$= 3.150 \text{ Kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 282.4 \cdot 3.15$$

$$= 889.569 \text{ Kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 11.7 + 18.533$$

$$= 30.233 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / Ar$$

$$= 889.6 / 30.2329$$



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 77 of 190

Saddle Calcs: Road Transport: Step: 20 10:30am Sep 9,2026

= 2.886 N./mm²

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	127.0	...	12.7	...	171.

Rib dimension [D]:

= Saddle Width - Web Thickness

= 127.0 - 10.0

= 117.000 mm

Distance to Centroid from Datum [ytot]:

= AY / A

= 0.0/30.233

= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2

= 127.0/2

= 63.500 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$

= $\sqrt{(170.7/30.233)}$

= 23.762 mm

Length of Outer Rib [L]:

= Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk

= 800.0 - cos(120.0/2)(400.0 + 18.0 + 10.0) - 12.0

= 574.000 mm

Intermediate Term [Cc]:

= $\sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$

= $\sqrt{(2 \cdot \pi^2 \cdot 0.19994\text{E}+09/260.0)}$

= 123.211

Slenderness ratio [KL/r]:

= KL/r

= 1 • 574.0/23.762

= 24.157

Bending Moment [Rm]:

= FI / (2 • Bplen) • e • L / 2

= 412.3 / (2 • 850.0) • 370.659 • 574.0/2

= 25.799 Kg-m.

Compressive Allowable, KL/r < Cc (24.1567 < 123.2105) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$

= $(1 - (24.16)^2 / (2 \cdot 123.21^2)) 260 /$

$(5/3 + 3 \cdot (24.16) / (8 \cdot 123.21) - (24.16^3) / (8 \cdot 123.21^3))$

= 146.6 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)

= Sc/Sca + (Rm • C1 / I) / Sba

= 2.89/146.61 + (25.8 • 63.5/1706986) / 173.33

= 0.074

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
--	---	---	---	---	----	----



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 78 of 190

Saddle Calcs: Road Transport: Step: 20 10:30am Sep 9, 2026

Rib	10.0	117.0	0.0	11.7	0.0	171.
Web	370.7	10.0	0.0	37.1	0.0	3.09
Totals	...	...	...	48.8	...	174.

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/48.766$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 127.0/2$$

$$= 63.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 800.0 - \cos(428.0/2) (12.0 + 0.0 + 0.0) - 0.0$$

$$= 360.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(173.7/48.766)}$$

$$= 18.873 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 360.0/18.873$$

$$= 19.075$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 412.27/(2 \cdot 850.0)$$

$$= 0.243 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.243 \cdot 370.659 \cdot 360.0$$

$$= 32.361 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 32.361/27354.967$$

$$= 11.601 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ (19.0746 < 123.2105) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (19.07)^2 / (2 \cdot 123.21^2)) 260 /$$

$$(5/3 + 3 \cdot (19.07) / (8 \cdot 123.21) - (19.07^3) / (8 \cdot 123.21^3))$$

$$= 149 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c/S_{ca} + (M_{base,c} \cdot C1/I) / S_{ba}$$

$$= 3.58/148.98 + (32.36 \cdot 63.5/173.704) / 173.33$$

$$= 0.091$$

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:

Wear Pad Width (152.40) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(127.0 + 1.56 \cdot \sqrt{409.0 \cdot 18.0}, 2 \cdot 520.0)$$



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 79 of 190

Saddle Calcs: Road Transport: Step: 20 10:30am Sep 9,2026

= 260.8513 mm

Input and Calculated Values:

Vessel Mean Radius	Rm	409.00	mm
Shell Thickness used in this Case	t	18.000	mm
Stiffened Vessel Length per 4.15.6	L	2776.00	mm
Distance from Saddle to Vessel tangent	a1 or a	520.00	mm
Saddle Width	b1 or b	127.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	205.95	mm
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	800.00	mm
Coefficient of Friction	mu	0.30	

Results for Horizontal Vessel Analysis:

Saddle Force Q for this Transportation Case		834.76	Kgf
Shipping Longitudinal Acceleration	gx	0.10	g
Shipping Transverse Acceleration	gz	0.10	g
Shipping Vertical Acceleration	gy	0.05	g
Vertical Acceleration Acting with Transverse	gw	1.00	g
Internal Pressure during Transport		0.00	KPa.
Transportation Wind Load Scalar Factor	wlsf	1.00	
Transportation Wind Velocity	wv	92.7	Km/hr

Horizontal Vessel Analysis Results:		Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	-0.11	-234.00	
Long. Stress at Top of Midspan	0.11	138.00	
Long. Stress at Bottom of Midspan	0.11	138.00	
Long. Stress at Top of Saddles	0.93	138.00	
Long. Stress at Bottom of Saddles	-0.52	-234.00	
Long. Stress at Bottom of Saddles	0.52	138.00	
Tangential Shear in Shell	0.74	110.40	
Circ. Stress at Horn of Saddle	2.80	207.00	
Circ. Compressive Stress in Shell	0.13	138.00	

Intermediate Results: Saddle Reaction Q due to Road Transport Load:

Transverse Force due to Wind during Transport [Ftw]:

$$= \text{Transverse Area} \cdot 0.00256 \cdot wv^2 \cdot wlsf$$

$$= 56.74 \cdot 0.00256 \cdot 57.6^2 \cdot 1.0$$

$$= 481.9 \text{ lbf}$$

$$= 218.601 \text{ Kgf}$$

Longitudinal Force due to Wind during Transport [Flw]:

$$= \pi \cdot D^2/4 \cdot 0.00256 \cdot wv^2 \cdot wlsf \cdot \text{Wind Dia. Mult}$$

$$= \pi \cdot 3.399^2/4 \cdot 0.00256 \cdot 57.6^2 \cdot 1.0 \cdot 1.2$$

$$= 92.5 \text{ lbf}$$

$$= 42.0 \text{ Kgf}$$

Transverse Force due to Lateral Acceleration [Ft]:

$$= gz \cdot \text{Shipping Weight} \cdot gw + Ftw$$

$$= 0.1 \cdot 3703.17 \cdot 1.0 + 218.6$$



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 80 of 190

Saddle Calcs: Road Transport: Step: 20 10:30am Sep 9, 2026

= 588.9 Kgf

Longitudinal Force due to Axial Acceleration [FI]:

= $g_x \cdot \text{Shipping Weight} \cdot g_w + F_{lw}$

= $0.1 \cdot 3703.17 \cdot 1.0 + 41.95$

= 412.3 Kgf

Saddle Reaction Force due to Transverse Load [Fwt]:

= $F_t / \text{Num of Saddles} \cdot B / E$

= $588.9/2 \cdot 800.0/741.3178$

= 317.8 Kgf

Saddle Reaction Force due to Longitudinal Load [Fwl]:

= $F_l \cdot B / L_s$

= $412 \cdot 800.0/1560.0$

= 211.4 Kgf

Saddle Reaction Force due to Vertical Transportation Acceleration [Frv]:

= Saddle Load $\cdot g_y$

= $492.4 \cdot 0.05$

= 24.619 Kgf

Load Combination Results due to Transport Loading [Q]:

= Saddle Load + Frv + max(Fwl, Fwt)

= $492 + 25 + \max(211, 318)$

= 834.8 Kgf

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)

910.60 Kgf

Transverse Shear Load

294.46 Kgf

Longitudinal Shear Load

412.27 Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

= $\min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m)$

= $\min(2(800.0/2 + 18.0 + 10.0) \sin(120.0/2), 2 \cdot 409.0)$

= 741.318 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066 K2 = 1.1707 K3 = 0.8799 K4 = 0.4011

K5 = 0.7603 K6 = 0.0529 K7 = 0.0529 K8 = 0.3405

K9 = 0.2711 K10 = 0.0581 K1* = 0.1923

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

= $-Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L)) / (1 + (4h_m)/3L)]$

= $-835 \cdot 520.0 [1 - (1 - 520.0/2776.0 + (409.0^2 - 205.95^2) /$

$(2 \cdot 520.0 \cdot 2776.0) / (1 + (4 \cdot 205.95) / (3 \cdot 2776.0))]$

= -96.0 Kg-m.

Moment per Equation 4.15.2 [M2]:

= $Q \cdot L / 4 ((1 + 2(R_m^2 - h_m^2)/(L^2)) / (1 + (4h_m^2)/(3L)) - 4a/L)$

= $835 \cdot 2776 / 4 ((1 + 2(409^2 - 206^2)/(2776^2)) / (1 + (4 \cdot 206) /$

$(3 \cdot 2776)) - 4 \cdot 520 / 2776)$

= 110.2 Kg-m.

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

= $P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 t)$

= $0.0 \cdot 409.0 / (2 \cdot 18.0) - 110.2 / (\pi \cdot 409.0^2 \cdot 18.0)$

= -0.11 N./mm²

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

= $P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 81 of 190

Saddle Calcs: Road Transport: Step: 20 10:30am Sep 9, 2026

$$= 0.0 \cdot 409.0 / (2 \cdot 18.0) + 110.2 / (\pi \cdot 409.0^2 \cdot 18.0)$$

$$= 0.11 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot R_m / (2t) - M1 / (K1 \cdot \pi \cdot R_m^2 t)$$

$$= 0.0 \cdot 409.0 / (2 \cdot 18.0) - 96.0 / (0.1066 \cdot \pi \cdot 409.0^2 \cdot 18.0)$$

$$= 0.93 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m / (2t) + M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 0.0 \cdot 409.0 / (2 \cdot 18.0) + 96.0 / (0.1923 \cdot \pi \cdot 409.0^2 \cdot 18.0)$$

$$= -0.52 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 835(2776.0 - 2 \cdot 520.0) / (2776.0 + (4 \cdot 205.95 / 3))$$

$$= 475.0 \text{ Kg}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 475.04 / (409.0 \cdot 18.0)$$

$$= 0.74 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(409.0 \cdot 18.0)}$$

$$= 66.926 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.7603 \cdot 835 \cdot 0.1 / (18.0(127.0 + 66.93 + 66.93))$$

$$= -0.13 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.23) [σ^*7]:

$$= -Q / (4 \cdot t(b + X1 + X2)) - 12 \cdot K7 \cdot Q \cdot R_m / (L \cdot t^2)$$

$$= -835 / (4 \cdot 18.0(127.0 + 66.926 + 66.926)) - 12 \cdot 0.053 \cdot 835 \cdot 409.0 / (2776.0 \cdot 18.0^2)$$

$$= -2.80 \text{ N./mm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(127.0 + 1.56 \cdot \sqrt{(409.0 \cdot 18.0)}, 2 \cdot 520.0)$$

$$= 260.85 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	850.0001	mm
Baseplate Thickness	Bpthk	12.0000	mm
Baseplate Width	Bpwid	152.4000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw, c	300.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(400.0 \cdot 18.0)} + 152.4)^3) / 12$$

$$= 13.840 \text{ cm}^4$$

Resolved Inertia for the Shell [I, shell]:



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 82 of 190

Saddle Calcs: Road Transport: Step: 20 10:30am Sep 9,2026

$$\begin{aligned}
 &= \text{ShellInertia} + A\text{Shell}(C1 - Y)^2 \\
 &= 13.84 + 51.259(128.369 - 9.0)^2 \\
 &= 7317.716 \text{ cm}^4
 \end{aligned}$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	284.8	18.0	9.0	51.3	46132.8	0.732E+04
Wearplate	152.4	10.0	23.0	15.2	35052.0	0.169E+04
Web	10.0	360.0	208.0	36.0	748799.8	0.617E+04
BasePlate	152.4	12.0	394.0	18.3	720547.1	0.129E+05
Totals	...	...	...	120.8	1550531.8	0.281E+05

Distance to Centroid [C1]:

$$\begin{aligned}
 &= AY / A \\
 &= 1550532/120.787 \\
 &= 128.369 \text{ mm}
 \end{aligned}$$

Angle [β]:

$$\begin{aligned}
 &= 180 - \text{Saddle Angle}/2 \\
 &= 180 - 120.0/2 \\
 &= 120.0
 \end{aligned}$$

Saddle Splitting Coefficient [K1]:

$$\begin{aligned}
 &= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta)\cos(\beta)) \\
 &= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0)\cos(120.0)) \\
 &= 0.2035
 \end{aligned}$$

Saddle Splitting Force [Fh]:

$$\begin{aligned}
 &= K1 \cdot Q \\
 &= 0.204 \cdot 834.762 \\
 &= 169.8921 \text{ Kgf}
 \end{aligned}$$

$$\begin{aligned}
 \text{Tension Stress, } St &= (Fh/As) = 0.2396 \text{ N./mm}^2 \\
 \text{Allowed Stress, } Sa &= 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2
 \end{aligned}$$

Saddle Splitting Dimension [d]:

$$\begin{aligned}
 &= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians}) \\
 &= 800.0 - 400.0 \cdot \sin(120.0/2) / 1.0472 \\
 &= 469.203 \text{ mm}
 \end{aligned}$$

$$\text{Bending Moment, } M = Fh \cdot d = 79.7154 \text{ Kg-m.}$$

$$\begin{aligned}
 \text{Bending Stress, } Sb &= (M \cdot C1 / I) = 0.3573 \text{ N./mm}^2 \\
 \text{Allowed Stress, } Sa &= 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2
 \end{aligned}$$

Minimum Thickness of Baseplate per Moss:

$$\begin{aligned}
 &= \sqrt{(3(Q + \text{SaddleWt})\text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress}))} \\
 &= \sqrt{(3(835 + 76)152.4 / (4 \cdot 850.0 \cdot 173.33))} \\
 &= 2.632 \text{ mm}
 \end{aligned}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$\begin{aligned}
 &= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk}) \\
 &= 2 \cdot \cos(90 - 120.0/2) (400.0 + 18.0 + 10.0) \\
 &= 741.318 \text{ mm}
 \end{aligned}$$

Distance between Ribs [e]:

$$\begin{aligned}
 &= \text{Web Length} / (N\text{ribs} - 1) \\
 &= 741.3177 / (3 - 1) \\
 &= 370.659 \text{ mm}
 \end{aligned}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 83 of 190

Saddle Calcs: Road Transport: Step: 20 10:30am Sep 9,2026

$$= 370.6589 \cdot 152.4/2$$

$$= 282.442 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 910.602 / (850.0 \cdot 152.4)$$

$$= 0.703 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= A_p \cdot B_p$$

$$= 282.4 \cdot 0.7$$

$$= 198.543 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 11.7 + 18.533$$

$$= 30.233 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / A_r$$

$$= 198.5 / 30.2329$$

$$= 0.644 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	127.0	...	12.7	...	171.

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 127.0 - 10.0$$

$$= 117.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0 / 30.233$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 127.0 / 2$$

$$= 63.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(170.7 / 30.233)}$$

$$= 23.762 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 800.0 - \cos(120.0/2) (400.0 + 18.0 + 10.0) - 12.0$$

$$= 574.000 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 0.19994\text{E}+09 / 260.0)}$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 574.0 / 23.762$$

$$= 24.157$$



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 84 of 190

Saddle Calcs: Road Transport: Step: 20 10:30am Sep 9, 2026

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 412.3 / (2 \cdot 850.0) \cdot 370.659 \cdot 574.0 / 2$$

$$= 25.799 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ (24.1567 < 123.2105) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (24.16)^2 / (2 \cdot 123.21^2)) 260 /$$

$$(5/3 + 3 \cdot (24.16) / (8 \cdot 123.21) - (24.16^3) / (8 \cdot 123.21^3))$$

$$= 146.6 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 0.64 / 146.61 + (25.8 \cdot 63.5 / 1706986) / 173.33$$

$$= 0.059$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	117.0	0.0	11.7	0.0	171.
Web	370.7	10.0	0.0	37.1	0.0	3.09
Totals	...	...	...	48.8	...	174.

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0 / 48.766$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 127.0 / 2$$

$$= 63.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - B_{pthk}$$

$$= 800.0 - \cos(428.0 / 2) (12.0 + 0.0 + 0.0) - 0.0$$

$$= 360.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (173.7 / 48.766) }$$

$$= 18.873 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 360.0 / 18.873$$

$$= 19.075$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 412.27 / (2 \cdot 850.0)$$

$$= 0.243 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.243 \cdot 370.659 \cdot 360.0$$

$$= 32.361 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [$\sigma_{B,base,c}$]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 32.361 / 27354.967$$

$$= 11.601 \text{ N./mm}^2$$



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 85 of 190

Saddle Calcs: Road Transport: Step: 20 10:30am Sep 9, 2026

Compressive Allowable, $KL/r < C_c$ (19.0746 < 123.2105) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (19.07)^2 / (2 \cdot 123.21^2)) 260 /$$

$$(5/3 + 3 \cdot (19.07) / (8 \cdot 123.21) - (19.07^3) / (8 \cdot 123.21^3))$$

$$= 149 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_1 / I) / S_{ba}$$

$$= 0.8 / 148.98 + (32.36 \cdot 63.5 / 173.704) / 173.33$$

$$= 0.072$$

Review notes about nozzle loadings and supports in the Warnings report.

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PV Elite 28 Licensee: SL User

FileName : E-0102

Page 86 of 190

Saddle Calcs: Operating Case: Step: 21 10:30am Sep 9,2026

ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Operating Case

Note:

Wear Pad Width (152.40) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
and less than $2a$. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

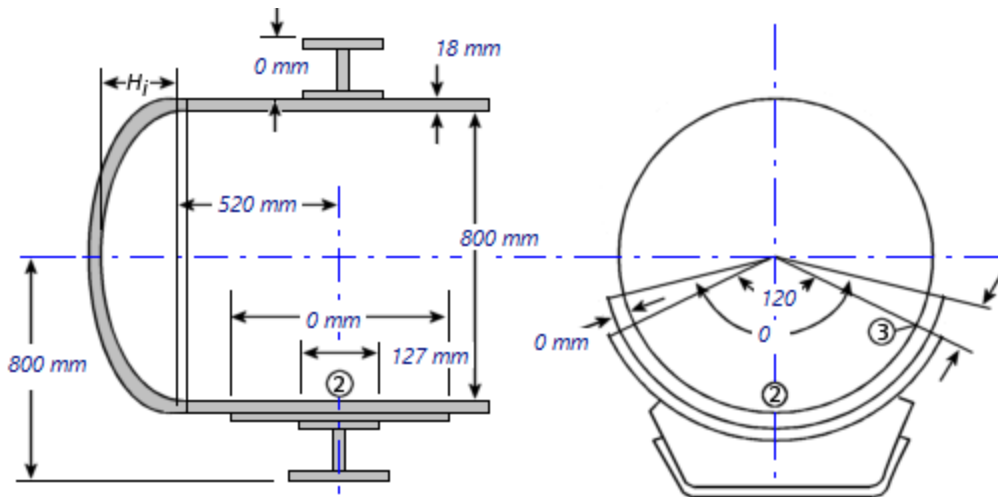
$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(127.0 + 1.56 \cdot \sqrt{410.6 \cdot 14.8}, 2 \cdot 520.0)$$

$$= 248.6088 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	410.60	mm
Shell Thickness used in this Case	t	18.000	mm
Stiffened Vessel Length per 4.15.6	L	2776.00	mm
Distance from Saddle to Vessel tangent	a1 or a	520.00	mm
Saddle Width	b1 or b	127.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	800.00	mm
Coefficient of Friction	mu	0.30	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case	7791.53	Kgf
Pressure used in Analysis	P	1203.921 KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	14.71	138.00
Long. Stress at Bottom of Midspan	18.69	138.00
Long. Stress at Top of Saddles	22.83	138.00
Long. Stress at Bottom of Saddles	13.30	138.00



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 87 of 190

Saddle Calcs: Operating Case: Step: 21 10:30am Sep 9, 2026

Tangential Shear in Shell	9.21	110.40
Circ. Stress at Horn of Saddle	37.92	172.50
Circ. Compressive Stress in Shell	1.58	138.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$$

$$= 3.0 (1356.2/2 + 0) \cdot 800.0/741.3178$$

$$= 2195.4 \text{ Kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(F_{l, \text{wind}}, \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(669.81, 0) \cdot 800.0/1560.0$$

$$= 343.5 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(F_{l, \text{eq}}, \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(379.81, 0) \cdot 800.0/1560.0$$

$$= 194.8 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$$

$$= 3.0 (380/2 + 0) \cdot 800.0/741.3178$$

$$= 614.8 \text{ Kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$$

$$= 5596 + \max(343, 2195, 195, 615)$$

$$= 7791.5 \text{ Kgf}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 4919.274 (0.0 + \pi/4 (1.036 \cdot 1.2)^2)$$

$$= 6568.540 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	7867.37	Kgf
Transverse Shear Load	678.12	Kgf
Longitudinal Shear Load	669.81	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \min(2 (\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m)$$

$$= \min(2 (800.0/2 + 18.0 + 10.0) \sin(120.0/2), 2 \cdot 410.6)$$

$$= 741.318 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L)) / (1 + (4h_m)/3L)]$$

$$= -7792 \cdot 520.0 [1 - (1 - 520.0/2776.0 + (410.6^2 - 0.0^2)/(2 \cdot 520.0 \cdot 2776.0)) / (1 + (4 \cdot 0.0)/(3 \cdot 2776.0))]$$

$$= -522.4 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4 ((1 + 2(R_m^2 - h_m^2)/(L^2)) / (1 + (4h_m^2)/(3L)) - 4a/L)$$



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 88 of 190

Saddle Calcs: Operating Case: Step: 21 10:30am Sep 9, 2026

$$= 7792 \cdot 2776 / 4 \cdot ((1 + 2(411^2 - 0^2) / (2776^2)) / (1 + (4 \cdot 0) / (3 \cdot 2776))) - 4 \cdot 520 / 2776$$

$$= 1592.4 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 1203.921 \cdot 410.6 / (2 \cdot 14.8) - 1592.4 / (\pi \cdot 410.6^2 \cdot 14.8)$$

$$= 14.71 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 1203.921 \cdot 410.6 / (2 \cdot 14.8) + 1592.4 / (\pi \cdot 410.6^2 \cdot 14.8)$$

$$= 18.69 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 1203.921 \cdot 410.6 / (2 \cdot 14.8) - 522.4 / (0.1066 \cdot \pi \cdot 410.6^2 \cdot 14.8)$$

$$= 22.83 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 1203.921 \cdot 410.6 / (2 \cdot 14.8) + 522.4 / (0.1923 \cdot \pi \cdot 410.6^2 \cdot 14.8)$$

$$= 13.30 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot h_m / 3))$$

$$= 7792(2776.0 - 2 \cdot 520.0) / (2776.0 + (4 \cdot 0.0 / 3))$$

$$= 4872.5 \text{ Kg}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 4872.51 / (410.6 \cdot 14.8)$$

$$= 9.21 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(410.6 \cdot 14.8)}$$

$$= 60.804 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K_5 \cdot Q \cdot k / (t(b + X_1 + X_2))$$

$$= -0.7603 \cdot 7792 \cdot 0.1 / (14.8(127.0 + 60.8 + 60.8))$$

$$= -1.58 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.23) [σ_{*7}]:

$$= -Q / (4 \cdot t(b + X_1 + X_2)) - 12 \cdot K_7 \cdot Q \cdot R_m / (L \cdot t^2)$$

$$= -7792 / (4 \cdot 14.8(127.0 + 60.804 + 60.804)) - 12 \cdot 0.053 \cdot 7792 \cdot 410.6 / (2776.0 \cdot 14.8^2)$$

$$= -37.92 \text{ N./mm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(127.0 + 1.56 \cdot \sqrt{(410.6 \cdot 14.8)}, 2 \cdot 520.0)$$

$$= 248.61 \text{ mm}$$

Distance between Saddle Supports [Ls]:

$$= 1560.0 \text{ mm}$$

Free Un-Restrained Thermal Expansion between the Saddles [Exp]:

$$= \alpha \cdot L_s(\text{Design Temperature} - \text{Ambient Temperature})$$

$$= 0.000012 \cdot 1560.0(82.0 - 21.1)$$

$$= 1.138 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length BpLen 850.0001 mm



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 89 of 190

Saddle Calcs: Operating Case: Step: 21 10:30am Sep 9, 2026

Baseplate Thickness	Bpthk	12.0000	mm
Baseplate Width	Bpwid	152.4000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	262.0	N./mm ²
Height of Web at Center	Hw,c	300.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t) + \text{WearPlateWidth}})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(403.2 \cdot 14.8) + 152.4})^3) / 12$$

$$= 7.373 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 7.373 + 40.39(137.56 - 7.4)^2$$

$$= 6850.169 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	272.9	14.8	7.4	40.4	29888.9	0.685E+04
Wearplate	152.4	10.0	19.8	15.2	30175.2	0.211E+04
Web	10.0	360.0	204.8	36.0	737279.9	0.552E+04
BasePlate	152.4	12.0	390.8	18.3	714694.9	0.117E+05
Totals	...	...	...	109.9	1512038.9	0.262E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 1512039 / 109.918$$

$$= 137.560 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120.0 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 7791.53$$

$$= 1585.7454 \text{ Kg}$$

$$\text{Tension Stress, } St = (Fh / As) = 2.2367 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 157.1972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 800.0 - 403.2 \cdot \sin(120.0/2) / 1.0472$$

$$= 466.556 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 739.8541 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 3.8078 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 174.6635 \text{ N./mm}^2$$



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 90 of 190

Saddle Calcs: Operating Case: Step: 21 10:30am Sep 9, 2026

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{\frac{3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}}{4 \cdot \text{BasePlateLength} \cdot \text{AllStress}}}$$

$$= \sqrt{\frac{3(7792 + 76)152.4}{4 \cdot 850.0 \cdot 174.663}}$$

$$= 7.707 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (400.0 + 18.0 + 10.0)$$

$$= 741.318 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 741.3177 / (3 - 1)$$

$$= 370.659 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 370.6589 \cdot 152.4 / 2$$

$$= 282.442 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 7867.371 / (850.0 \cdot 152.4)$$

$$= 6.073 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= \text{Ap} \cdot \text{Bp}$$

$$= 282.4 \cdot 6.07$$

$$= 1715.359 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 11.7 + 18.533$$

$$= 30.233 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / \text{Ar}$$

$$= 1715.4 / 30.2329$$

$$= 5.564 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	127.0	...	12.7	...	171.

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 127.0 - 10.0$$

$$= 117.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= \text{AY} / \text{A}$$

$$= 0.0 / 30.233$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 127.0 / 2$$

$$= 63.500 \text{ mm}$$



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 91 of 190

Saddle Calcs: Operating Case: Step: 21 10:30am Sep 9,2026

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{170.7/30.233}$$

$$= 23.762 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\text{theta}/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 800.0 - \cos(120.0/2) (400.0 + 18.0 + 10.0) - 12.0$$

$$= 574.000 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$$

$$= \sqrt{2 \cdot \pi^2 \cdot 0.19994\text{E}+09/262.0}$$

$$= 122.739$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 574.0/23.762$$

$$= 24.157$$

Bending Moment [Rm]:

$$= FI / (2 \cdot Bplen) \cdot e \cdot L / 2$$

$$= 669.8 / (2 \cdot 850.0) \cdot 370.659 \cdot 574.0/2$$

$$= 41.915 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < Cc$ (24.1567 < 122.7394) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (24.16)^2 / (2 \cdot 122.74^2)) 262 /$$

$$(5/3 + 3 \cdot (24.16) / (8 \cdot 122.74) - (24.16^3) / (8 \cdot 122.74^3))$$

$$= 147.7 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 5.56/147.7 + (41.92 \cdot 63.5/1706986) / 174.66$$

$$= 0.125$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	117.0	0.0	11.7	0.0	171.
Web	370.7	10.0	0.0	37.1	0.0	3.09
Totals	...	...	...	48.8	...	174.

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/48.766$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 127.0/2$$

$$= 63.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 800.0 - \cos(428.0/2) (12.0 + 0.0 + 0.0) - 0.0$$

$$= 360.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{173.7/48.766}$$

$$= 18.873 \text{ mm}$$



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 92 of 190

Saddle Calcs: Operating Case: Step: 21 10:30am Sep 9, 2026

Slenderness ratio $[KL/r]$:

$$= KL/r$$

$$= 1 \cdot 360.0 / 18.873$$

$$= 19.075$$

Unit Force $[Force, u]$:

$$= F_l / (2 \cdot \text{Baseplate Length})$$

$$= 669.815 / (2 \cdot 850.0)$$

$$= 0.394 \text{ Kg/mm}$$

Moment at base of inner Rib $[M_{base, c}]$:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.394 \cdot 370.659 \cdot 360.0$$

$$= 52.576 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load $[\sigma_B, base, c]$:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 52.576 / 27354.967$$

$$= 18.848 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($19.0746 < 122.7394$) per AISC E2-1 $[Sca]$:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (19.07)^2 / (2 \cdot 122.74^2)) 262 /$$

$$(5/3 + 3 \cdot (19.07) / (8 \cdot 122.74) - (19.07^3) / (8 \cdot 122.74^3))$$

$$= 150.1 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base, c} \cdot C_1 / I) / S_{ba}$$

$$= 6.9 / 150.09 + (52.58 \cdot 63.5 / 173.704) / 174.66$$

$$= 0.154$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-307 A
Bolt Allowable Stress	Stba	138.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.0 mm
User Defined Area in a Single Bolt	UserBoltArea	2.4000 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	1.6469 cm ²
Saddle Load Q0 (Weight)	Q0	5672.0 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	343.5 Kg
Maximum Transverse Force	Ft	678.1 Kg
Maximum Longitudinal Force	F1	669.8 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction $[\tau_b, l]$:

$$= F_l / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 669.8 / (1.647 \cdot 4)$$

$$= 10.0 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction $[\tau_b, t]$:

$$= F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 678.1 / (1.647 \cdot 4)$$

$$= 10.1 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load $[Bltarearl]$:

$$= 0.0 \text{ (} Q_0 > Q_L \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load $[Bltarears]$:



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 93 of 190

Saddle Calcs: Operating Case: Step: 21 10:30am Sep 9, 2026

= $F_l / (\text{BoltShearAllowable} \cdot \text{Nbolts})$
 = $669.81 / (82.8 \cdot 4.0)$
 = 0.1983 cm^2

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 = $B \cdot F_t + \text{Sum of X Moments}$
 = $800.0 \cdot 678.12 + 0.0$
 = 542.51 Kg-m .

Eccentricity (e):
 = R_{mom} / QO
 = $542.51 / 5671.97$
 = $95.65 \text{ mm} < B_{plen} / 6 \rightarrow$ No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
 = 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
 = $\max[Bltarearl, Bltarears, Bltareart]$
 = $\max[0.0, 0.1983, 0.0]$
 = 0.1983 cm^2

MDMT Calculations for: LH saddle, Saddle to Wearplate Junction, Curve: B

Govrn. thk, $t_g = 10.0$, $c = 3.2 \text{ mm}$, $E^* = 1.0$
 Thickness Ratio = $tr \cdot E^* / (t_g - c) = 0.299$, Temp. Reduction = 78°C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:
 Wear Pad Width (152.40) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
 and less than $2a$. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:
 = $\min(b + 1.56 \cdot \sqrt{ R_m \cdot t }, 2a)$
 = $\min(127.0 + 1.56 \cdot \sqrt{ 410.6 \cdot 14.8 }, 2 \cdot 520.0)$
 = 248.6088 mm

Input and Calculated Values:

Vessel Mean Radius	Rm	410.60	mm
Shell Thickness used in this Case	t	18.000	mm
Stiffened Vessel Length per 4.15.6	L	2776.00	mm
Distance from Saddle to Vessel tangent	a1 or a	520.00	mm
Saddle Width	b1 or b	127.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	207.55	mm
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	800.00	mm
Coefficient of Friction	mu	0.30	



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 94 of 190

Saddle Calcs: Operating Case: Step: 21 10:30am Sep 9,2026

Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case 3261.18 Kgf
Pressure used in Analysis P 1203.921 KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	16.16	138.00
Long. Stress at Bottom of Midspan	17.24	138.00
Long. Stress at Top of Saddles	21.11	138.00
Long. Stress at Bottom of Saddles	14.26	138.00
Tangential Shear in Shell	3.50	110.40
Circ. Stress at Horn of Saddle	15.87	172.50
Circ. Compressive Stress in Shell	0.66	138.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
= Ftr(Ft/Num of Saddles + Z Force Load) • B / E
= 3.0(1356.2/2 + 0) • 800.0/741.3178
= 2195.4 Kgf

Longitudinal Saddle Reaction Force [Fwl]:
= max(Fl,wind; Sum of X Forces) • B / Ls
= max(669.81, 0) • 800.0/1560.0
= 343.5 Kgf

Saddle Reaction Force due to Earthquake Fl [Fsl]:
= max(Fl,eq; Sum of X Forces) • B / Ls
= max(379.81, 0) • 800.0/1560.0
= 194.8 Kgf

Saddle Reaction Force due to Earthquake Ft [Fst]:
= Ftr(Ft/Num of Saddles + Z Force Load) • B / E
= 3.0(380/2 + 0) • 800.0/741.3178
= 614.8 Kgf

Load Combination Results for Q + Wind or Seismic [Q]:
= Saddle Load + max(Fwl, Fwt, Fsl, Fst)
= 1066 + max(343, 2195, 195, 615)
= 3261.2 Kgf

Longitudinal Wind Force [Fl]:
= WindScalar • WindPress(Platform Area + ($\pi/4$ (OD • WindDiaMult)²))
= 1.1 • 4919.274(0.0 + $\pi/4$ (1.036 • 1.2)²)
= 6568.540 N

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 3337.02 Kgf
Transverse Shear Load 678.12 Kgf
Longitudinal Shear Load 669.81 Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:
= min(2(ShellID/2 + t + WearPadThickness)sin($\theta/2$), 2•Rm)
= min(2(800.0/2 + 18.0 + 10.0)sin(120.0/2), 2•410.6)
= 741.318 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066 K2 = 1.1707 K3 = 0.8799 K4 = 0.4011
K5 = 0.7603 K6 = 0.0529 K7 = 0.0529 K8 = 0.3405



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 95 of 190

Saddle Calcs: Operating Case: Step: 21 10:30am Sep 9, 2026

$$K9 = 0.2711 \quad K10 = 0.0581 \quad K1* = 0.1923$$

Note: Dimension a is greater than or equal to $Rm/2$.

Moment per Equation 4.15.1 [M1]:

$$\begin{aligned} &= -Q \cdot a [1 - (1 - a/L + (Rm^2 \cdot hm^2)/(2a \cdot L))/(1 + (4hm)/3L)] \\ &= -3261 \cdot 520.0 [1 - (1 - 520.0/2776.0 + (410.6^2 \cdot 207.55^2)/(2 \cdot 520.0 \cdot 2776.0))/(1 + (4 \cdot 207.55)/(3 \cdot 2776.0))] \\ &= -375.6 \text{ Kg-m.} \end{aligned}$$

Moment per Equation 4.15.2 [M2]:

$$\begin{aligned} &= Q \cdot L/4 ((1 + 2(Rm^2 \cdot hm^2)/(L^2))/(1 + (4hm^2)/(3L)) - 4a/L) \\ &= 3261 \cdot 2776/4 ((1 + 2(411^2 \cdot 208^2)/(2776^2))/(1 + (4 \cdot 208)/(3 \cdot 2776)) - 4 \cdot 520/2776) \\ &= 429.3 \text{ Kg-m.} \end{aligned}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$\begin{aligned} &= P \cdot Rm/(2t) - M2/(\pi \cdot Rm^2 \cdot t) \\ &= 1203.921 \cdot 410.6/(2 \cdot 14.8) - 429.3/(\pi \cdot 410.6^2 \cdot 14.8) \\ &= 16.16 \text{ N./mm}^2 \end{aligned}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$\begin{aligned} &= P \cdot Rm/(2t) + M2/(\pi \cdot Rm^2 \cdot t) \\ &= 1203.921 \cdot 410.6/(2 \cdot 14.8) + 429.3/(\pi \cdot 410.6^2 \cdot 14.8) \\ &= 17.24 \text{ N./mm}^2 \end{aligned}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$\begin{aligned} &= P \cdot Rm/(2t) - M1/(K1 \cdot \pi \cdot Rm^2 \cdot t) \\ &= 1203.921 \cdot 410.6/(2 \cdot 14.8) - 375.6/(0.1066 \cdot \pi \cdot 410.6^2 \cdot 14.8) \\ &= 21.11 \text{ N./mm}^2 \end{aligned}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$\begin{aligned} &= P \cdot Rm/(2t) + M1/(K1^* \cdot \pi \cdot Rm^2 \cdot t) \\ &= 1203.921 \cdot 410.6/(2 \cdot 14.8) + 375.6/(0.1923 \cdot \pi \cdot 410.6^2 \cdot 14.8) \\ &= 14.26 \text{ N./mm}^2 \end{aligned}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$\begin{aligned} &= Q(L - 2a)/(L + (4 \cdot hm/3)) \\ &= 3261(2776.0 - 2 \cdot 520.0)/(2776.0 + (4 \cdot 207.55/3)) \\ &= 1854.5 \text{ Kgf} \end{aligned}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$\begin{aligned} &= K2 \cdot T / (Rm \cdot t) \\ &= 1.1707 \cdot 1854.54/(410.6 \cdot 14.8) \\ &= 3.50 \text{ N./mm}^2 \end{aligned}$$

Decay Length (4.15.20) [x_1, x_2]:

$$\begin{aligned} &= 0.78 \cdot \sqrt{(Rm \cdot t)} \\ &= 0.78 \cdot \sqrt{(410.6 \cdot 14.8)} \\ &= 60.804 \text{ mm} \end{aligned}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$\begin{aligned} &= -K5 \cdot Q \cdot k / (t(b + X1 + X2)) \\ &= -0.7603 \cdot 3261 \cdot 0.1/(14.8(127.0 + 60.8 + 60.8)) \\ &= -0.66 \text{ N./mm}^2 \end{aligned}$$

Circ. Comp. Stress at Horn of Saddle, $L < 8Rm$ (4.15.23) [σ^*7]:

$$\begin{aligned} &= -Q/(4 \cdot t(b + X1 + X2)) - 12 \cdot K7 \cdot Q \cdot Rm/(L \cdot t^2) \\ &= -3261/(4 \cdot 14.8(127.0 + 60.804 + 60.804)) - 12 \cdot 0.053 \cdot 3261 \cdot 410.6/(2776.0 \cdot 14.8^2) \\ &= -15.87 \text{ N./mm}^2 \end{aligned}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$\begin{aligned} &= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a) \\ &= \min(127.0 + 1.56 \cdot \sqrt{(410.6 \cdot 14.8)}, 2 \cdot 520.0) \end{aligned}$$



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 96 of 190

Saddle Calcs: Operating Case: Step: 21 10:30am Sep 9,2026

= 248.61 mm

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	850.0001	mm
Baseplate Thickness	Bpthk	12.0000	mm
Baseplate Width	Bpwid	152.4000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	262.0	N./mm ²
Height of Web at Center	Hw,c	300.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3)/12$$

$$= ((1.56 \cdot \sqrt{(403.2 \cdot 14.8)} + 152.4)14.8^3)/12$$

$$= 7.373 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 7.373 + 40.39(137.56 - 7.4)^2$$

$$= 6850.169 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	272.9	14.8	7.4	40.4	29888.9	0.685E+04
Wearplate	152.4	10.0	19.8	15.2	30175.2	0.211E+04
Web	10.0	360.0	204.8	36.0	737279.9	0.552E+04
BasePlate	152.4	12.0	390.8	18.3	714694.9	0.117E+05
Totals	...	...	...	109.9	1512038.9	0.262E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 1512039/109.918$$

$$= 137.560 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120.0/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta)\cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0)\cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 3261.181$$

$$= 663.7210 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh/As) = 0.9362 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 157.1972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$$

$$= 800.0 - 403.2 \cdot \sin(120.0/2) / 1.0472$$

$$= 466.556 \text{ mm}$$



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 97 of 190

Saddle Calcs: Operating Case: Step: 21 10:30am Sep 9, 2026

Bending Moment, $M = F_h \cdot d = 309.6693 \text{ Kg-m.}$

Bending Stress, $S_b = (M \cdot C_1 / I) = 1.5938 \text{ N./mm}^2$

Allowed Stress, $S_a = 2/3 \cdot \text{Yield Str} = 174.6635 \text{ N./mm}^2$

Minimum Thickness of Baseplate per Moss:

$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress}))}$

$= \sqrt{(3(3261 + 76)152.4 / (4 \cdot 850.0 \cdot 174.663))}$

$= 5.019 \text{ mm}$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$

$= 2 \cdot \cos(90 - 120.0/2) (400.0 + 18.0 + 10.0)$

$= 741.318 \text{ mm}$

Distance between Ribs [e]:

$= \text{Web Length} / (\text{Nr ribs} - 1)$

$= 741.3177 / (3 - 1)$

$= 370.659 \text{ mm}$

Baseplate Pressure Area [Ap]:

$= e \cdot \text{Bpwid} / 2$

$= 370.6589 \cdot 152.4 / 2$

$= 282.442 \text{ cm}^2$

Bearing Pressure [Bp]:

$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$

$= 3337.021 / (850.0 \cdot 152.4)$

$= 2.576 \text{ Kg/cm}^2$

Axial Load [P]:

$= A_p \cdot B_p$

$= 282.4 \cdot 2.58$

$= 727.586 \text{ Kg}$

Area of the Rib and Web [Ar]:

$= \text{Rib Area} + \text{Web Area}$

$= 11.7 + 18.533$

$= 30.233 \text{ cm}^2$

Compressive Stress [Sc]:

$= P / A_r$

$= 727.6 / 30.2329$

$= 2.360 \text{ N./mm}^2$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	127.0	...	12.7	...	171.

Rib dimension [D]:

$= \text{Saddle Width} - \text{Web Thickness}$

$= 127.0 - 10.0$

$= 117.000 \text{ mm}$

Distance to Centroid from Datum [ytot]:

$= AY / A$

$= 0.0 / 30.233$

$= 0.000 \text{ mm}$



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 98 of 190

Saddle Calcs: Operating Case: Step: 21 10:30am Sep 9, 2026

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 127.0 / 2$$

$$= 63.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{170.7 / 30.233}$$

$$= 23.762 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 800.0 - \cos(120.0/2) (400.0 + 18.0 + 10.0) - 12.0$$

$$= 574.000 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 0.19994\text{E}+09 / 262.0)}$$

$$= 122.739$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 574.0 / 23.762$$

$$= 24.157$$

Bending Moment [Rm]:

$$= FI / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$$

$$= 669.8 / (2 \cdot 850.0) \cdot 370.659 \cdot 574.0 / 2$$

$$= 41.915 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < Cc$ ($24.1567 < 122.7394$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (24.16)^2 / (2 \cdot 122.74^2)) 262 /$$

$$(5/3 + 3 \cdot (24.16) / (8 \cdot 122.74) - (24.16)^3 / (8 \cdot 122.74^3))$$

$$= 147.7 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 2.36 / 147.7 + (41.92 \cdot 63.5 / 1706986) / 174.66$$

$$= 0.104$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	117.0	0.0	11.7	0.0	171.
Web	370.7	10.0	0.0	37.1	0.0	3.09
Totals	...	...	...	48.8	...	174.

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0 / 48.766$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 127.0 / 2$$

$$= 63.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 800.0 - \cos(428.0/2) (12.0 + 0.0 + 0.0) - 0.0$$

$$= 360.000 \text{ mm}$$



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 99 of 190

Saddle Calcs: Operating Case: Step: 21 10:30am Sep 9, 2026

Radius of Gyration [r]:
 $= \sqrt{\text{Total Inertia} / \text{Total Area}}$
 $= \sqrt{(173.7/48.766)}$
 $= 18.873 \text{ mm}$

Slenderness ratio [KL/r]:
 $= KL/r$
 $= 1 \cdot 360.0/18.873$
 $= 19.075$

Unit Force [Force,u]:
 $= FI / (2 \cdot \text{Baseplate Length})$
 $= 669.815 / (2 \cdot 850.0)$
 $= 0.394 \text{ Kgf/mm}$

Moment at base of inner Rib [Mbase,c]:
 $= \text{Unit Force} \cdot e \cdot L$
 $= 0.394 \cdot 370.659 \cdot 360.0$
 $= 52.576 \text{ Kg-m.}$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:
 $= \text{Bending Moment} / \text{Section Modulus}$
 $= 52.576/27354.967$
 $= 18.848 \text{ N./mm}^2$

Compressive Allowable, $KL/r < C_c$ ($19.0746 < 122.7394$) per AISC E2-1 [Sca]:
 $= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$
 $= (1 - (19.07)^2 / (2 \cdot 122.74^2)) 262 /$
 $(5/3 + 3 \cdot (19.07) / (8 \cdot 122.74) - (19.07)^3 / (8 \cdot 122.74^3))$
 $= 150.1 \text{ N./mm}^2$

AISC Unity Check of Inside Ribs (must be ≤ 1)
 $= S_c / S_{ca} + (M_{base,c} \cdot C1/I) / S_{ba}$
 $= 2.93/150.09 + (52.58 \cdot 63.5/173.704) / 174.66$
 $= 0.127$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-307 A
Bolt Allowable Stress	Stba	138.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.0 mm
User Defined Area in a Single Bolt	UserBoltArea	2.4000 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	1.6469 cm ²
Saddle Load Q0 (Weight)	Q0	1141.6 Kgf
Saddle Load QL (Wind/Seismic contribution)	QL	343.5 Kgf
Maximum Transverse Force	Ft	678.1 Kgf
Maximum Longitudinal Force	F1	669.8 Kgf
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:
 $= FI / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 $= 669.8 / (1.647 \cdot 4)$
 $= 10.0 \text{ N./mm}^2$. Must be less than 82.8 N./mm^2 .

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:
 $= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 $= 678.1 / (1.647 \cdot 4)$
 $= 10.1 \text{ N./mm}^2$. Must be less than 82.8 N./mm^2 .

Bolt Area Calculation per Dennis R. Moss



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 100 of 190

Saddle Calcs: Operating Case: Step: 21 10:30am Sep 9, 2026

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
= $F_l / (BoltShearAllowable \cdot Nbolts)$
= $669.81 / (82.8 \cdot 4.0)$
= 0.1983 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
= $B \cdot F_t + \text{Sum of X Moments}$
= $800.0 \cdot 678.12 + 0.0$
= 542.51 Kg-m.

Eccentricity (e):
= $Rmom / QO$
= $542.51 / 1141.62$
= 475.20 mm > $Bplen / 6$ --> Uplift in Transverse direction

$f = Bplen / 2 - Edgedis$
= $850.0 / 2 - 50.0$
= 375.00 mm

Modular Ratio Of Steel/Concrete (n1):
= ES / EC
= $203402.5 / 21526.32$
= 9.45

$K1 = 3 (e - 0.5 \cdot Bplen)$
= $3 (475.2 - 0.5 \cdot 850.0)$
= 150.60 mm

$K2 = 6 \cdot n1 \cdot At / Bpwid \cdot (f + e)$
= $6 \cdot 9.45 \cdot 3.29 / 152.4 \cdot (375.0 + 475.2)$
= 104174.97 mm²

$K3 = -K2 \cdot (0.5 \cdot Bplen + f)$
= $-104174.97 \cdot (0.5 \cdot 850.0 + 375.0)$
= -83339984.49 mm³

Iteratively Solving for the Effective Bearing Length:
 $Y^3 + K1 \cdot Y^2 + K2 \cdot Y + K3 = 0$
 $Y^3 + 150.6 \cdot Y^2 + 104174.97 \cdot Y + -0.8E+08 = 0$
 $Y = 323.55 \text{ mm}$

$Num = (Bplen / 2 - Y / 3 - e)$
= $(850.0 / 2 - 323.55 / 3 - 475.2)$
= -158.05

$Denom = (Bplen / 2 - Y / 3 + f)$
= $(850.0 / 2 - 323.55 / 3 + 375.0)$
= 692.15

Total Bolt Tension Force [Tforce]:
= $-QO \cdot Num / Denom$
= $-1141.62 \cdot -158.05 / 692.15$
= 260.68 Kg

Bolt Area Required due to Transverse Load [Bltareart]:
= $Tforce / (Stba \cdot Nbt)$
= $260.68 / (138.0 \cdot 2.0)$
= 0.0926 cm²



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 101 of 190

Saddle Calcs: Operating Case: Step: 21 10:30am Sep 9, 2026

Required Area of a Single Bolt [Bltarear]:
 = max[Bltarearl, Bltarears, Bltareart]
 = max[0.0, 0.1983, 0.0926]
 = 0.1983 cm²

Baseplate Thickness Calculation per D. Moss:

Bearing Pressure (fc)
 = $2(QO + Tforce) / (Y \cdot Bpwid)$
 = $2(1141.62 + 260.68) / (323.55 \cdot 152.4)$
 = 557.77 KPa.

Distance from Baseplate Edge to the Web [ADIST]:
 = $(Bplen - WebIngh) / 2$
 = $(850.0 - 799.2) / 2$
 = 25.4000 mm

Overturning Moment due To Bolt Tension [Mt]:
 = $Tforce \cdot Adist$
 = $260.68 \cdot 25.4$
 = 6.62 Kg-m.

Equivalent Bearing Pressure (f1):
 = $fc \cdot (Y - Adist) / Y$
 = $557.77 \cdot (323.55 - 25.4) / 323.55$
 = 513.98 KPa.

Overturning Moment due to Bearing Pressure [Mc]:
 = $(Adist^2 \cdot Bpwid / 6) \cdot (f1 + 2 \cdot fc)$
 = $(25.4^2 \cdot 152.4 / 6) \cdot (513.98 + 2 \cdot 557.77)$
 = 2.72 Kg-m.

Baseplate Required Thickness [Treq]:
 = $\sqrt{(6 \cdot \max(Mt, Mc) / (Bpwid \cdot Sba))}$
 = $\sqrt{(6 \cdot \max(6.62, 2.72) / (152.4 \cdot 207.0))}$
 = 3.5143 mm

MDMT Calculations for: Right saddle, Saddle to Wearplate Junction, Curve: B

Govrn. thk, tg = 10.0, c = 3.2 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E* / (tg - c) = 0.299$, Temp. Reduction = 78 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Review notes about nozzle loadings and supports in the Warnings report.

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PV Elite 28 Licensee: SL User

FileName : E-0102

Page 102 of 190

Saddle Calcs: Test Case:

Step: 22 10:30am Sep 9, 2026

ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Test Case

Note:

Wear Pad Width (152.40) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
and less than $2a$. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

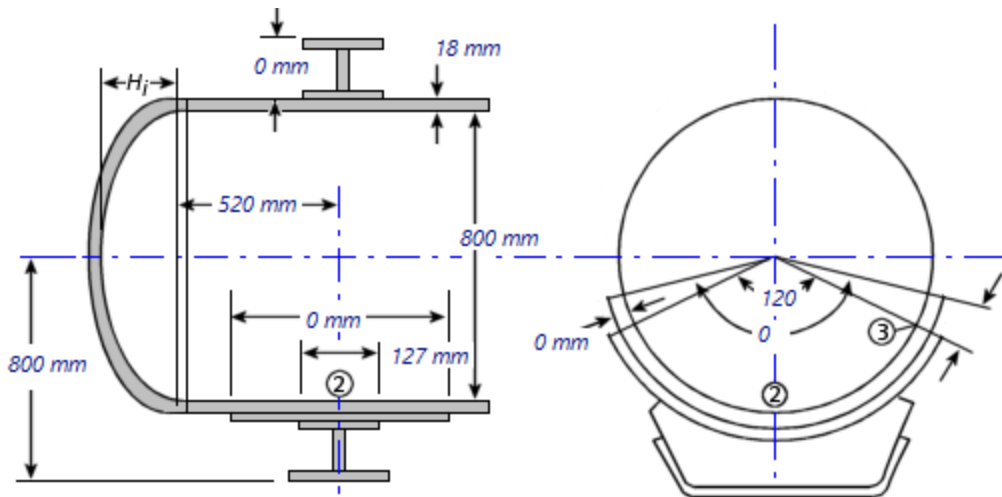
$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(127.0 + 1.56 \cdot \sqrt{410.6 \cdot 14.8}, 2 \cdot 520.0)$$

$$= 248.6088 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	410.60	mm
Shell Thickness used in this Case	t	18.000	mm
Stiffened Vessel Length per 4.15.6	L	2776.00	mm
Distance from Saddle to Vessel tangent	a1 or a	520.00	mm
Saddle Width	b1 or b	127.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Shell Allowable Stress used in Calculation		234.00	N./mm ²
Head Allowable Stress used in Calculation		234.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	800.00	mm
Coefficient of Friction	mu	0.30	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		6224.96	Kgf
Pressure used in Analysis	P	1564.025	KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	20.11	234.00
Long. Stress at Bottom of Midspan	23.29	234.00
Long. Stress at Top of Saddles	26.59	234.00
Long. Stress at Bottom of Saddles	18.98	234.00



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 103 of 190

Saddle Calcs: Test Case:

Step: 22 10:30am Sep 9, 2026

Tangential Shear in Shell	7.35	138.58
Circ. Stress at Horn of Saddle	30.29	350.99
Circ. Compressive Stress in Shell	1.26	234.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0 (678.1/2 + 0) \cdot 800.0/741.3178$
 $= 1097.7 \text{ Kg}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l, \text{wind}}, \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(334.91, 0) \cdot 800.0/1560.0$
 $= 171.7 \text{ Kg}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$
 $= 5127 + \max(172, 1098, 0, 0)$
 $= 6225.0 \text{ Kg}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 4919.274 (0.0 + \pi/4 (1.036 \cdot 1.2)^2)$
 $= 6568.540 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	6300.80	Kgf
Transverse Shear Load	339.06	Kgf
Longitudinal Shear Load	334.91	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m)$
 $= \min(2(800.0/2 + 18.0 + 10.0) \sin(120.0/2), 2 \cdot 410.6)$
 $= 741.318 \text{ mm}$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K11* = 0.1923	

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L)) / (1 + (4h_m)/3L)]$
 $= -6225 \cdot 520.0 [1 - (1 - 520.0/2776.0 + (410.6^2 - 0.0^2) / (2 \cdot 520.0 \cdot 2776.0)) / (1 + (4 \cdot 0.0)/(3 \cdot 2776.0))]$
 $= -417.3 \text{ Kg-m.}$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L/4 ((1 + 2(R_m^2 - h_m^2)/(L^2)) / (1 + (4h_m^2)/(3L)) - 4a/L)$
 $= 6225 \cdot 2776/4 ((1 + 2(411^2 - 0^2)/(2776^2)) / (1 + (4 \cdot 0)/(3 \cdot 2776)) - 4 \cdot 520/2776)$
 $= 1272.2 \text{ Kg-m.}$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:
 $= P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 t)$
 $= 1564.025 \cdot 410.6 / (2 \cdot 14.8) - 1272.2 / (\pi \cdot 410.6^2 \cdot 14.8)$
 $= 20.11 \text{ N./mm}^2$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 104 of 190

Saddle Calcs: Test Case:

Step: 22 10:30am Sep 9, 2026

$$= P \cdot R_m / (2t) + M2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 1564.025 \cdot 410.6 / (2 \cdot 14.8) + 1272.2 / (\pi \cdot 410.6^2 \cdot 14.8)$$

$$= 23.29 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot R_m / (2t) - M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 1564.025 \cdot 410.6 / (2 \cdot 14.8) - 417.3 / (0.1066 \cdot \pi \cdot 410.6^2 \cdot 14.8)$$

$$= 26.59 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m / (2t) + M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 1564.025 \cdot 410.6 / (2 \cdot 14.8) + 417.3 / (0.1923 \cdot \pi \cdot 410.6^2 \cdot 14.8)$$

$$= 18.98 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 6225(2776.0 - 2 \cdot 520.0) / (2776.0 + (4 \cdot 0.0 / 3))$$

$$= 3892.8 \text{ Kg}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 3892.84 / (410.6 \cdot 14.8)$$

$$= 7.35 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(410.6 \cdot 14.8)}$$

$$= 60.804 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.7603 \cdot 6225 \cdot 0.1 / (14.8(127.0 + 60.8 + 60.8))$$

$$= -1.26 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.23) [σ^*7]:

$$= -Q / (4 \cdot t(b + X1 + X2)) - 12 \cdot K7 \cdot Q \cdot R_m / (L \cdot t^2)$$

$$= -6225 / (4 \cdot 14.8(127.0 + 60.804 + 60.804)) -$$

$$12 \cdot 0.053 \cdot 6225 \cdot 410.6 / (2776.0 \cdot 14.8^2)$$

$$= -30.29 \text{ N./mm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(127.0 + 1.56 \cdot \sqrt{(410.6 \cdot 14.8)}, 2 \cdot 520.0)$$

$$= 248.61 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	850.0001	mm
Baseplate Thickness	Bpthk	12.0000	mm
Baseplate Width	Bpwid	152.4000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw, c	300.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(403.2 \cdot 14.8)} + 152.4)^3) / 12$$

$$= 7.373 \text{ cm}^4$$

Resolved Inertia for the Shell [I, shell]:



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 105 of 190

Saddle Calcs: Test Case:

Step: 22 10:30am Sep 9, 2026

$$\begin{aligned}
 &= \text{ShellInertia} + A\text{Shell}(C1 - Y)^2 \\
 &= 7.373 + 40.39(137.56 - 7.4)^2 \\
 &= 6850.169 \text{ cm}^4
 \end{aligned}$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	272.9	14.8	7.4	40.4	29888.9	0.685E+04
Wearplate	152.4	10.0	19.8	15.2	30175.2	0.211E+04
Web	10.0	360.0	204.8	36.0	737279.9	0.552E+04
BasePlate	152.4	12.0	390.8	18.3	714694.9	0.117E+05
Totals	...	...	...	109.9	1512038.9	0.262E+05

Distance to Centroid [C1]:

$$\begin{aligned}
 &= AY / A \\
 &= 1512039 / 109.918 \\
 &= 137.560 \text{ mm}
 \end{aligned}$$

Angle [β]:

$$\begin{aligned}
 &= 180 - \text{Saddle Angle}/2 \\
 &= 180 - 120.0/2 \\
 &= 120.0
 \end{aligned}$$

Saddle Splitting Coefficient [K1]:

$$\begin{aligned}
 &= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta)\cos(\beta)) \\
 &= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0)\cos(120.0)) \\
 &= 0.2035
 \end{aligned}$$

Saddle Splitting Force [Fh]:

$$\begin{aligned}
 &= K1 \cdot Q \\
 &= 0.204 \cdot 6224.958 \\
 &= 1266.9139 \text{ Kgf}
 \end{aligned}$$

$$\begin{aligned}
 \text{Tension Stress, } St &= (Fh / As) = 1.7870 \text{ N./mm}^2 \\
 \text{Allowed Stress, } Sa &= 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2
 \end{aligned}$$

Saddle Splitting Dimension [d]:

$$\begin{aligned}
 &= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians}) \\
 &= 800.0 - 403.2 \cdot \sin(120.0/2) / 1.0472 \\
 &= 466.556 \text{ mm}
 \end{aligned}$$

$$\text{Bending Moment, } M = Fh \cdot d = 591.0983 \text{ Kg-m.}$$

$$\begin{aligned}
 \text{Bending Stress, } Sb &= (M \cdot C1 / I) = 3.0422 \text{ N./mm}^2 \\
 \text{Allowed Stress, } Sa &= 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2
 \end{aligned}$$

Minimum Thickness of Baseplate per Moss:

$$\begin{aligned}
 &= \sqrt{(3(Q + \text{SaddleWt})\text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})} \\
 &= \sqrt{(3(6225 + 76)152.4 / (4 \cdot 850.0 \cdot 173.33))} \\
 &= 6.924 \text{ mm}
 \end{aligned}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$\begin{aligned}
 &= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk}) \\
 &= 2 \cdot \cos(90 - 120.0/2) (400.0 + 18.0 + 10.0) \\
 &= 741.318 \text{ mm}
 \end{aligned}$$

Distance between Ribs [e]:

$$\begin{aligned}
 &= \text{Web Length} / (N\text{ribs} - 1) \\
 &= 741.3177 / (3 - 1) \\
 &= 370.659 \text{ mm}
 \end{aligned}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 106 of 190

Saddle Calcs: Test Case:

Step: 22 10:30am

Sep 9, 2026

$$= 370.6589 \cdot 152.4/2$$

$$= 282.442 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 6300.798 / (850.0 \cdot 152.4)$$

$$= 4.864 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= A_p \cdot B_p$$

$$= 282.4 \cdot 4.86$$

$$= 1373.792 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 11.7 + 18.533$$

$$= 30.233 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / A_r$$

$$= 1373.8 / 30.2329$$

$$= 4.456 \text{ N/mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	127.0	...	12.7	...	171.

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 127.0 - 10.0$$

$$= 117.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0 / 30.233$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 127.0 / 2$$

$$= 63.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(170.7 / 30.233)}$$

$$= 23.762 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 800.0 - \cos(120.0/2) (400.0 + 18.0 + 10.0) - 12.0$$

$$= 574.000 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 0.19994\text{E}+09 / 260.0)}$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 574.0 / 23.762$$

$$= 24.157$$



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 107 of 190

Saddle Calcs: Test Case: Step: 22 10:30am Sep 9, 2026

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 334.9 / (2 \cdot 850.0) \cdot 370.659 \cdot 574.0 / 2$$

$$= 20.958 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ (24.1567 < 123.2105) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (24.16)^2 / (2 \cdot 123.21^2)) 260 / (5/3 + 3 \cdot (24.16) / (8 \cdot 123.21) - (24.16^3) / (8 \cdot 123.21^3))$$

$$= 146.6 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 4.46 / 146.61 + (20.96 \cdot 63.5 / 1706986) / 173.33$$

$$= 0.075$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	117.0	0.0	11.7	0.0	171.
Web	370.7	10.0	0.0	37.1	0.0	3.09
Totals	...	...	...	48.8	...	174.

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0 / 48.766$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 127.0 / 2$$

$$= 63.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - B_{pthk}$$

$$= 800.0 - \cos(428.0 / 2) (12.0 + 0.0 + 0.0) - 0.0$$

$$= 360.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (173.7 / 48.766) }$$

$$= 18.873 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 360.0 / 18.873$$

$$= 19.075$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 334.907 / (2 \cdot 850.0)$$

$$= 0.197 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.197 \cdot 370.659 \cdot 360.0$$

$$= 26.288 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [$\sigma_{B,base,c}$]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 26.288 / 27354.967$$

$$= 9.424 \text{ N./mm}^2$$



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 108 of 190

Saddle Calcs: Test Case: Step: 22 10:30am Sep 9, 2026

Compressive Allowable, $KL/r < C_c$ (19.0746 < 123.2105) per AISC E2-1 [Sca]:
 $= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$
 $= (1 - (19.07)^2 / (2 \cdot 123.21^2)) 260 /$
 $(5/3 + 3 \cdot (19.07) / (8 \cdot 123.21) - (19.07^3) / (8 \cdot 123.21^3))$
 $= 149 \text{ N./mm}^2$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$= S_c / S_{ca} + (M_{base,c} \cdot C_1 / I) / S_{ba}$
 $= 5.53 / 148.98 + (26.29 \cdot 63.5 / 173.704) / 173.33$
 $= 0.091$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Material Specification		SA-307 A
Bolt Allowable Stress	Stba	48.30 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.0 mm
User Defined Area in a Single Bolt	UserBoltArea	2.4000 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	1.6469 cm ²
Saddle Load Q0 (Weight)	Q0	5203.1 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	171.7 Kg
Maximum Transverse Force	Ft	339.1 Kg
Maximum Longitudinal Force	F1	334.9 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:
 $= F_1 / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 $= 334.9 / (1.647 \cdot 4)$
 $= 5.0 \text{ N./mm}^2$. Must be less than 29.0 N./mm².

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:
 $= F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 $= 339.1 / (1.647 \cdot 4)$
 $= 5.0 \text{ N./mm}^2$. Must be less than 29.0 N./mm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
 $= 0.0$ (Q0 > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
 $= F_1 / (\text{BoltShearAllowable} \cdot \text{Nbolts})$
 $= 334.91 / (28.98 \cdot 4.0)$
 $= 0.2833 \text{ cm}^2$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 $= B \cdot F_t + \text{Sum of X Moments}$
 $= 800.0 \cdot 339.06 + 0.0$
 $= 271.25 \text{ Kg-m}$.

Eccentricity (e):
 $= R_{mom} / Q_0$
 $= 271.25 / 5203.1$
 $= 52.13 \text{ mm} < B_{plen} / 6$ --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
 $= 0$ (No Uplift)

Required Area of a Single Bolt [Bltarear]:



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 109 of 190

Saddle Calcs: Test Case:

Step: 22 10:30am

Sep 9, 2026

= max[Bltarearl, Bltarears, Bltareart]
= max[0.0, 0.2833, 0.0]
= 0.2833 cm²

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:

Wear Pad Width (152.40) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

= $\min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$
= $\min(127.0 + 1.56 \cdot \sqrt{410.6 \cdot 14.8}, 2 \cdot 520.0)$
= 248.6088 mm

Input and Calculated Values:

Vessel Mean Radius	Rm	410.60	mm
Shell Thickness used in this Case	t	18.000	mm
Stiffened Vessel Length per 4.15.6	L	2776.00	mm
Distance from Saddle to Vessel tangent	a1 or a	520.00	mm
Saddle Width	b1 or b	127.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	207.55	mm
Shell Allowable Stress used in Calculation		234.00	N./mm ²
Head Allowable Stress used in Calculation		234.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	800.00	mm
Coefficient of Friction	mu	0.30	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		1882.32	Kgf
Pressure used in Analysis	P	1564.025	KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	21.39	234.00
Long. Stress at Bottom of Midspan	22.01	234.00
Long. Stress at Top of Saddles	24.24	234.00
Long. Stress at Bottom of Saddles	20.29	234.00
Tangential Shear in Shell	2.02	138.58
Circ. Stress at Horn of Saddle	9.16	350.99
Circ. Compressive Stress in Shell	0.38	234.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

= $F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
= $3.0 (678.1/2 + 0) \cdot 800.0 / 741.3178$
= 1097.7 Kgf

Longitudinal Saddle Reaction Force [Fwl]:

= $\max(F_{l, \text{wind}}, \text{Sum of X Forces}) \cdot B / L_s$
= $\max(334.91, 0) \cdot 800.0 / 1560.0$
= 171.7 Kgf



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 110 of 190

Saddle Calcs: Test Case:

Step: 22 10:30am Sep 9, 2026

Load Combination Results for Q + Wind or Seismic [Q]:

= Saddle Load + max(Fwl, Fwt, Fsl, Fst)

= 785 + max(172, 1098, 0, 0)

= 1882.3 Kgf

Longitudinal Wind Force [F_L]:

= WindScalar • WindPress(Platform Area + ($\pi/4$ (OD • WindDiaMult)²))

= 1.1 • 4919.274(0.0 + $\pi/4$ (1.036 • 1.2)²)

= 6568.540 N

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 1958.16 Kgf

Transverse Shear Load 339.06 Kgf

Longitudinal Shear Load 334.91 Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

= min(2(ShellID/2 + t + WearPadThickness)sin($\theta/2$), 2•Rm)

= min(2(800.0/2 + 18.0 + 10.0)sin(120.0/2), 2•410.6)

= 741.318 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066 K2 = 1.1707 K3 = 0.8799 K4 = 0.4011

K5 = 0.7603 K6 = 0.0529 K7 = 0.0529 K8 = 0.3405

K9 = 0.2711 K10 = 0.0581 K1* = 0.1923

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

= -Q•a [1 - (1 - a/L + (Rm²-hm²)/(2a•L))/(1+(4hm)/3L)]

= -1882•520.0[1 - (1 - 520.0/2776.0 + (410.6²-207.55²)/(2•520.0•2776.0))/(1+(4•207.55)/(3•2776.0))]

= -216.8 Kg-m.

Moment per Equation 4.15.2 [M2]:

= Q•L/4((1+2(Rm²-hm²)/(L²))/(1+(4hm²)/(3L)) - 4a/L)

= 1882•2776/4((1+2(411²-208²)/(2776²))/(1+(4•208)/(3•2776)) - 4•520/2776)

= 247.8 Kg-m.

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

= P • Rm/(2t) - M2/(π •Rm²t)

= 1564.025 • 410.6/(2•14.8) - 247.8/(π •410.6²•14.8)

= 21.39 N./mm²

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

= P • Rm/(2t) + M2/(π • Rm² • t)

= 1564.025 • 410.6/(2 • 14.8) + 247.8/(π • 410.6² • 14.8)

= 22.01 N./mm²

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_3]:

= P • Rm/(2t) - M1/(K1• π •Rm²t)

= 1564.025•410.6/(2•14.8)--216.8/(0.1066• π •410.6²•14.8)

= 24.24 N./mm²

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_4]:

= P • Rm/(2t) + M1/(K1* • π • Rm² • t)

= 1564.025•410.6/(2•14.8)+216.8/(0.1923• π •410.6²•14.8)

= 20.29 N./mm²

Maximum Shear Force in the Saddle (4.15.3) [T]:

= Q(L-2a)/(L+(4•hm/3))

= 1882(2776.0 - 2 • 520.0)/(2776.0 + (4 • 207.55/3))



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 111 of 190

Saddle Calcs: Test Case:

Step: 22 10:30am Sep 9, 2026

= 1070.4 Kgf

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 1070.42 / (410.6 \cdot 14.8)$$

$$= 2.02 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(410.6 \cdot 14.8)}$$

$$= 60.804 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K_5 \cdot Q \cdot k / (t(b + X_1 + X_2))$$

$$= -0.7603 \cdot 1882 \cdot 0.1 / (14.8(127.0 + 60.8 + 60.8))$$

$$= -0.38 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.23) [σ_7^*]:

$$= -Q / (4 \cdot t(b + X_1 + X_2)) - 12 \cdot K_7 \cdot Q \cdot R_m / (L \cdot t^2)$$

$$= -1882 / (4 \cdot 14.8(127.0 + 60.804 + 60.804)) -$$

$$12 \cdot 0.053 \cdot 1882 \cdot 410.6 / (2776.0 \cdot 14.8^2)$$

$$= -9.16 \text{ N./mm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(127.0 + 1.56 \cdot \sqrt{(410.6 \cdot 14.8)}, 2 \cdot 520.0)$$

$$= 248.61 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	850.0001	mm
Baseplate Thickness	Bpthk	12.0000	mm
Baseplate Width	Bpwid	152.4000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	300.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(403.2 \cdot 14.8)} + 152.4)^3) / 12$$

$$= 7.373 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 7.373 + 40.39(137.56 - 7.4)^2$$

$$= 6850.169 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	272.9	14.8	7.4	40.4	29888.9	0.685E+04
Wearplate	152.4	10.0	19.8	15.2	30175.2	0.211E+04
Web	10.0	360.0	204.8	36.0	737279.9	0.552E+04
BasePlate	152.4	12.0	390.8	18.3	714694.9	0.117E+05
Totals	...	...	...	109.9	1512038.9	0.262E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 1512039 / 109.918$$



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 112 of 190

Saddle Calcs: Test Case: Step: 22 10:30am Sep 9, 2026

$$= 137.560 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120.0/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 1882.32$$

$$= 383.0929 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 0.5403 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 800.0 - 403.2 \cdot \sin(120.0/2) / 1.0472$$

$$= 466.556 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 178.7379 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 0.9199 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(1882 + 76) \cdot 152.4) / (4 \cdot 850.0 \cdot 173.33)}$$

$$= 3.860 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (400.0 + 18.0 + 10.0)$$

$$= 741.318 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (N_{\text{ribs}} - 1)$$

$$= 741.3177 / (3 - 1)$$

$$= 370.659 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 370.6589 \cdot 152.4 / 2$$

$$= 282.442 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 1958.16 / (850.0 \cdot 152.4)$$

$$= 1.512 \text{ Kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 282.4 \cdot 1.51$$

$$= 426.947 \text{ Kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 11.7 + 18.533$$



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 113 of 190

Saddle Calcs: Test Case: Step: 22 10:30am Sep 9, 2026

$$= 30.233 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P/A_r$$

$$= 426.9/30.2329$$

$$= 1.385 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	127.0	...	12.7	...	171.

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 127.0 - 10.0$$

$$= 117.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/30.233$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 127.0/2$$

$$= 63.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(170.7/30.233)}$$

$$= 23.762 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 800.0 - \cos(120.0/2) (400.0 + 18.0 + 10.0) - 12.0$$

$$= 574.000 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 0.19994\text{E}+09/260.0)}$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 574.0/23.762$$

$$= 24.157$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 334.9 / (2 \cdot 850.0) \cdot 370.659 \cdot 574.0/2$$

$$= 20.958 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ ($24.1567 < 123.2105$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (24.16)^2 / (2 \cdot 123.21^2)) 260 /$$

$$(5/3 + 3 \cdot (24.16) / (8 \cdot 123.21) - (24.16^3) / (8 \cdot 123.21^3))$$

$$= 146.6 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 1.38/146.61 + (20.96 \cdot 63.5/1706986) / 173.33$$

$$= 0.054$$



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 114 of 190

Saddle Calcs: Test Case:

Step: 22 10:30am Sep 9, 2026

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	117.0	0.0	11.7	0.0	171.
Web	370.7	10.0	0.0	37.1	0.0	3.09
Totals	...	...	...	48.8	...	174.

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/48.766$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 127.0/2$$

$$= 63.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 800.0 - \cos(428.0/2)(12.0 + 0.0 + 0.0) - 0.0$$

$$= 360.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(173.7/48.766)}$$

$$= 18.873 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 360.0/18.873$$

$$= 19.075$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 334.907/(2 \cdot 850.0)$$

$$= 0.197 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.197 \cdot 370.659 \cdot 360.0$$

$$= 26.288 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 26.288/27354.967$$

$$= 9.424 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($19.0746 < 123.2105$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (19.07)^2 / (2 \cdot 123.21^2)) 260 /$$

$$(5/3 + 3 \cdot (19.07) / (8 \cdot 123.21) - (19.07^3) / (8 \cdot 123.21^3))$$

$$= 149 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C1/I) / S_{ba}$$

$$= 1.72/148.98 + (26.29 \cdot 63.5/173.704) / 173.33$$

$$= 0.066$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 115 of 190

Saddle Calcs: Test Case:

Step: 22 10:30am

Sep 9, 2026

Bolt Material Specification		SA-307 A
Bolt Allowable Stress	Stba	48.30 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.0 mm
User Defined Area in a Single Bolt	UserBoltArea	2.4000 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	1.6469 cm ²
Saddle Load QO (Weight)	QO	860.5 Kgf
Saddle Load QL (Wind/Seismic contribution)	QL	171.7 Kgf
Maximum Transverse Force	Ft	339.1 Kgf
Maximum Longitudinal Force	Fl	334.9 Kgf
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:

$$= Fl / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 334.9 / (1.647 \cdot 4)$$

$$= 5.0 \text{ N./mm}^2. \text{ Must be less than } 29.0 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 339.1 / (1.647 \cdot 4)$$

$$= 5.0 \text{ N./mm}^2. \text{ Must be less than } 29.0 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} QO > QL \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= Fl / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 334.91 / (28.98 \cdot 4.0)$$

$$= 0.2833 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 800.0 \cdot 339.06 + 0.0$$

$$= 271.25 \text{ Kg-m.}$$

Eccentricity (e):

$$= Rmom / QO$$

$$= 271.25 / 860.46$$

$$= 315.24 \text{ mm} > Bplen / 6 \text{ --> Uplift in Transverse direction}$$

f = Bplen / 2 - Edgedis

$$= 850.0 / 2 - 50.0$$

$$= 375.00 \text{ mm}$$

Modular Ratio Of Steel/Concrete (n1):

$$= ES / EC$$

$$= 203402.5 / 21526.32$$

$$= 9.45$$

K1 = 3 (e - 0.5 • Bplen)

$$= 3 (315.24 - 0.5 \cdot 850.0)$$

$$= -329.29 \text{ mm}$$

K2 = 6 • n1 • At / Bpwid • (f + e)

$$= 6 \cdot 9.45 \cdot 3.29 / 152.4 \cdot (375.0 + 315.24)$$

$$= 84574.68 \text{ mm}^2$$

K3 = -K2 • (0.5 • Bplen + f)

$$= -84574.68 \cdot (0.5 \cdot 850.0 + 375.0)$$

$$= -67659751.90 \text{ mm}^3$$



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 116 of 190

Saddle Calcs: Test Case: Step: 22 10:30am Sep 9, 2026

Iteratively Solving for the Effective Bearing Length:

$$Y^3 + K1 * Y^2 + K2 * Y + K3 = 0$$

$$Y^3 + -329.29 * Y^2 + 84574.68 * Y + -0.7E+08 = 0$$

$$Y = 462.62 \text{ mm}$$

$$\text{Num} = (\text{Bplen} / 2 - Y / 3 - e)$$

$$= (850.0 / 2 - 462.62 / 3 - 315.24)$$

$$= -44.44$$

$$\text{Denom} = (\text{Bplen} / 2 - Y / 3 + f)$$

$$= (850.0 / 2 - 462.62 / 3 + 375.0)$$

$$= 645.79$$

Total Bolt Tension Force [Tforce]:

$$= -QO * \text{Num} / \text{Denom}$$

$$= -860.46 * -44.44 / 645.79$$

$$= 59.22 \text{ Kgf}$$

Bolt Area Required due to Transverse Load [Bltareart]:

$$= \text{Tforce} / (\text{Stba} * \text{Nbt})$$

$$= 59.22 / (48.3 * 2.0)$$

$$= 0.0601 \text{ cm}^2$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[\text{Bltarearl}, \text{Bltarears}, \text{Bltareart}]$$

$$= \max[0.0, 0.2833, 0.0601]$$

$$= 0.2833 \text{ cm}^2$$

Baseplate Thickness Calculation per D. Moss:

Bearing Pressure (fc)

$$= 2(QO + \text{Tforce}) / (Y * \text{Bpwid})$$

$$= 2(860.46 + 59.22) / (462.62 * 152.4)$$

$$= 255.84 \text{ KPa.}$$

Distance from Baseplate Edge to the Web [ADIST]:

$$= (\text{Bplen} - \text{WebIngh}) / 2$$

$$= (850.0 - 799.2) / 2$$

$$= 25.4000 \text{ mm}$$

Overturning Moment due To Bolt Tension [Mt]:

$$= \text{Tforce} * \text{Adist}$$

$$= 59.22 * 25.4$$

$$= 1.50 \text{ Kg-m.}$$

Equivalent Bearing Pressure (f1):

$$= fc * (Y - \text{Adist}) / Y$$

$$= 255.84 * (462.62 - 25.4) / 462.62$$

$$= 241.79 \text{ KPa.}$$

Overturning Moment due to Bearing Pressure [Mc]:

$$= (\text{Adist}^2 * \text{Bpwid} / 6) * (f1 + 2 * fc)$$

$$= (25.4^2 * 152.4 / 6) * (241.79 + 2 * 255.84)$$

$$= 1.26 \text{ Kg-m.}$$

Baseplate Required Thickness [Treq]:

$$= \sqrt{(6 * \max(\text{Mt}, \text{Mc}) / (\text{Bpwid} * \text{Sba}))}$$

$$= \sqrt{(6 * \max(1.5, 1.26) / (152.4 * 207.0))}$$

$$= 1.6750 \text{ mm}$$

Review notes about nozzle loadings and supports in the Warnings report.



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FileName : E-0102

Page 117 of 190

Saddle Calcs: Test Case:

Step: 22

10:30am

Sep 9, 2026

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PV Elite 28 Licensee: SL User

FileName : E-0102

Page 118 of 190

Saddle Calcs: Ext. Pressure:

Step: 23 10:30am

Sep 9, 2026

ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : External Pressure

Note:

Wear Pad Width (152.40) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
and less than $2a$. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

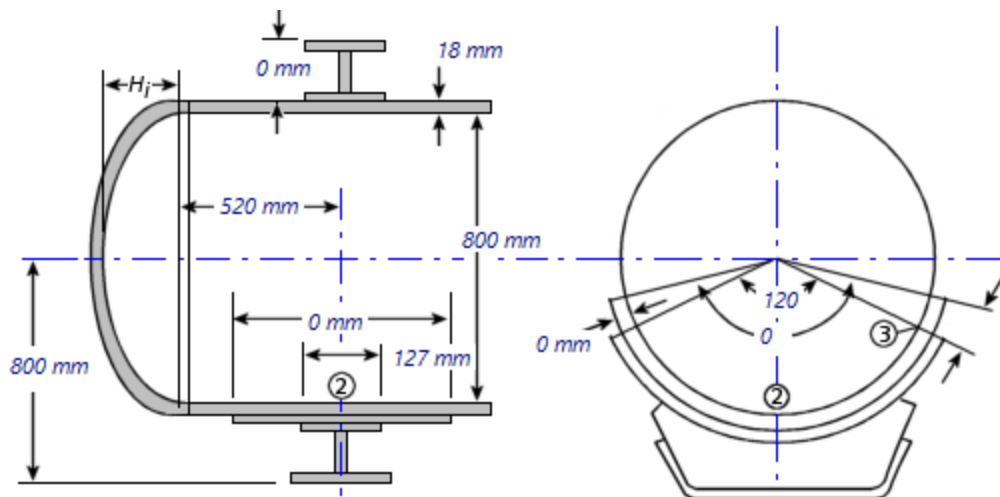
$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(127.0 + 1.56 \cdot \sqrt{410.6 \cdot 14.8}, 2 \cdot 520.0)$$

$$= 248.6088 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	410.60	mm
Shell Thickness used in this Case	t	18.000	mm
Stiffened Vessel Length per 4.15.6	L	2776.00	mm
Distance from Saddle to Vessel tangent	a1 or a	520.00	mm
Saddle Width	b1 or b	127.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	800.00	mm
Coefficient of Friction	mu	0.30	



Results for Horizontal Vessel Analysis:

Saddle Force Q, External Pressure Case		7791.53	Kgf
Pressure used in Analysis	P	-101.353	KPa.

Horizontal Vessel Analysis Results:		Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top	of Midspan	-3.40	-218.34
Long. Stress at Top	of Midspan	3.40	138.00
Long. Stress at Bottom	of Midspan	0.59	138.00
Long. Stress at Top	of Saddles	4.72	138.00



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 119 of 190

Saddle Calcs: Ext. Pressure: Step: 23 10:30am Sep 9, 2026

Long. Stress at Bottom of Saddles	-4.80	-218.34
Long. Stress at Bottom of Saddles	4.80	138.00
Tangential Shear in Shell	9.21	110.40
Circ. Stress at Horn of Saddle	37.92	172.50
Circ. Compressive Stress in Shell	1.58	138.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3.0(1356.2/2 + 0) \cdot 800.0/741.3178$$

$$= 2195.4 \text{ Kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / Ls$$

$$= \max(669.81, 0) \cdot 800.0/1560.0$$

$$= 343.5 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot B / Ls$$

$$= \max(379.81, 0) \cdot 800.0/1560.0$$

$$= 194.8 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3.0(380/2 + 0) \cdot 800.0/741.3178$$

$$= 614.8 \text{ Kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 5596 + \max(343, 2195, 195, 615)$$

$$= 7791.5 \text{ Kgf}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 4919.274(0.0 + \pi/4(1.036 \cdot 1.2)^2)$$

$$= 6568.540 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	7867.37	Kgf
Transverse Shear Load	678.12	Kgf
Longitudinal Shear Load	669.81	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot Rm)$$

$$= \min(2(800.0/2 + 18.0 + 10.0)\sin(120.0/2), 2 \cdot 410.6)$$

$$= 741.318 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm)/3L)]$$

$$= -7792 \cdot 520.0 [1 - (1 - 520.0/2776.0 + (410.6^2 - 0.0^2)/$$

$$(2 \cdot 520.0 \cdot 2776.0)/(1 + (4 \cdot 0.0)/(3 \cdot 2776.0))]$$

$$= -522.4 \text{ Kg-m.}$$



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 120 of 190

Saddle Calcs: Ext. Pressure: Step: 23 10:30am Sep 9, 2026

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 \cdot ((1 + 2(Rm^2 - hm^2) / (L^2)) / (1 + (4hm^2) / (3L)) - 4a/L)$$

$$= 7792 \cdot 2776 / 4 \cdot ((1 + 2(411^2 - 0^2) / (2776^2)) / (1 + (4 \cdot 0) / (3 \cdot 2776)) - 4 \cdot 520 / 2776)$$

$$= 1592.4 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 \cdot t)$$

$$= -101.353 \cdot 410.6 / (2 \cdot 14.8) - 1592.4 / (\pi \cdot 410.6^2 \cdot 14.8)$$

$$= -3.40 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t)$$

$$= -101.353 \cdot 410.6 / (2 \cdot 14.8) + 1592.4 / (\pi \cdot 410.6^2 \cdot 14.8)$$

$$= 0.59 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t)$$

$$= -101.353 \cdot 410.6 / (2 \cdot 14.8) - 522.4 / (0.1066 \cdot \pi \cdot 410.6^2 \cdot 14.8)$$

$$= 4.72 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot Rm / (2t) + M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t)$$

$$= -101.353 \cdot 410.6 / (2 \cdot 14.8) + 522.4 / (0.1923 \cdot \pi \cdot 410.6^2 \cdot 14.8)$$

$$= -4.80 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot hm / 3))$$

$$= 7792(2776.0 - 2 \cdot 520.0) / (2776.0 + (4 \cdot 0.0 / 3))$$

$$= 4872.5 \text{ Kg}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (Rm \cdot t)$$

$$= 1.1707 \cdot 4872.51 / (410.6 \cdot 14.8)$$

$$= 9.21 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)}$$

$$= 0.78 \cdot \sqrt{(410.6 \cdot 14.8)}$$

$$= 60.804 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.7603 \cdot 7792 \cdot 0.1 / (14.8(127.0 + 60.8 + 60.8))$$

$$= -1.58 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8Rm$ (4.15.23) [σ_{*7}]:

$$= -Q / (4 \cdot t(b + X1 + X2)) - 12 \cdot K7 \cdot Q \cdot Rm / (L \cdot t^2)$$

$$= -7792 / (4 \cdot 14.8(127.0 + 60.804 + 60.804)) - 12 \cdot 0.053 \cdot 7792 \cdot 410.6 / (2776.0 \cdot 14.8^2)$$

$$= -37.92 \text{ N./mm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a)$$

$$= \min(127.0 + 1.56 \cdot \sqrt{(410.6 \cdot 14.8)}, 2 \cdot 520.0)$$

$$= 248.61 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	850.0001	mm
Baseplate Thickness	Bpthk	12.0000	mm
Baseplate Width	Bpwid	152.4000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	



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FileName : E-0102

Page 121 of 190

Saddle Calcs: Ext. Pressure: Step: 23 10:30am Sep 9, 2026

Saddle Design Temperature		21.1 °C
Saddle Yield Stress	Sy	262.0 N./mm²
Height of Web at Center	Hw, c	300.0 mm
Friction Coefficient	mu	0.300

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t) + \text{WearPlateWidth}})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(403.2 \cdot 14.8) + 152.4})^3) / 12$$

$$= 7.373 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 7.373 + 40.39 (137.56 - 7.4)^2$$

$$= 6850.169 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm²	AY mm³	I + AD²(2) cm⁴
Shell	272.9	14.8	7.4	40.4	29888.9	0.685E+04
Wearplate	152.4	10.0	19.8	15.2	30175.2	0.211E+04
Web	10.0	360.0	204.8	36.0	737279.9	0.552E+04
BasePlate	152.4	12.0	390.8	18.3	714694.9	0.117E+05
Totals	...	...	...	109.9	1512038.9	0.262E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 1512039 / 109.918$$

$$= 137.560 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120.0 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 7791.53$$

$$= 1585.7454 \text{ Kgf}$$

Tension Stress, St = (Fh/As)	=	2.2367	N./mm²
Allowed Stress, Sa = 0.6 * Yield Str	=	157.1972	N./mm²

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$$

$$= 800.0 - 403.2 \cdot \sin(120.0/2) / 1.0472$$

$$= 466.556 \text{ mm}$$

Bending Moment, M = Fh * d	=	739.8541	Kg-m.
----------------------------	---	----------	-------

Bending Stress, Sb = (M * C1 / I)	=	3.8078	N./mm²
Allowed Stress, Sa = 2/3 * Yield Str	=	174.6635	N./mm²

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(7792 + 76)152.4) / (4 \cdot 850.0 \cdot 174.663)}$$

$$= 7.707 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:



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FileName : E-0102

Page 122 of 190

Saddle Calcs: Ext. Pressure: Step: 23 10:30am Sep 9,2026

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (400.0 + 18.0 + 10.0)$$

$$= 741.318 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 741.3177 / (3 - 1)$$

$$= 370.659 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 370.6589 \cdot 152.4 / 2$$

$$= 282.442 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 7867.371 / (850.0 \cdot 152.4)$$

$$= 6.073 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= \text{Ap} \cdot \text{Bp}$$

$$= 282.4 \cdot 6.07$$

$$= 1715.359 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 11.7 + 18.533$$

$$= 30.233 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / \text{Ar}$$

$$= 1715.4 / 30.2329$$

$$= 5.564 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	127.0	...	12.7	...	171.

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 127.0 - 10.0$$

$$= 117.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= \text{AY} / \text{A}$$

$$= 0.0 / 30.233$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 127.0 / 2$$

$$= 63.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(170.7 / 30.233)}$$

$$= 23.762 \text{ mm}$$

Length of Outer Rib [L]:



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 123 of 190

Saddle Calcs: Ext. Pressure: Step: 23 10:30am Sep 9, 2026

$$\begin{aligned}
 &= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk} \\
 &= 800.0 - \cos(120.0/2) (400.0 + 18.0 + 10.0) - 12.0 \\
 &= 574.000 \text{ mm}
 \end{aligned}$$

Intermediate Term [Cc]:

$$\begin{aligned}
 &= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})} \\
 &= \sqrt{(2 \cdot \pi^2 \cdot 0.19994\text{E}+09 / 262.0)} \\
 &= 122.739
 \end{aligned}$$

Slenderness ratio [KL/r]:

$$\begin{aligned}
 &= KL/r \\
 &= 1 \cdot 574.0 / 23.762 \\
 &= 24.157
 \end{aligned}$$

Bending Moment [Rm]:

$$\begin{aligned}
 &= FI / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2 \\
 &= 669.8 / (2 \cdot 850.0) \cdot 370.659 \cdot 574.0 / 2 \\
 &= 41.915 \text{ Kg-m.}
 \end{aligned}$$

Compressive Allowable, $KL/r < Cc$ ($24.1567 < 122.7394$) per AISC E2-1 [Sca]:

$$\begin{aligned}
 &= (1 - (KL/r)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3)) \\
 &= (1 - (24.16)^2 / (2 \cdot 122.74^2)) 262 / \\
 &\quad (5/3 + 3 \cdot (24.16) / (8 \cdot 122.74) - (24.16)^3 / (8 \cdot 122.74^3)) \\
 &= 147.7 \text{ N./mm}^2
 \end{aligned}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$\begin{aligned}
 &= Sc/Sca + (Rm \cdot C1 / I) / Sba \\
 &= 5.56 / 147.7 + (41.92 \cdot 63.5 / 1706986) / 174.66 \\
 &= 0.125
 \end{aligned}$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	117.0	0.0	11.7	0.0	171.
Web	370.7	10.0	0.0	37.1	0.0	3.09
Totals	...	...	...	48.8	...	174.

Distance to Centroid from Datum [ytot]:

$$\begin{aligned}
 &= AY / A \\
 &= 0.0 / 48.766 \\
 &= 0.000 \text{ mm}
 \end{aligned}$$

Distance to Centroid [C1]:

$$\begin{aligned}
 &= \text{Saddle Width} / 2 \\
 &= 127.0 / 2 \\
 &= 63.500 \text{ mm}
 \end{aligned}$$

Length of Inner Rib [L]:

$$\begin{aligned}
 &= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk} \\
 &= 800.0 - \cos(428.0/2) (12.0 + 0.0 + 0.0) - 0.0 \\
 &= 360.000 \text{ mm}
 \end{aligned}$$

Radius of Gyration [r]:

$$\begin{aligned}
 &= \sqrt{(\text{Total Inertia} / \text{Total Area})} \\
 &= \sqrt{(173.7 / 48.766)} \\
 &= 18.873 \text{ mm}
 \end{aligned}$$

Slenderness ratio [KL/r]:

$$\begin{aligned}
 &= KL/r \\
 &= 1 \cdot 360.0 / 18.873 \\
 &= 19.075
 \end{aligned}$$

Unit Force [Force,u]:



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 124 of 190

Saddle Calcs: Ext. Pressure: Step: 23 10:30am Sep 9, 2026

$$= F_l / (2 \cdot \text{Baseplate Length})$$

$$= 669.815 / (2 \cdot 850.0)$$

$$= 0.394 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.394 \cdot 370.659 \cdot 360.0$$

$$= 52.576 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 52.576 / 27354.967$$

$$= 18.848 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($19.0746 < 122.7394$) per AISC E2-1 [Sca]:

$$= (1 - (Kl/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (Kl/r) / (8 \cdot C_c) - (Kl/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (19.07)^2 / (2 \cdot 122.74^2)) 262 /$$

$$(5/3 + 3 \cdot (19.07) / (8 \cdot 122.74) - (19.07^3) / (8 \cdot 122.74^3))$$

$$= 150.1 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_1 / I) / S_{ba}$$

$$= 6.9 / 150.09 + (52.58 \cdot 63.5 / 173.704) / 174.66$$

$$= 0.154$$

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:

Wear Pad Width (152.40) is less than $1.56 \cdot \sqrt{r_m \cdot t}$ and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(127.0 + 1.56 \cdot \sqrt{410.6 \cdot 14.8}, 2 \cdot 520.0)$$

$$= 248.6088 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	410.60	mm
Shell Thickness used in this Case	t	18.000	mm
Stiffened Vessel Length per 4.15.6	L	2776.00	mm
Distance from Saddle to Vessel tangent	a1 or a	520.00	mm
Saddle Width	b1 or b	127.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	207.55	mm
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	800.00	mm
Coefficient of Friction	mu	0.30	

Results for Horizontal Vessel Analysis:

Saddle Force Q, External Pressure Case		3261.18	Kgf
Pressure used in Analysis	P	-101.353	KPa.

Horizontal Vessel Analysis Results:		Actual	Allowable
		N./mm ²	N./mm ²
Long. Stress at Top	of Midspan	-1.94	-218.34



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 125 of 190

Saddle Calcs: Ext. Pressure: Step: 23 10:30am Sep 9, 2026

Long. Stress at Top of Midspan	1.94	138.00
Long. Stress at Bottom of Midspan	-0.87	-218.34
Long. Stress at Bottom of Midspan	0.87	138.00
Long. Stress at Top of Saddles	3.00	138.00
Long. Stress at Bottom of Saddles	-3.85	-218.34
Long. Stress at Bottom of Saddles	3.85	138.00
Tangential Shear in Shell	3.50	110.40
Circ. Stress at Horn of Saddle	15.87	172.50
Circ. Compressive Stress in Shell	0.66	138.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= Ftr(Ft/Num of Saddles + Z Force Load) \cdot B / E$$

$$= 3.0(1356.2/2 + 0) \cdot 800.0/741.3178$$

$$= 2195.4 \text{ Kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(Fl,wind; \text{Sum of X Forces}) \cdot B / Ls$$

$$= \max(669.81, 0) \cdot 800.0/1560.0$$

$$= 343.5 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(Fl,eq; \text{Sum of X Forces}) \cdot B / Ls$$

$$= \max(379.81, 0) \cdot 800.0/1560.0$$

$$= 194.8 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= Ftr(Ft/Num of Saddles + Z Force Load) \cdot B / E$$

$$= 3.0(380/2 + 0) \cdot 800.0/741.3178$$

$$= 614.8 \text{ Kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(Fwl, Fwt, Fsl, Fst)$$

$$= 1066 + \max(343, 2195, 195, 615)$$

$$= 3261.2 \text{ Kgf}$$

Longitudinal Wind Force [Fl]:

$$= WindScalar \cdot WindPress(Platform Area + (\pi/4(OD \cdot WindDiaMult)^2))$$

$$= 1.1 \cdot 4919.274(0.0 + \pi/4(1.036 \cdot 1.2)^2)$$

$$= 6568.540 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	3337.02	Kgf
Transverse Shear Load	678.12	Kgf
Longitudinal Shear Load	669.81	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \min(2(ShellID/2 + t + WearPadThickness) \sin(\theta/2), 2 \cdot Rm)$$

$$= \min(2(800.0/2 + 18.0 + 10.0) \sin(120.0/2), 2 \cdot 410.6)$$

$$= 741.318 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2) / (2a \cdot L)) / (1 + (4hm) / (3L))]$$



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 126 of 190

Saddle Calcs: Ext. Pressure: Step: 23 10:30am Sep 9, 2026

$$= -3261 \cdot 520.0 [1 - (1 - 520.0 / 2776.0 + (410.6^2 - 207.55^2) / (2 \cdot 520.0 \cdot 2776.0)) / (1 + (4 \cdot 207.55) / (3 \cdot 2776.0))] \\ = -375.6 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 \cdot ((1 + 2(Rm^2 - hm^2) / (L^2)) / (1 + (4hm^2) / (3L)) - 4a / L) \\ = 3261 \cdot 2776 / 4 \cdot ((1 + 2(411^2 - 208^2) / (2776^2)) / (1 + (4 \cdot 208) / (3 \cdot 2776)) - 4 \cdot 520 / 2776) \\ = 429.3 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 \cdot t) \\ = -101.353 \cdot 410.6 / (2 \cdot 14.8) - 429.3 / (\pi \cdot 410.6^2 \cdot 14.8) \\ = -1.94 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t) \\ = -101.353 \cdot 410.6 / (2 \cdot 14.8) + 429.3 / (\pi \cdot 410.6^2 \cdot 14.8) \\ = -0.87 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t) \\ = -101.353 \cdot 410.6 / (2 \cdot 14.8) - 375.6 / (0.1066 \cdot \pi \cdot 410.6^2 \cdot 14.8) \\ = 3.00 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot Rm / (2t) + M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t) \\ = -101.353 \cdot 410.6 / (2 \cdot 14.8) + 375.6 / (0.1923 \cdot \pi \cdot 410.6^2 \cdot 14.8) \\ = -3.85 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot hm / 3)) \\ = 3261(2776.0 - 2 \cdot 520.0) / (2776.0 + (4 \cdot 207.55 / 3)) \\ = 1854.5 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (Rm \cdot t) \\ = 1.1707 \cdot 1854.54 / (410.6 \cdot 14.8) \\ = 3.50 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)} \\ = 0.78 \cdot \sqrt{(410.6 \cdot 14.8)} \\ = 60.804 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2)) \\ = -0.7603 \cdot 3261 \cdot 0.1 / (14.8(127.0 + 60.8 + 60.8)) \\ = -0.66 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8Rm$ (4.15.23) [σ^*7]:

$$= -Q / (4 \cdot t(b + X1 + X2)) - 12 \cdot K7 \cdot Q \cdot Rm / (L \cdot t^2) \\ = -3261 / (4 \cdot 14.8(127.0 + 60.804 + 60.804)) - 12 \cdot 0.053 \cdot 3261 \cdot 410.6 / (2776.0 \cdot 14.8^2) \\ = -15.87 \text{ N./mm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a) \\ = \min(127.0 + 1.56 \cdot \sqrt{(410.6 \cdot 14.8)}, 2 \cdot 520.0) \\ = 248.61 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	850.0001	mm
Baseplate Thickness	Bpthk	12.0000	mm



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 127 of 190

Saddle Calcs: Ext. Pressure: Step: 23 10:30am Sep 9, 2026

Baseplate Width	Bpwid	152.4000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	262.0	N./mm ²
Height of Web at Center	Hw,c	300.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3)/12$$

$$= ((1.56 \cdot \sqrt{(403.2 \cdot 14.8)} + 152.4)14.8^3)/12$$

$$= 7.373 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 7.373 + 40.39(137.56 - 7.4)^2$$

$$= 6850.169 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	272.9	14.8	7.4	40.4	29888.9	0.685E+04
Wearplate	152.4	10.0	19.8	15.2	30175.2	0.211E+04
Web	10.0	360.0	204.8	36.0	737279.9	0.552E+04
BasePlate	152.4	12.0	390.8	18.3	714694.9	0.117E+05
Totals	...	...	...	109.9	1512038.9	0.262E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 1512039/109.918$$

$$= 137.560 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120.0/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta)\cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0)\cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 3261.181$$

$$= 663.7210 \text{ Kg}$$

$$\text{Tension Stress, } St = (Fh/As) = 0.9362 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 157.1972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 800.0 - 403.2 \cdot \sin(120.0/2) / 1.0472$$

$$= 466.556 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 309.6693 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 1.5938 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 174.6635 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 128 of 190

Saddle Calcs: Ext. Pressure: Step: 23 10:30am Sep 9, 2026

$$= \sqrt{\frac{3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}}{4 \cdot \text{BasePlateLength} \cdot \text{AllStress}}}$$

$$= \sqrt{\frac{3(3261 + 76) \cdot 152.4}{4 \cdot 850.0 \cdot 174.663}}$$

$$= 5.019 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (400.0 + 18.0 + 10.0)$$

$$= 741.318 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 741.3177 / (3 - 1)$$

$$= 370.659 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 370.6589 \cdot 152.4 / 2$$

$$= 282.442 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 3337.021 / (850.0 \cdot 152.4)$$

$$= 2.576 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= \text{Ap} \cdot \text{Bp}$$

$$= 282.4 \cdot 2.58$$

$$= 727.586 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 11.7 + 18.533$$

$$= 30.233 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / \text{Ar}$$

$$= 727.6 / 30.2329$$

$$= 2.360 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	127.0	...	12.7	...	171.

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 127.0 - 10.0$$

$$= 117.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= \text{AY} / \text{A}$$

$$= 0.0 / 30.233$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 127.0 / 2$$

$$= 63.500 \text{ mm}$$

Radius of Gyration [r]:



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 129 of 190

Saddle Calcs: Ext. Pressure: Step: 23 10:30am Sep 9, 2026

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{(170.7/30.233)}$$

$$= 23.762 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 800.0 - \cos(120.0/2) (400.0 + 18.0 + 10.0) - 12.0$$

$$= 574.000 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 0.19994\text{E}+09/262.0)}$$

$$= 122.739$$

Slenderness ratio [KL/r]:

$$= \text{KL}/r$$

$$= 1 \cdot 574.0/23.762$$

$$= 24.157$$

Bending Moment [Rm]:

$$= \text{FI} / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$$

$$= 669.8 / (2 \cdot 850.0) \cdot 370.659 \cdot 574.0/2$$

$$= 41.915 \text{ Kg-m.}$$

Compressive Allowable, $\text{KL}/r < \text{Cc}$ ($24.1567 < 122.7394$) per AISC E2-1 [Sca]:

$$= (1 - (\text{KL}/r)^2 / (2 \cdot \text{Cc}^2)) \text{Fy} / (5/3 + 3 \cdot (\text{KL}/r) / (8 \cdot \text{Cc}) - (\text{KL}/r)^3 / (8 \cdot \text{Cc}^3))$$

$$= (1 - (24.16)^2 / (2 \cdot 122.74^2)) 262 / (5/3 + 3 \cdot (24.16) / (8 \cdot 122.74) - (24.16^3) / (8 \cdot 122.74^3))$$

$$= 147.7 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= \text{Sc}/\text{Sca} + (\text{Rm} \cdot \text{C1} / \text{I}) / \text{Sba}$$

$$= 2.36/147.7 + (41.92 \cdot 63.5/1706986) / 174.66$$

$$= 0.104$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	117.0	0.0	11.7	0.0	171.
Web	370.7	10.0	0.0	37.1	0.0	3.09
Totals	...	...	...	48.8	...	174.

Distance to Centroid from Datum [ytot]:

$$= \text{AY} / \text{A}$$

$$= 0.0/48.766$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 127.0/2$$

$$= 63.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 800.0 - \cos(428.0/2) (12.0 + 0.0 + 0.0) - 0.0$$

$$= 360.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{(173.7/48.766)}$$

$$= 18.873 \text{ mm}$$

Slenderness ratio [KL/r]:



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 130 of 190

Saddle Calcs: Ext. Pressure: Step: 23 10:30am Sep 9, 2026

$$= KL/r$$

$$= 1 \cdot 360.0/18.873$$

$$= 19.075$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 669.815 / (2 \cdot 850.0)$$

$$= 0.394 \text{ Kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.394 \cdot 370.659 \cdot 360.0$$

$$= 52.576 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 52.576/27354.967$$

$$= 18.848 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($19.0746 < 122.7394$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (19.07)^2 / (2 \cdot 122.74^2)) 262 / (5/3 + 3 \cdot (19.07) / (8 \cdot 122.74) - (19.07^3) / (8 \cdot 122.74^3))$$

$$= 150.1 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (M_{base,c} \cdot C1/I) / Sba$$

$$= 2.93/150.09 + (52.58 \cdot 63.5/173.704) / 174.66$$

$$= 0.127$$

Review notes about nozzle loadings and supports in the Warnings report.

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PV Elite 28 Licensee: SL User

FileName : E-0102

Page 131 of 190

Nozzle Calcs.: N1-Tube inlet

Nozl: 9 10:30am Sep 9,2026

Input, Nozzle Desc: N1-Tube inlet

From: 20

Pressure for Reinforcement Calculations	P	1057.842	KPa.
Temperature for Internal Pressure	Temp	230	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	800.00	mm
Shell Finished (Minimum) Thickness	t	14.0000	mm
Shell Internal Corrosion Allowance	c	3.2000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		350.00	mm
User Entered Minimum Design Metal Temperature		-45.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-333 6 +IT	
Material UNS Number		K03006	
Material Specification/Type	Smls. & wld. pipe		
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		80.0000	mm
Size and Thickness Basis		Nominal	
Nominal Thickness		80	
Flange Material		SA-350 LF2 +IT	
Hub Height of Integral Nozzle	h	30.0000	mm
Height of Beveled Transition	L'	9.4570	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		24.0000	mm
Corrosion Allowance	can	3.2000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		12.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



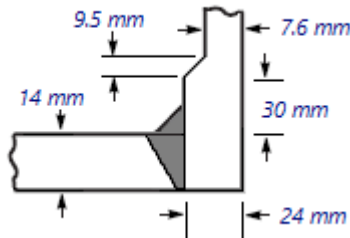
PV Elite 28 Licensee: SL User

FileName : E-0102

Page 132 of 190

Nozzle Calcs.: N1-Tube inlet

Noz1: 9 10:30am Sep 9,2026



Reinforcement CALCULATION, Description: N1-Tube inlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	73.660 mm.
Actual Thickness Used in Calculation	7.620 mm

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1057.84 \cdot 403.2 / (138 \cdot 1.0 - 0.6 \cdot 1057.84)$
 $= 3.1053 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1057.84 \cdot 40.03 / (118 \cdot 1.0 - 0.6 \cdot 1057.84)$
 $= 0.3608 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 16.38 / 0.5773$
 $= 28.3710 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	160.1200 mm
Parallel to Vessel Wall, opening length	d	80.0600 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	27.0000 mm

Taking a UG-36(c)(3)(a) exemption for nozzle: N1-Tube inlet.

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.5608 mm
Wall Thickness per UG16(b),	tr16b	= 4.7000 mm
Wall Thickness, shell/head, internal pressure	trb1	= 6.3053 mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 6.3053 mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.7000 mm
Wall Thickness per table UG-45	tb3	= 8.0000 mm

Determine Nozzle Thickness candidate [tb]:

$= \min[tb3, \max(tb1, tb2)]$
 $= \min[8.0, \max(6.3053, 4.7)]$



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FileName : E-0102

Page 133 of 190

Nozzle Calcs.: N1-Tube inlet

Noz1: 9 10:30am Sep 9, 2026

= 6.3053 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= max(ta, tb)

= max(3.5608, 6.3053)

= 6.3053 mm

Available Nozzle Neck Thickness = $0.875 \times 7.620 = 6.668$ mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	53.1,	Allowable	:	118.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	241.9 N./mm ²	Passed
Occasional	:	4.5,	Allowable	:	156.9 N./mm ²	Passed
Shear	:	38.9,	Allowable	:	82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Calculated Minimum Design Metal Temperature -104 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 14.0, tr = 3.105, c = 3.2 mm, E* = 1.0

Thickness Ratio = $tr \times E^* / (tg - c) = 0.288$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification -46 °C

Flange MDMT with Temp reduction per UCS-66(i)(2) -104 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :

Design Pressure/Ambient Rating = $1057.84 / 5100.00 = 0.207$

Weld Size Calculations, Description: N1-Tube inlet

Intermediate Calc. for nozzle/shell Welds Tmin 10.8000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 4.6524 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 218.6524 mm

Input Echo, WRC297 Item 1, Description: N1-Tube inlet :



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FileName : E-0102

Page 134 of 190

Nozzle Calcs.: N1-Tube inlet

Nozl: 9 10:30am Sep 9, 2026

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.2000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	800.000	mm
Shell Thickness	14.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.2000	mm
Nozzle Diameter	73.660	mm
Nozzle Thickness	24.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-333 6	
Nozzle Cold S.I. Allowable	SNmc 118.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 118.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	1057.84	KPa.
Radial Load	P	-611.87	Kgf
Circumferential Shear	Vc	407.91	Kgf
Longitudinal Shear	Vl	407.91	Kgf
Circumferential Moment	Mc	91.34	Kg-m.
Longitudinal Moment	Ml	91.34	Kg-m.
Torsional Moment	Mt	130.48	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Tn (5.069) is < 10
check the limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$
 $= 121.66 + 2 \cdot 1.65 \cdot \sqrt{408.6(14.0 - 3.2)}$
 $= 340.877 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	817.200	mm
Nozzle Outside Diameter	(d) =	121.660	mm
Vessel Thickness used	(T) =	10.800	mm
Nozzle Thickness used	(t) =	20.800	mm
T / t	=	0.519	
d / t	=	10.000	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.295$$

Nr/P	=	0.091
Mr/P	=	0.189



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 135 of 190

Nozzle Calcs.: N1-Tube inlet

Nozl: 9 10:30am Sep 9, 2026

MØ/P = 0.055
 NØ/P = 0.119
 MrD/Mc = 0.302
 NrDL/Mc = 0.097
 MØd/Mc = 0.094
 NØDL/Mc = 0.051
 MrD/Ml = 0.163
 NrDL/Ml = 0.082
 MØD/Ml = 0.053
 NØDL/Ml = 0.130

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	17	-17	17	-17
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-8	-8	8	8
Outplane Bending (ML)	-20	20	20	-20
Normal Pressure Stress	38	40	38	40
Outplane Stress Summary	33	40	90	17

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	4	4	4	4
Inplane Bending (P)	58	-58	58	-58
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-5	-5	5	5
Inplane Bending (ML)	-61	61	61	-61
Inplane Pressure Stress	19	19	19	19
Inplane Stress Summary	15	22	149	-90

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	5	5	5	5
Shear Stress Summary	7	7	3	3

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	36	43	149	107

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	6	6	6	6



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 136 of 190

Nozzle Calcs.: N1-Tube inlet Nozl: 9 10:30am Sep 9,2026

Outplane Bending (P)	17	-17	17	-17
Outplane Membrane (Mc)	-3	-3	3	3
Outplane Bending (Mc)	-35	35	35	-35
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	19	19	19	19
Outplane Stress Summary	3	40	81	-23

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	4	4	4	4
Inplane Bending (P)	58	-58	58	-58
Inplane Membrane (Mc)	-6	-6	6	6
Inplane Bending (Mc)	-114	114	114	-114
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	38	40	38	40
Inplane Stress Summary	-18	94	222	-121

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	5	5	5	5
Shear Stress Summary	3	3	7	7

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress	23	94	222	122

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Hoop Membrane (P)	6	6	6	6
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-8	-8	8	8
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	1	3	1	3
Hoop Stress Summary	0	1	16	17

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	11	-11	11	-11
Axial Membrane (Mc)	0	0	0	0



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 137 of 190

Nozzle Calcs.: N1-Tube inlet Nozl: 9 10:30am Sep 9,2026

Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-6	-6	6	6
Axial Bending (ML)	-12	12	12	-12
Axial Pressure Stress	0	0	0	0

Axial Stress Summary	-4	-3	32	-16
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	2	2	2	2

Shear Stress Summary	3	3	1	1
----------------------	---	---	---	---

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	8	8	32	33
------------------------	---	---	----	----

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	6	6	6	6
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-3	-3	3	3
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	1	3	1	3

Hoop Stress Summary	4	6	11	12
---------------------	---	---	----	----

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	11	-11	11	-11
Axial Membrane (Mc)	-6	-6	6	6
Axial Bending (Mc)	-25	25	25	-25
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	0	0	0	0

Axial Stress Summary	-18	9	45	-29
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	2	2	2	2

Shear Stress Summary	1	1	3	3
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
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FileName : E-0102

Page 138 of 190

Nozzle Calcs.: N1-Tube inlet

Nozl: 9 10:30am Sep 9,2026

(Stress Intensities Outside Inside Outside Inside

Two * Max Shear Stress 23 10 46 42

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	38	40	38	40	38	40	38	40
Circ. Pl (SUS)	-2	-2	14	14	-2	-2	10	10
Circ. Q (SUS)	-3	3	37	-37	-56	56	172	-172
Long. Pm (SUS)	19	19	19	19	19	19	19	19
Long. Pl (SUS)	-1	-1	9	9	3	3	9	9
Long. Q (SUS)	-3	3	119	-119	-18	18	52	-52
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	5	5	5	5	5	5	5	5
Pm (SUS)	38.0	40.0	38.0	40.0	38.0	40.0	38.0	40.0
Pm+Pl (SUS)	36.1	38.0	52.0	54.0	36.1	38.1	48.0	50.0
Pm+Pl+Q (Total)	34.8	42.7	147.3	108.3	25.3	94.3	220.3	122.4

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	40.00	138.00	Passed
Pm+Pl (SUS)	54.04	207.00	Passed
Pm+Pl+Q (TOTAL)	220.26	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	1	3	1	3	1	3	1	3
Circ. Pl (SUS)	-2	-2	14	14	3	3	9	9
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	0	0	0	0	0	0	0	0
Long. Pl (SUS)	-6	-6	6	6	-6	-6	6	6
Long. Q (SUS)	-1	1	23	-23	-14	14	36	-36
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	1.0	3.0	1.0	3.0	1.0	3.0	1.0	3.0
Pm+Pl (SUS)	6.0	7.0	15.0	17.0	10.0	12.0	10.0	12.0



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FileName : E-0102

Page 139 of 190

Nozzle Calcs.: N1-Tube inlet

Nozl: 9 10:30am

Sep 9, 2026

Pm+Pl+Q (Total)	7.6	7.2	29.3	34.2	24.3	9.2	42.1	42.2
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Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	3.00	118.00	Passed
Pm+Pl (SUS)	17.00	177.00	Passed
Pm+Pl+Q (TOTAL)	42.19	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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FileName : E-0102

Page 140 of 190

Nozzle Calcs.: N1-Tube Outlet

Nozl: 10 10:30am Sep 9,2026

Input, Nozzle Desc: N1-Tube Outlet

From: 20

Pressure for Reinforcement Calculations	P	1050.000	KPa.
Temperature for Internal Pressure	Temp	230	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	800.00	mm
Shell Finished (Minimum) Thickness	t	14.0000	mm
Shell Internal Corrosion Allowance	c	3.2000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		350.00	mm
User Entered Minimum Design Metal Temperature		-45.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-333 6 +IT	
Material UNS Number		K03006	
Material Specification/Type	Smls. & wld. pipe		
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		150.0000	mm
Size and Thickness Basis		Nominal	
Nominal Thickness		80	
Flange Material		SA-350 LF2 +IT	
Hub Height of Integral Nozzle	h	25.0000	mm
Height of Beveled Transition	L'	7.5213	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		24.0000	mm
Corrosion Allowance	can	3.2000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	12.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		12.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



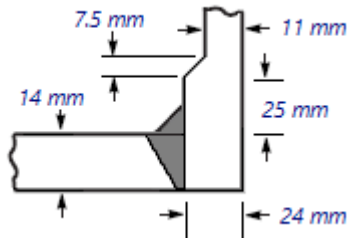
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FileName : E-0102

Page 141 of 190

Nozzle Calcs.: N1-Tube Outlet

Nozl: 10 10:30am Sep 9,2026



Reinforcement CALCULATION, Description: N1-Tube Outlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	146.329 mm.
Actual Thickness Used in Calculation	10.973 mm

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1050.0 \cdot 403.2 / (138 \cdot 1.0 - 0.6 \cdot 1050.0)$
 $= 3.0821 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1050.0 \cdot 76.36 / (118 \cdot 1.0 - 0.6 \cdot 1050.0)$
 $= 0.6832 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 13.027 / 0.5773$
 $= 22.5638 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	305.4588 mm
Parallel to Vessel Wall, opening length	d	152.7294 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	27.0000 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.855, 1.0)$
 $= 0.855$



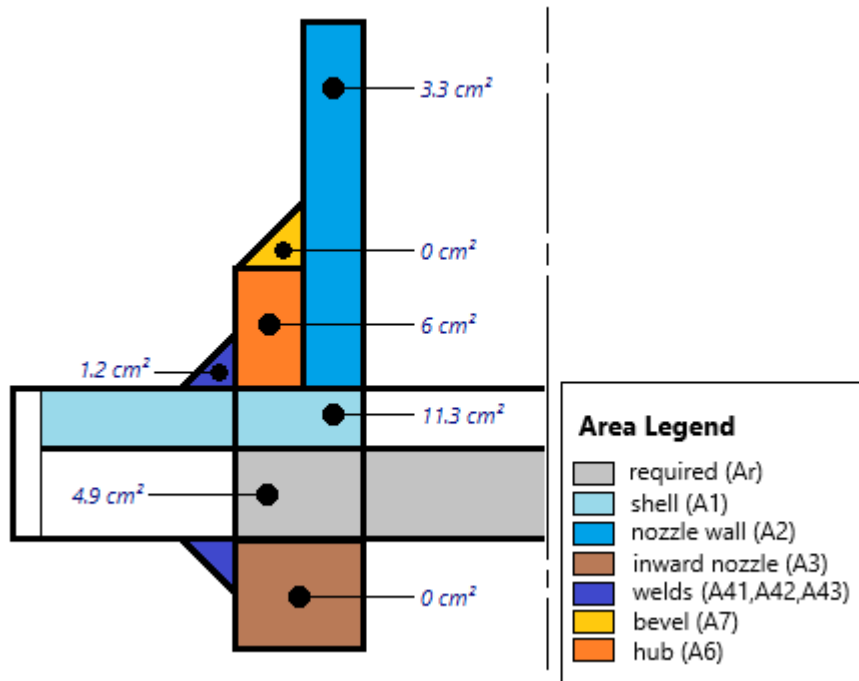
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FileName : E-0102

Page 142 of 190

Nozzle Calcs.: N1-Tube Outlet

Nozl: 10 10:30am Sep 9,2026



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	4.893	NA	NA
Area in Shell	A1	11.322	NA	NA
Area in Nozzle Wall	A2	3.274	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.231	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	5.956	NA	NA
TOTAL AREA AVAILABLE	Atot	21.783	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (152.7294 \cdot 3.0821 \cdot 1.0 + 2 \cdot 20.8 \cdot 3.0821 \cdot 1.0 \cdot (1 - 0.86))$$

$$= 4.893 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 152.729 (1.0 \cdot 10.8 - 1.0 \cdot 3.082) - 2 \cdot 20.8$$

$$(1.0 \cdot 10.8 - 1.0 \cdot 3.0821) \cdot (1 - 0.855)$$

$$= 11.322 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tnp) (tn - trn) fr2$$

$$= (2 \cdot 27.0) (7.77 - 0.68) 0.8551$$

$$= 3.274 \text{ cm}^2$$



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FileName : E-0102

Page 143 of 190

Nozzle Calcs.: N1-Tube Outlet

Noz1: 10 10:30am

Sep 9, 2026

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i - can / 0.707)^2 \cdot fr_2$$

$$= 12.0^2 \cdot 0.8551 + (0.0)^2 \cdot 0.8551$$

$$= 1.231 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(27.0, 250.0, 25.0)) \cdot (24.0 - 11.0) \cdot 0.855$$

$$= 5.956 \text{ cm}^2 \text{ Includes Bevel Area of } 0.386 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.8832 mm
Wall Thickness per UG16(b),	tr16b = 4.7000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 6.2821 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 6.2821 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.7000 mm
Wall Thickness per table UG-45	tb3 = 9.4200 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[9.42, \max(6.2821, 4.7)]$$

$$= 6.2821 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(3.8832, 6.2821)$$

$$= 6.2821 \text{ mm}$$

Available Nozzle Neck Thickness = $0.875 \cdot 10.973 = 9.601 \text{ mm} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 31.7,	Allowable	: 118.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 263.3 N./mm ²	Passed
Occasional	: 4.9,	Allowable	: 156.9 N./mm ²	Passed
Shear	: 21.5,	Allowable	: 82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Calculated Minimum Design Metal Temperature -104 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 14.0, tr = 3.082, c = 3.2 mm, E* = 1.0

Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.285$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-18 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 144 of 190

Nozzle Calcs.: N1-Tube Outlet Nozl: 10 10:30am Sep 9, 2026

MDMT of ASME B16.5/47 flange per Matl. Specification -46 °C
Flange MDMT with Temp reduction per UCS-66(i)(2) -104 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :
Design Pressure/Ambient Rating = 1050.00/5100.00 = 0.206

Weld Size Calculations, Description: N1-Tube Outlet

Intermediate Calc. for nozzle/shell Welds Tmin 10.8000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	8.4840 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (4.8931 - 11.3221 + 2 \cdot 20.8 \cdot 0.8551 \cdot (1.0 \cdot 10.8 - 3.0821)) \cdot 138)$
 $= \max(0, -5183.52) \text{ Kg}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A4-(W_i - \text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (9.2295 + 1.2313 - 0.0 \cdot 0.86) \cdot 138$
 $= 14719.95 \text{ Kg}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (9.2295 + 0.0 + 1.2313 + (3.8417)) \cdot 138$
 $= 20125.78 \text{ Kg}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (9.2295 + 0.0 + 1.2313 + 0.0 + (3.8417)) \cdot 138$
 $= 20125.78 \text{ Kg}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 194.3294 \cdot 12.0 \cdot 0.49 \cdot 118$
 $= 21596. \text{ Kg}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 86.7647) \cdot (24.0 - 3.2) \cdot 0.7 \cdot 118$
 $= 47753. \text{ Kg}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$
 $= (3.1416/2.0) \cdot 194.3294 \cdot (14.0 - 3.2) \cdot 0.74 \cdot 138$
 $= 34329. \text{ Kg}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SONW} + \text{SNW}) = (21596 + 47753) = 69349 \text{ Kg}$



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 145 of 190

Nozzle Calcs.: N1-Tube Outlet

Nozl: 10 10:30am

Sep 9, 2026

Failure Path [Path 2-2]:

= (Sonw + Tpgw + Tngw + Sinw)

= (21596 + 0 + 34329 + 0) = 55925 Kgf

Failure Path [Path 3-3]:

= (Sonw + Tngw + Sinw)

= (21596 + 34329 + 0) = 55925 Kgf

Summary of Failure Path Calculations:

Path 1-1 = 69349 Kgf, must exceed W = 0 Kgf or W1 = 14719 Kgf

Path 2-2 = 55925 Kgf, must exceed W = 0 Kgf or W2 = 20125 Kgf

Path 3-3 = 55925 Kgf, must exceed W = 0 Kgf or W3 = 20125 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 2850 KPa.

The Drop for this Nozzle is : 11.9806 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 275.9806 mm

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 152.73 mm

Input Echo, WRC297 Item 1, Description: N1-Tube Outlet:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.2000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	800.000	mm
Shell Thickness	14.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.2000	mm
Nozzle Diameter	146.329	mm
Nozzle Thickness	24.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-333 6	
Nozzle Cold S.I. Allowable	SNmc 118.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 118.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	1050.00	KPa.
Radial Load	P	-1147.26	Kgf
Circumferential Shear	Vc	764.84	Kgf
Longitudinal Shear	Vl	764.84	Kgf
Circumferential Moment	Mc	321.11	Kg-m.
Longitudinal Moment	Ml	321.11	Kg-m.
Torsional Moment	Mt	458.73	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 146 of 190

Nozzle Calcs.: N1-Tube Outlet

Nozl: 10 10:30am

Sep 9, 2026

*Warning - The ratio, Dn/Tn (8.097) is < 10
check the limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 194.329 + 2 \cdot 1.65 \cdot \sqrt{408.6(14.0 - 3.2)}$$

$$= 413.547 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	817.200 mm
Nozzle Outside Diameter	(d) =	194.329 mm
Vessel Thickness used	(T) =	10.800 mm
Nozzle Thickness used	(t) =	20.800 mm
T / t	=	0.519
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.069$$

Nr/P	=	0.068
Mr/P	=	0.137
M0/P	=	0.040
N0/P	=	0.097
MrD/Mc	=	0.281
NrDL/Mc	=	0.102
M0d/Mc	=	0.087
N0DL/Mc	=	0.070
MrD/Ml	=	0.110
NrDL/Ml	=	0.064
M0D/Ml	=	0.036
N0DL/Ml	=	0.121

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	9	9	9	9
Outplane Bending (P)	23	-23	23	-23
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-16	-16	16	16
Outplane Bending (ML)	-29	29	29	-29
Normal Pressure Stress	38	39	38	39
Outplane Stress Summary	24	38	117	12

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	6	6	6	6
Inplane Bending (P)	79	-79	79	-79
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-8	-8	8	8
Inplane Bending (ML)	-91	91	91	-91
Inplane Pressure Stress	19	19	19	19
Inplane Stress Summary	4	29	206	-136



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 147 of 190

Nozzle Calcs.: N1-Tube Outlet

Nozl: 10 10:30am

Sep 9, 2026

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE			Au	Al	Bu	B1
(Shear stress normal to longitudinal plane)			Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)			2	2	-2	-2
Outplane Shear (Vl)			0	0	0	0
Outplane Shear (Mt)			7	7	7	7
Shear Stress Summary			9	9	4	4

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE			Au	Al	Bu	B1
(Stress Intensities)			Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress			28	44	206	149

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	D1
(Stresses Normal to circumferential plane)			Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)			9	9	9	9
Outplane Bending (P)			23	-23	23	-23
Outplane Membrane (Mc)			-9	-9	9	9
Outplane Bending (Mc)			-72	72	72	-72
Outplane Membrane (ML)			0	0	0	0
Outplane Bending (ML)			0	0	0	0
Normal Pressure Stress			19	19	19	19
Outplane Stress Summary			-30	68	134	-57

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	D1
(Stresses parallel to circumferential plane)			Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)			6	6	6	6
Inplane Bending (P)			79	-79	79	-79
Inplane Membrane (Mc)			-14	-14	14	14
Inplane Bending (Mc)			-233	233	233	-233
Inplane Membrane (ML)			0	0	0	0
Inplane Bending (ML)			0	0	0	0
Inplane Pressure Stress			38	39	38	39
Inplane Stress Summary			-123	186	372	-252

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	D1
(Shear stress normal to circumferential plane)			Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)			0	0	0	0
Outplane Shear (Vl)			-2	-2	2	2
Torsional Shear (Mt)			7	7	7	7
Shear Stress Summary			4	4	9	9

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	D1
(Stress Intensities)			Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress			123	186	373	253



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 148 of 190

Nozzle Calcs.: N1-Tube Outlet

Nozl: 10 10:30am Sep 9,2026

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	9	9	9	9
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-16	-16	16	16
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	4	5	4	5
Hoop Stress Summary	-3	-2	30	31

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	16	-16	16	-16
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-7	-7	7	7
Axial Bending (ML)	-17	17	17	-17
Axial Pressure Stress	2	2	2	2
Axial Stress Summary	-5	-2	43	-23

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	1	1	-1	-1
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	3	3	3	3
Shear Stress Summary	4	4	2	2

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	9	9	44	55

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	9	9	9	9
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-9	-9	9	9
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	4	5	4	5
Hoop Stress Summary	3	4	23	24

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to	Cu Left	Cl Left	Du Right	Dl Right
--	------------	------------	-------------	-------------



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 149 of 190

Nozzle Calcs.: N1-Tube Outlet

Nozl: 10 10:30am

Sep 9,2026

pipe cross-section)	Outside	Inside	Outside	Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	16	-16	16	-16
Axial Membrane (Mc)	-7	-7	7	7
Axial Bending (Mc)	-51	51	51	-51
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	2	2	2	2
Axial Stress Summary	-39	31	77	-57

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-1	-1	1	1
Shear due to Torsion	3	3	3	3
Shear Stress Summary	2	2	4	4

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	43	31	78	82

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	38	39	38	39	38	39	38	39
Circ. Pl (SUS)	-7	-7	25	25	-8	-8	20	20
Circ. Q (SUS)	-6	6	52	-52	-154	154	312	-312
Long. Pm (SUS)	19	19	19	19	19	19	19	19
Long. Pl (SUS)	-2	-2	14	14	0	0	18	18
Long. Q (SUS)	-12	12	170	-170	-49	49	95	-95
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	7	7	7	7	7	7	7	7
Pm (SUS)	38.0	39.0	38.0	39.0	38.0	39.0	38.0	39.0
Pm+Pl (SUS)	31.3	32.3	63.1	64.1	30.4	31.3	58.2	59.2
Pm+Pl+Q (Total)	28.5	43.6	203.3	149.3	124.3	185.2	370.3	253.4

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	39.00	138.00	Passed
Pm+Pl (SUS)	64.13	207.00	Passed
Pm+Pl+Q (TOTAL)	370.34	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



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FileName : E-0102

Page 150 of 190

Nozzle Calcs.: N1-Tube Outlet

Nozl: 10 10:30am

Sep 9, 2026

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	4	5	4	5	4	5	4	5
Circ. Pl (SUS)	-7	-7	25	25	0	0	18	18
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	2	2	2	2	2	2	2	2
Long. Pl (SUS)	-7	-7	7	7	-7	-7	7	7
Long. Q (SUS)	-1	1	33	-33	-35	35	67	-67
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	4.0	5.0	4.0	5.0	4.0	5.0	4.0	5.0
Pm+Pl (SUS)	5.4	5.3	29.0	30.0	9.2	10.2	22.1	23.1
Pm+Pl+Q (Total)	8.8	8.2	42.3	54.1	44.2	30.2	76.3	81.4

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	5.00	118.00	Passed
Pm+Pl (SUS)	30.05	177.00	Passed
Pm+Pl+Q (TOTAL)	81.39	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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FileName : E-0102

Page 151 of 190

Nozzle Calcs.: N3-Shell inlet

Nozl: 11 10:30am Sep 9,2026

Input, Nozzle Desc: N3-Shell inlet

From: 50

Pressure for Reinforcement Calculations	P	1200.000	KPa.
Temperature for Internal Pressure	Temp	82	°C
Design External Pressure	Pext	101.35	KPa.
Temperature for External Pressure	Tempex	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	800.00	mm
Design Length of Section	L	2716.6667	mm
Shell Finished (Minimum) Thickness	t	18.0000	mm
Shell Internal Corrosion Allowance	c	3.2000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		1570.00	mm
User Entered Minimum Design Metal Temperature		-30.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-333 6 +IT	
Material UNS Number		K03006	
Material Specification/Type	Smls. & wld. pipe		
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		450.0000	mm
Size and Thickness Basis		Nominal	
Nominal Thickness		80	
Flange Material		SA-350 LF2 +IT	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.2000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	12.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	12.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	657.2000	mm
Thickness of Pad	te	16.0000	mm
Weld leg size between Pad and Shell	Wp	10.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	16.0000	mm
Reinforcing Pad Width		100.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



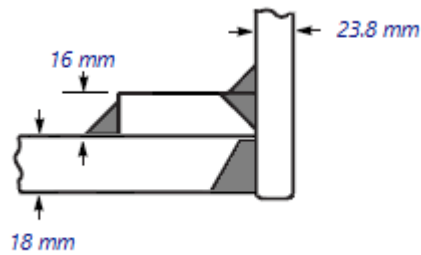
PV Elite 28 Licensee: SL User

FileName : E-0102

Page 152 of 190

Nozzle Calcs.: N3-Shell inlet

Nozl: 11 10:30am Sep 9, 2026



Reinforcement CALCULATION, Description: N3-Shell inlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	409.600 mm.
Actual Thickness Used in Calculation	23.800 mm

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1200.0 \cdot 403.2 / (138 \cdot 1.0 - 0.6 \cdot 1200.0)$
 $= 3.5247 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1200.0 \cdot 208.0 / (118 \cdot 1.0 - 0.6 \cdot 1200.0)$
 $= 2.1284 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.1358 mm

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	832.0007 mm
Parallel to Vessel Wall, opening length	d	416.0004 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		37.0000 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.855, 1.0)$
 $= 0.855$



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 153 of 190

Nozzle Calcs.: N3-Shell inlet

Nozl: 11 10:30am Sep 9,2026

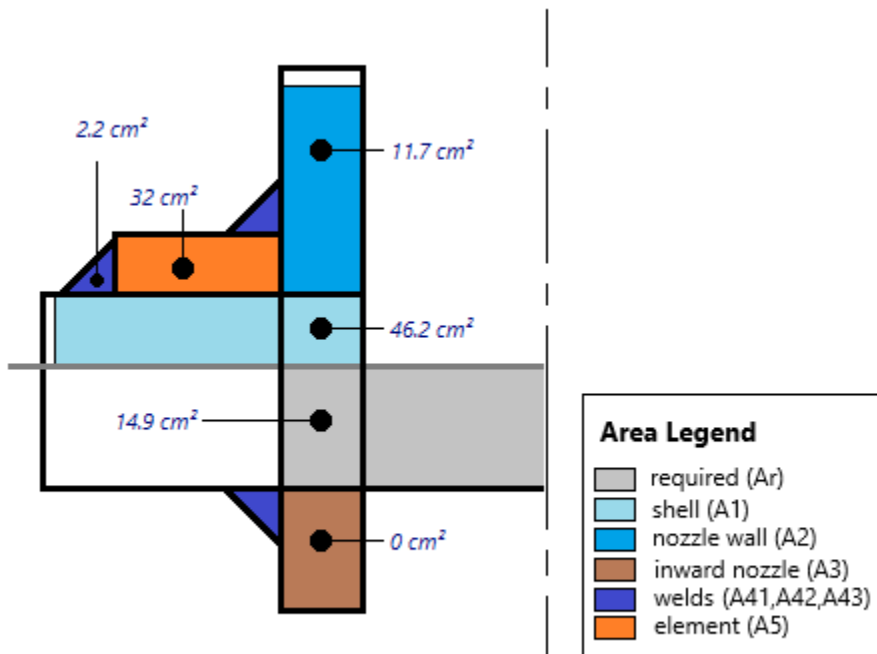


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	14.873	9.169	NA
Area in Shell	A1	46.232	42.864	NA
Area in Nozzle Wall	A2	11.688	12.316	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	2.231	2.231	NA
Area in Element	A5	32.000	32.000	NA
TOTAL AREA AVAILABLE	Atot	92.151	89.412	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

The area available with the given pad is Sufficient.

Area Required [A]:

$$\begin{aligned}
 &= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)} \\
 &= (416.0004 \cdot 3.5247 \cdot 1.0 + 2 \cdot 20.5998 \cdot 3.5247 \cdot 1.0 \cdot (1 - 0.86)) \\
 &= 14.873 \text{ cm}^2
 \end{aligned}$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$\begin{aligned}
 &= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1) \\
 &= 416.0(1.0 \cdot 14.8 - 1.0 \cdot 3.525) - 2 \cdot 20.6 \\
 &\quad (1.0 \cdot 14.8 - 1.0 \cdot 3.5247) \cdot (1 - 0.855) \\
 &= 46.232 \text{ cm}^2
 \end{aligned}$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$\begin{aligned}
 &= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2 \\
 &= (2 \cdot 37.0) \cdot (20.6 - 2.13) \cdot 0.8551
 \end{aligned}$$



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 154 of 190

Nozzle Calcs.: N3-Shell inlet

Nozl: 11 10:30am

Sep 9, 2026

$$= 11.688 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= W_o^2 \cdot fr_3 + (W_i - can/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$$

$$= 12.0^2 \cdot 0.86 + (0.0)^2 \cdot 0.86 + 10.0^2 \cdot 1.0$$

$$= 2.231 \text{ cm}^2$$

Area Available in Element [A5]:

$$= (\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, T_{lwp}, t_e)) \cdot fr_4$$

$$= (657.2 - 457.2) \cdot 16.0 \cdot 1.0$$

$$= 32.000 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $t_a = 5.3284 \text{ mm}$

Wall Thickness per UG16(b), $tr_{16b} = 4.7000 \text{ mm}$

Wall Thickness, shell/head, internal pressure $trb_1 = 6.7247 \text{ mm}$

Wall Thickness $tb_1 = \max(trb_1, tr_{16b}) = 6.7247 \text{ mm}$

Wall Thickness, shell/head, external pressure $trb_2 = 3.4963 \text{ mm}$

Wall Thickness $tb_2 = \max(trb_2, tr_{16b}) = 4.7000 \text{ mm}$

Wall Thickness per table UG-45 $tb_3 = 11.5312 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb_3, \max(tb_1, tb_2)]$$

$$= \min[11.531, \max(6.7247, 4.7)]$$

$$= 6.7247 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(5.3284, 6.7247)$$

$$= 6.7247 \text{ mm}$$

Available Nozzle Neck Thickness = $0.875 \cdot 23.800 = 20.825 \text{ mm} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 18.2,	Allowable	: 118.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 276.8 N./mm ²	Passed
Occasional	: 5.8,	Allowable	: 156.9 N./mm ²	Passed
Shear	: 9.6,	Allowable	: 82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Calculated Minimum Design Metal Temperature -104 °C

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Calculated Minimum Design Metal Temperature -104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 155 of 190

Nozzle Calcs.: N3-Shell inlet

Nozl: 11 10:30am

Sep 9, 2026

Govrn. thk, tg = 16.0, c = 3.2 mm, E* = 1.0

Thickness Ratio = $tr * E^* / (tg - c) = 0.238$, Temp. Reduction = 78 °C

Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 16.0, c = 3.2 mm, E* = 1.0

Thickness Ratio = $tr * E^* / (tg - c) = 0.238$, Temp. Reduction = 78 °C

Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 18.0, tr = 3.525, c = 3.2 mm, E* = 1.0

Thickness Ratio = $tr * E^* / (tg - c) = 0.238$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-11 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -104 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification	-46 °C
Flange MDMT with Temp reduction per UCS-66(i)(2)	-67 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :

Design Pressure/Ambient Rating = $1200.00 / 1960.01 = 0.612$

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	624.0005 mm
Parallel to Vessel Wall, opening length	d	312.0003 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		37.0000 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Reinforcement Areas for Large Nozzle (Per App 1-7(a)): (cm²)

AREA AVAILABLE, A1 to A6		No Pad	With Pad
Area Required [Internal Pressure]	AR	9.916	9.916
Area Available in Shell	A1	22.779	22.779
Area Available in Nozzle Wall	A2	11.688	11.688
Area Available in Inward Nozzle	A3	0.000	0.000
Area Available in Welds	A4	1.231	1.231
Area Available in Pad	A5	0.000	26.688
Area Available in Hub	A6	0.000	0.000
TOTAL AREA AVAILABLE	Atot	35.698	62.387

The area available without a pad is Sufficient.

The area available with the given pad is Sufficient.

The Nozzle does not fall within the scope of the diameter requirements of App. 1-7(b). Therefore, the Membrane and Bending stress checks



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 156 of 190

Nozzle Calcs.: N3-Shell inlet Nozl: 11 10:30am Sep 9, 2026

are not required.

Weld Size Calculations, Description: N3-Shell inlet

Intermediate Calc. for nozzle/shell Welds	Tmin	16.0000 mm
Intermediate Calc. for pad/shell Welds	TminPad	14.8000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	8.4840 = 0.7 * Wo mm
<i>Pad Weld</i>	<i>7.4000 = 0.5*TminPad</i>	<i>7.0700 = 0.7 * Wp mm</i>

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t_{tr})) \cdot S_v)$
 $= \max(0, (14.8734 - 46.2319 + 2 \cdot 20.5998 \cdot 0.8551 \cdot (1.0 \cdot 14.8 - 3.5247)) \cdot 138)$
 $= \max(0, -38536.86) \text{ Kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i \cdot \text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (11.6878 + 32.0 + 2.2313 - 0.0 \cdot 0.86) \cdot 138$
 $= 64615.45 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (11.6878 + 0.0 + 1.2313 + (5.2138)) \cdot 138$
 $= 25515.96 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (11.6878 + 0.0 + 2.2313 + 32.0 + (5.2138)) \cdot 138$
 $= 71952.14 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 457.2 \cdot 12.0 \cdot 0.49 \cdot 118$
 $= 50810. \text{ Kgf}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$
 $= (3.1416/2.0) \cdot 657.2 \cdot 10.0 \cdot 0.49 \cdot 138$
 $= 71180. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 218.3001) \cdot (23.7998 - 3.2) \cdot 0.7 \cdot 118$
 $= 118990. \text{ Kgf}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$
 $= (3.1416/2) \cdot 457.2 \cdot 16.0 \cdot 0.74 \cdot 138$
 $= 119652. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng}$
 $= (3.1416/2.0) \cdot 457.2 \cdot (12.0 - 0.0) \cdot 0.74 \cdot 138$
 $= 89739. \text{ Kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 157 of 190

Nozzle Calcs.: N3-Shell inlet

Nozl: 11 10:30am

Sep 9, 2026

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SPEW} + \text{SNW}) = (71180 + 118990) = 190170 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\ = (50810 + 119652 + 89739 + 0) = 260201 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (\text{Spew} + \text{Tngw} + \text{Sinw}) \\ = (71180 + 89739 + 0) = 160919 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 190169 Kgf, must exceed W = 0 Kgf	or W1 = 64615 Kgf
Path 2-2 = 260201 Kgf, must exceed W = 0 Kgf	or W2 = 25515 Kgf
Path 3-3 = 160918 Kgf, must exceed W = 0 Kgf	or W3 = 71952 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 4168 KPa.

Nozzle is O.K. for the External Pressure: 101 KPa.

The Drop for this Nozzle is : 71.7592 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 339.7592 mm

Warning:

Review Appendix 1-7 (b)(3) for notes regarding externally applied loads to large nozzles and U-2(g).

Input Echo, WRC297 Item 1, Description: N3-Shell inlet:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.2000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	800.000	mm
Shell Thickness	18.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.2000	mm
Nozzle Diameter	409.600	mm
Nozzle Thickness	23.7998	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-333 6	
Nozzle Cold S.I. Allowable	SNmc 118.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 118.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	657.200	mm
Thickness of Reinforcing Pad	16.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:
These loads are assumed to be SUSained loads.



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 158 of 190

Nozzle Calcs.: N3-Shell inlet Nozl: 11 10:30am Sep 9, 2026

Design Internal Pressure	Dp	1200.00	KPa.
Radial Load	P	-3441.77	Kgf
Circumferential Shear	Vc	2294.51	Kgf
Longitudinal Shear	Vl	2294.51	Kgf
Circumferential Moment	Mc	2889.98	Kg-m.
Longitudinal Moment	Ml	2889.98	Kg-m.
Torsional Moment	Mt	4128.54	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Dv (.547) is > 0.5
check the limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 457.2 + 2 \cdot 1.65 \cdot \sqrt{418.6(18.0 - 3.2)}$$

$$= 716.943 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	837.200 mm
Nozzle Outside Diameter	(d) =	457.200 mm
Vessel Thickness used	(T) =	30.800 mm
Nozzle Thickness used	(t) =	20.600 mm
T / t	=	1.495
d / t	=	22.194

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.847$$

Nr/P	=	0.060
Mr/P	=	0.088
M0/P	=	0.038
N0/P	=	0.131
MrD/Mc	=	0.215
NrDL/Mc	=	0.108
M0d/Mc	=	0.108
N0DL/Mc	=	0.135
MrD/Ml	=	0.111
NrDL/Ml	=	0.057
M0D/Ml	=	0.056
N0DL/Ml	=	0.199

Vessel Stresses (N/mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	8	-8	8	-8
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	-12	-12	12	12
Outplane Bending (Ml)	-22	22	22	-22
Normal Pressure Stress	15	16	15	16
Outplane Stress Summary	-7	21	62	3

Vessel Stresses (N/mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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PV Elite 28 Licensee: SL User

FileName : E-0102

Page 159 of 190

Nozzle Calcs.: N3-Shell inlet

Nozl: 11 10:30am

Sep 9,2026

Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	18	-18	18	-18
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-3	-3	3	3
Inplane Bending (ML)	-43	43	43	-43
Inplane Pressure Stress	7	7	7	7

Inplane Stress Summary	-18	30	75	-48
------------------------	-----	----	----	-----

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE				
(Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside

Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3

Shear Stress Summary	5	5	2	2
----------------------	---	---	---	---

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stress Intensities	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside

Two * Max Shear Stress	20	32	76	52
------------------------	----	----	----	----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside

Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	8	-8	8	-8
Outplane Membrane (Mc)	-8	-8	8	8
Outplane Bending (Mc)	-42	42	42	-42
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	7	7	7	7

Outplane Stress Summary	-30	37	71	-29
-------------------------	-----	----	----	-----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside

Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	18	-18	18	-18
Inplane Membrane (Mc)	-7	-7	7	7
Inplane Bending (Mc)	-84	84	84	-84
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	15	16	15	16

Inplane Stress Summary	-55	77	127	-77
------------------------	-----	----	-----	-----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside

Outplane Shear (Vc)	0	0	0	0
---------------------	---	---	---	---



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 160 of 190

Nozzle Calcs.: N3-Shell inlet Nozl: 11 10:30am Sep 9,2026

Outplane Shear (V1)	-1	-1	1	1
Torsional Shear (Mt)	3	3	3	3

Shear Stress Summary	2	2	5	5
----------------------	---	---	---	---

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
---	-----------------------	----------------------	------------------------	-----------------------

Two * Max Shear Stress	55	77	127	78
------------------------	----	----	-----	----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
---	----------------------	---------------------	-------------------------	------------------------

Hoop Membrane (P)	4	4	4	4
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-12	-12	12	12
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	13	15	13	15

Hoop Stress Summary	5	6	31	32
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Axial Membrane (P)	1	1	1	1
Axial Bending (P)	27	-27	27	-27
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-9	-9	9	9
Axial Bending (ML)	-72	72	72	-72
Axial Pressure Stress	6	6	6	6

Axial Stress Summary	-46	43	117	-81
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
--------------------------------------	---------------	--------------	---------------	--------------

Shear due to (Vc)	1	1	-1	-1
Shear due to (V1)	0	0	0	0
Shear due to Torsion	5	5	5	5

Shear Stress Summary	7	7	4	4
----------------------	---	---	---	---

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
--	---------------	--------------	---------------	--------------

Two * Max Shear Stress	53	44	117	115
------------------------	----	----	-----	-----

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Hoop Membrane (P)	4	4	4	4
Hoop Bending (P)	0	0	0	0



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 161 of 190

Nozzle Calcs.: N3-Shell inlet Nozl: 11 10:30am Sep 9,2026

Hoop Membrane (Mc)	-8	-8	8	8
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	13	15	13	15

Hoop Stress Summary	9	11	27	28
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	1	1	1	1
Axial Bending (P)	27	-27	27	-27
Axial Membrane (Mc)	-9	-9	9	9
Axial Bending (Mc)	-141	141	141	-141
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	6	6	6	6

Axial Stress Summary	-115	112	186	-151
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-1	-1	1	1
Shear due to Torsion	5	5	5	5

Shear Stress Summary	4	4	7	7
----------------------	---	---	---	---

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	125	112	186	180
------------------------	-----	-----	-----	-----

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 457.2 + 2 \cdot 1.65 \cdot \sqrt{410.6(18.0 - 3.2)}$$

$$= 714.449 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	821.200 mm
Nozzle Outside Diameter	(d) =	657.200 mm
Vessel Thickness used	(T) =	14.800 mm
Nozzle Thickness used	(t) =	120.600 mm
T / t	=	0.123
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 5.961$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.044
Mr/P	=	0.070
M0/P	=	0.020
N0/P	=	0.034
MrD/Mc	=	0.241



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 162 of 190

Nozzle Calcs.: N3-Shell inlet

Nozl: 11 10:30am Sep 9,2026

NrDL/Mc = 0.091
MØd/Mc = 0.070
NØDL/Mc = 0.061
MrD/Ml = 0.046
NrDL/Ml = 0.039
MØD/Ml = 0.015
NØDL/Ml = 0.030

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	18	-18	18	-18
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-5	-5	5	5
Outplane Bending (ML)	-17	17	17	-17
Normal Pressure Stress	32	33	32	33
Outplane Stress Summary	32	31	79	8

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	6	6	6	6
Inplane Bending (P)	64	-64	64	-64
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-7	-7	7	7
Inplane Bending (ML)	-54	54	54	-54
Inplane Pressure Stress	16	16	16	16
Inplane Stress Summary	25	5	149	-88

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	4	4	4	4
Shear Stress Summary	5	5	2	2

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	35	32	149	96

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	18	-18	18	-18
Outplane Membrane (Mc)	-12	-12	12	12
Outplane Bending (Mc)	-82	82	82	-82



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 163 of 190

Nozzle Calcs.: N3-Shell inlet Nozl: 11 10:30am Sep 9,2026

Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	16	16	16	16

Outplane Stress Summary	-54	73	134	-67
-------------------------	-----	----	-----	-----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	6	6	6	6
Inplane Bending (P)	64	-64	64	-64
Inplane Membrane (Mc)	-17	-17	17	17
Inplane Bending (Mc)	-284	284	284	-284
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	32	33	32	33

Inplane Stress Summary	-199	242	406	-291
------------------------	------	-----	-----	------

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	4	4	4	4

Shear Stress Summary	2	2	5	5
----------------------	---	---	---	---

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside

Two * Max Shear Stress	199	242	406	291
------------------------	-----	-----	-----	-----

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	15	16	15	16	15	16	15	16
Circ. Pl (SUS)	-8	-8	16	16	-5	-5	9	9
Circ. Q (SUS)	-14	14	30	-30	-66	66	102	-102
Long. Pm (SUS)	7	7	7	7	7	7	7	7
Long. Pl (SUS)	-1	-1	5	5	-4	-4	12	12
Long. Q (SUS)	-25	25	61	-61	-34	34	50	-50
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	15.0	16.0	15.0	16.0	15.0	16.0	15.0	16.0
Pm+Pl (SUS)	7.6	8.4	31.1	32.0	10.1	11.1	24.2	25.2
Pm+Pl+Q (Total)	20.2	32.5	73.3	51.2	56.2	77.1	126.3	77.3



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 164 of 190

Nozzle Calcs.: N3-Shell inlet

Nozl: 11 10:30am

Sep 9,2026

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	16.00	138.00	Passed
Pm+Pl (SUS)	32.05	207.00	Passed
Pm+Pl+Q (TOTAL)	126.28	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	32	33	32	33	32	33	32	33
Circ. Pl (SUS)	0	0	10	10	-11	-11	23	23
Circ. Q (SUS)	1	-1	35	-35	-220	220	348	-348
Long. Pm (SUS)	16	16	16	16	16	16	16	16
Long. Pl (SUS)	-1	-1	13	13	-7	-7	17	17
Long. Q (SUS)	10	-10	118	-118	-64	64	100	-100
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	32.0	33.0	32.0	33.0	32.0	33.0	32.0	33.0
Pm+Pl (SUS)	32.1	33.1	42.1	43.1	21.1	22.1	55.0	56.0
Pm+Pl+Q (Total)	35.4	32.9	147.1	97.2	199.1	242.1	403.1	292.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	33.00	138.00	Passed
Pm+Pl (SUS)	56.04	207.00	Passed
Pm+Pl+Q (TOTAL)	403.09	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	13	15	13	15	13	15	13	15
Circ. Pl (SUS)	-8	-8	16	16	-4	-4	12	12
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	6	6	6	6	6	6	6	6
Long. Pl (SUS)	-8	-8	10	10	-8	-8	10	10
Long. Q (SUS)	-45	45	99	-99	-114	114	168	-168



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FileName : E-0102

Page 165 of 190

Nozzle Calcs.: N3-Shell inlet

Nozl: 11 10:30am

Sep 9, 2026

Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	5	5	5	5	5	5	5	5
Pm (SUS)	13.0	15.0	13.0	15.0	13.0	15.0	13.0	15.0
Pm+Pl (SUS)	7.3	9.2	29.1	31.1	11.2	13.2	25.1	27.1
Pm+Pl+Q (Total)	53.4	44.0	115.2	114.3	125.3	112.2	184.2	179.4

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	15.00	118.00	Passed
Pm+Pl (SUS)	31.07	177.00	Passed
Pm+Pl+Q (TOTAL)	184.23	353.99	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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PV Elite 28 Licensee: SL User

FileName : E-0102

Page 166 of 190

Nozzle Calcs.: N4-Shell outlet

Nozl: 12 10:31am

Sep 9, 2026

Input, Nozzle Desc: N4-Shell outlet

From: 50

Pressure for Reinforcement Calculations	P	1207.842	KPa.
Temperature for Internal Pressure	Temp	82	°C
Design External Pressure	Pext	101.35	KPa.
Temperature for External Pressure	Tempex	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	800.00	mm
Design Length of Section	L	2716.6667	mm
Shell Finished (Minimum) Thickness	t	18.0000	mm
Shell Internal Corrosion Allowance	c	3.2000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		2570.00	mm
User Entered Minimum Design Metal Temperature		-30.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-333 6 +IT	
Material UNS Number		K03006	
Material Specification/Type	Smls. & wld. pipe		
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		450.0000	mm
Size and Thickness Basis		Nominal	
Nominal Thickness		80	
Flange Material		SA-350 LF2 +IT	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.2000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	12.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	657.2000	mm
Thickness of Pad	te	16.0000	mm
Weld leg size between Pad and Shell	Wp	14.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	16.0000	mm
Reinforcing Pad Width		100.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



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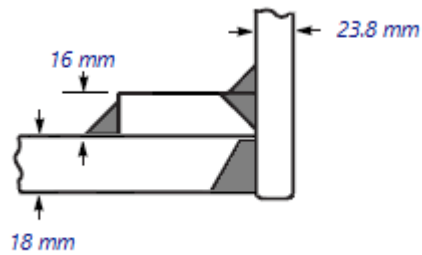
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Page 167 of 190

Nozzle Calcs.: N4-Shell outlet

Nozl: 12 10:31am

Sep 9, 2026



Reinforcement CALCULATION, Description: N4-Shell outlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	409.600 mm.
Actual Thickness Used in Calculation	23.800 mm

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1207.84 \cdot 403.2 / (138 \cdot 1.0 - 0.6 \cdot 1207.84)$
 $= 3.5479 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1207.84 \cdot 208.0 / (118 \cdot 1.0 - 0.6 \cdot 1207.84)$
 $= 2.1424 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.0307 mm

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	832.0007 mm
Parallel to Vessel Wall, opening length	d	416.0004 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		37.0000 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.855, 1.0)$
 $= 0.855$



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 168 of 190

Nozzle Calcs.: N4-Shell outlet

Nozl: 12 10:31am

Sep 9, 2026

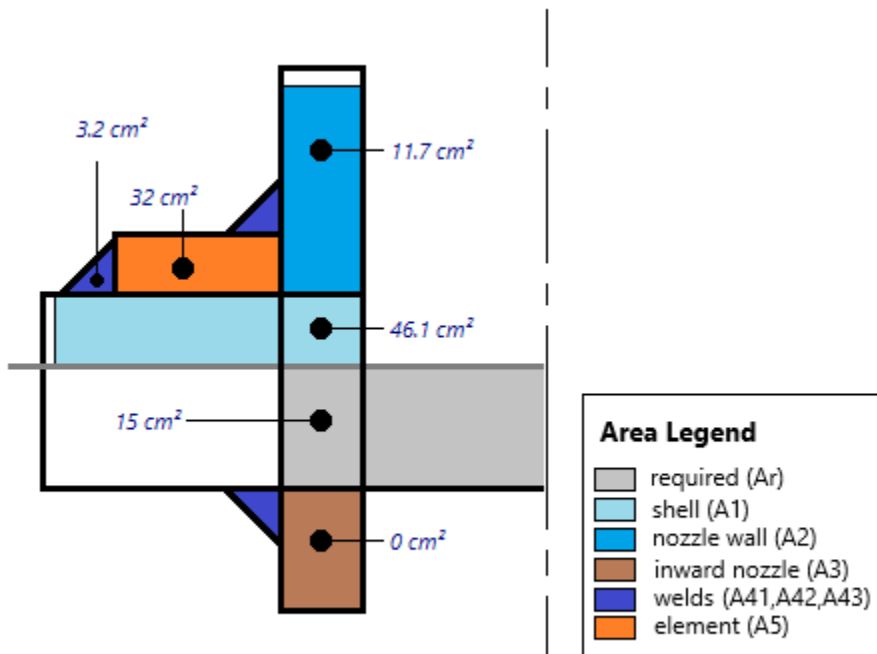


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	14.971	9.169	NA
Area in Shell	A1	46.137	42.864	NA
Area in Nozzle Wall	A2	11.679	12.382	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	3.191	3.191	NA
Area in Element	A5	32.000	32.000	NA
TOTAL AREA AVAILABLE	Atot	93.007	90.438	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

The area available with the given pad is Sufficient.

Area Required [A]:

$$\begin{aligned}
 &= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)} \\
 &= (416.0004 \cdot 3.5479 \cdot 1.0 + 2 \cdot 20.5998 \cdot 3.5479 \cdot 1.0 \cdot (1 - 0.86)) \\
 &= 14.971 \text{ cm}^2
 \end{aligned}$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$\begin{aligned}
 &= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1) \\
 &= 416.0(1.0 \cdot 14.8 - 1.0 \cdot 3.548) - 2 \cdot 20.6 \\
 &\quad (1.0 \cdot 14.8 - 1.0 \cdot 3.5479) \cdot (1 - 0.855) \\
 &= 46.137 \text{ cm}^2
 \end{aligned}$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$\begin{aligned}
 &= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2 \\
 &= (2 \cdot 37.0) \cdot (20.6 - 2.14) \cdot 0.8551
 \end{aligned}$$



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 169 of 190

Nozzle Calcs.: N4-Shell outlet Nozl: 12 10:31am Sep 9, 2026

$$= 11.679 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= W_o^2 \cdot fr_3 + (W_i - can / 0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$$

$$= 12.0^2 \cdot 0.86 + (0.0)^2 \cdot 0.86 + 14.0^2 \cdot 1.0$$

$$= 3.191 \text{ cm}^2$$

Area Available in Element [A5]:

$$= (\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, T_{lwp}, t_e)) \cdot fr_4$$

$$= (657.2 - 457.2) \cdot 16.0 \cdot 1.0$$

$$= 32.000 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 5.3424 \text{ mm}$
Wall Thickness per UG16(b),	$tr_{16b} = 4.7000 \text{ mm}$
Wall Thickness, shell/head, internal pressure	$trb_1 = 6.7479 \text{ mm}$
Wall Thickness	$tb_1 = \max(trb_1, tr_{16b}) = 6.7479 \text{ mm}$
Wall Thickness, shell/head, external pressure	$trb_2 = 3.4963 \text{ mm}$
Wall Thickness	$tb_2 = \max(trb_2, tr_{16b}) = 4.7000 \text{ mm}$
Wall Thickness per table UG-45	$tb_3 = 11.5312 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb_3, \max(tb_1, tb_2)]$$

$$= \min[11.531, \max(6.7479, 4.7)]$$

$$= 6.7479 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(5.3424, 6.7479)$$

$$= 6.7479 \text{ mm}$$

Available Nozzle Neck Thickness = $0.875 \cdot 23.800 = 20.825 \text{ mm} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 18.2,	Allowable	: 118.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 276.8 N./mm ²	Passed
Occasional	: 5.8,	Allowable	: 156.9 N./mm ²	Passed
Shear	: 9.6,	Allowable	: 82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 170 of 190

Nozzle Calcs.: N4-Shell outlet Nozl: 12 10:31am Sep 9,2026

Govrn. thk, tg = 16.0, c = 3.2 mm, E* = 1.0
 Thickness Ratio = $tr * E^*/(tg - c) = 0.24$, Temp. Reduction = 78 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 16.0, c = 3.2 mm, E* = 1.0
 Thickness Ratio = $tr * E^*/(tg - c) = 0.24$, Temp. Reduction = 78 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 18.0, tr = 3.548, c = 3.2 mm, E* = 1.0
 Thickness Ratio = $tr * E^*/(tg - c) = 0.24$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-11 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -104 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification	-46 °C
Flange MDMT with Temp reduction per UCS-66(i)(2)	-67 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :
 Design Pressure/Ambient Rating = $1207.84/1960.01 = 0.616$

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	624.0005 mm
Parallel to Vessel Wall, opening length	d	312.0003 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		37.0000 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Reinforcement Areas for Large Nozzle (Per App 1-7(a)): (cm²)

AREA AVAILABLE, A1 to A6		No Pad	With Pad
Area Required [Internal Pressure]	AR	9.981	9.981
Area Available in Shell	A1	22.733	22.733
Area Available in Nozzle Wall	A2	11.679	11.679
Area Available in Inward Nozzle	A3	0.000	0.000
Area Available in Welds	A4	1.231	1.231
Area Available in Pad	A5	0.000	26.688
Area Available in Hub	A6	0.000	0.000
TOTAL AREA AVAILABLE	Atot	35.643	62.331

The area available without a pad is Sufficient.
 The area available with the given pad is Sufficient.

The Nozzle does not fall within the scope of the diameter requirements of App. 1-7(b). Therefore, the Membrane and Bending stress checks



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 171 of 190

Nozzle Calcs.: N4-Shell outlet Nozl: 12 10:31am Sep 9, 2026

are not required.

Weld Size Calculations, Description: N4-Shell outlet

Intermediate Calc. for nozzle/shell Welds	Tmin	16.0000 mm
Intermediate Calc. for pad/shell Welds	TminPad	14.8000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	8.4840 = 0.7 * Wo mm
Pad Weld	7.4000 = 0.5*TminPad	9.8980 = 0.7 * Wp mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t_{tr})) \cdot S_v)$$

$$= \max(0, (14.9711 - 46.1369 + 2 \cdot 20.5998 \cdot 0.8551 \cdot (1.0 \cdot 14.8 - 3.5479)) \cdot 138)$$

$$= \max(0, -38277.25) \text{ Kgf}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2+A5+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$$

$$= (11.679 + 32.0 + 3.1913 - 0.0 \cdot 0.86) \cdot 138$$

$$= 65953.87 \text{ Kgf}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$$

$$= (11.679 + 0.0 + 1.2313 + (5.2138)) \cdot 138$$

$$= 25503.50 \text{ Kgf}$$

Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$$

$$= (11.679 + 0.0 + 3.1913 + 32.0 + (5.2138)) \cdot 138$$

$$= 73290.55 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 457.2 \cdot 12.0 \cdot 0.49 \cdot 118$$

$$= 50810. \text{ Kgf}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$$

$$= (3.1416/2.0) \cdot 657.2 \cdot 14.0 \cdot 0.49 \cdot 138$$

$$= 99652. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 218.3001) \cdot (23.7998 - 3.2) \cdot 0.7 \cdot 118$$

$$= 118990. \text{ Kgf}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$$

$$= (3.1416/2) \cdot 457.2 \cdot 16.0 \cdot 0.74 \cdot 138$$

$$= 119652. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 457.2 \cdot (18.0 - 3.2) \cdot 0.74 \cdot 138$$

$$= 110678. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 172 of 190

Nozzle Calcs.: N4-Shell outlet

Nozl: 12 10:31am

Sep 9,2026

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SPEW} + \text{SNW}) = (99652 + 118990) = 218642 \text{ Kg}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\ = (50810 + 119652 + 110678 + 0) = 281140 \text{ Kg}$$

Failure Path [Path 3-3]:

$$= (\text{Spew} + \text{Tngw} + \text{Sinw}) \\ = (99652 + 110678 + 0) = 210330 \text{ Kg}$$

Summary of Failure Path Calculations:

Path 1-1 = 218641 Kg, must exceed W = 0 Kg	or W1 = 65953 Kg
Path 2-2 = 281140 Kg, must exceed W = 0 Kg	or W2 = 25503 Kg
Path 3-3 = 210329 Kg, must exceed W = 0 Kg	or W3 = 73290 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 4197 KPa.

Nozzle is O.K. for the External Pressure: 101 KPa.

The Drop for this Nozzle is : 71.7592 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 289.7592 mm

Warning:

Review Appendix 1-7 (b)(3) for notes regarding externally applied loads to large nozzles and U-2(g).

Input Echo, WRC297 Item 1, Description: N4-Shell outle:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.2000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	800.000	mm
Shell Thickness	18.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.2000	mm
Nozzle Diameter	409.600	mm
Nozzle Thickness	23.7998	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-333 6	
Nozzle Cold S.I. Allowable	SNmc 118.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 118.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	657.200	mm
Thickness of Reinforcing Pad	16.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:
These loads are assumed to be SUSained loads.



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 173 of 190

Nozzle Calcs.: N4-Shell outlet Nozl: 12 10:31am Sep 9, 2026

Design Internal Pressure	Dp	1207.84	KPa.
Radial Load	P	-3441.77	Kgf
Circumferential Shear	Vc	2294.51	Kgf
Longitudinal Shear	Vl	2294.51	Kgf
Circumferential Moment	Mc	2889.98	Kg-m.
Longitudinal Moment	Ml	2889.98	Kg-m.
Torsional Moment	Mt	4128.54	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Dv (.547) is > 0.5
check the limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 457.2 + 2 \cdot 1.65 \cdot \sqrt{418.6(18.0 - 3.2)}$$

$$= 716.943 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	837.200 mm
Nozzle Outside Diameter	(d) =	457.200 mm
Vessel Thickness used	(T) =	30.800 mm
Nozzle Thickness used	(t) =	20.600 mm
T / t	=	1.495
d / t	=	22.194

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.847$$

Nr/P	=	0.060
Mr/P	=	0.088
M0/P	=	0.038
N0/P	=	0.131
MrD/Mc	=	0.215
NrDL/Mc	=	0.108
M0d/Mc	=	0.108
N0DL/Mc	=	0.135
MrD/Ml	=	0.111
NrDL/Ml	=	0.057
M0D/Ml	=	0.056
N0DL/Ml	=	0.199

Vessel Stresses (N/mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	8	-8	8	-8
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	-12	-12	12	12
Outplane Bending (Ml)	-22	22	22	-22
Normal Pressure Stress	15	16	15	16
Outplane Stress Summary	-6	21	63	3

Vessel Stresses (N/mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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PV Elite 28 Licensee: SL User

FileName : E-0102

Page 174 of 190

Nozzle Calcs.: N4-Shell outlet

Nozl: 12 10:31am

Sep 9,2026

Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	18	-18	18	-18
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-3	-3	3	3
Inplane Bending (ML)	-43	43	43	-43
Inplane Pressure Stress	7	7	7	7

Inplane Stress Summary	-18	30	75	-48
------------------------	-----	----	----	-----

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE				
(Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside

Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3

Shear Stress Summary	5	5	2	2
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stress Intensities	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside

Two * Max Shear Stress	20	33	76	52
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside

Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	8	-8	8	-8
Outplane Membrane (Mc)	-8	-8	8	8
Outplane Bending (Mc)	-42	42	42	-42
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	7	7	7	7

Outplane Stress Summary	-30	37	71	-29
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside

Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	18	-18	18	-18
Inplane Membrane (Mc)	-7	-7	7	7
Inplane Bending (Mc)	-84	84	84	-84
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	15	16	15	16

Inplane Stress Summary	-55	77	127	-77
------------------------	-----	----	-----	-----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside

Outplane Shear (Vc)	0	0	0	0
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PV Elite 28 Licensee: SL User

FileName : E-0102

Page 175 of 190

Nozzle Calcs.: N4-Shell outlet Nozl: 12 10:31am Sep 9,2026

Outplane Shear (V1)	-1	-1	1	1
Torsional Shear (Mt)	3	3	3	3

Shear Stress Summary	2	2	5	5
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Two * Max Shear Stress	55	77	127	78
------------------------	----	----	-----	----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
---	----------------------	---------------------	-------------------------	------------------------

Hoop Membrane (P)	4	4	4	4
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-12	-12	12	12
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	13	15	13	15

Hoop Stress Summary	5	7	31	33
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Axial Membrane (P)	1	1	1	1
Axial Bending (P)	27	-27	27	-27
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-9	-9	9	9
Axial Bending (ML)	-72	72	72	-72
Axial Pressure Stress	6	6	6	6

Axial Stress Summary	-46	43	117	-81
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
--------------------------------------	---------------	--------------	---------------	--------------

Shear due to (Vc)	1	1	-1	-1
Shear due to (V1)	0	0	0	0
Shear due to Torsion	5	5	5	5

Shear Stress Summary	7	7	4	4
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
--	---------------	--------------	---------------	--------------

Two * Max Shear Stress	53	44	117	115
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Hoop Membrane (P)	4	4	4	4
Hoop Bending (P)	0	0	0	0



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 176 of 190

Nozzle Calcs.: N4-Shell outlet Nozl: 12 10:31am Sep 9, 2026

Hoop Membrane (Mc)	-8	-8	8	8
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	13	15	13	15

Hoop Stress Summary	9	11	27	28
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to pipe cross-section)	Left Outside	Left Inside	Right Outside	Right Inside

Axial Membrane (P)	1	1	1	1
Axial Bending (P)	27	-27	27	-27
Axial Membrane (Mc)	-9	-9	9	9
Axial Bending (Mc)	-141	141	141	-141
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	6	6	6	6

Axial Stress Summary	-115	112	186	-151
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-1	-1	1	1
Shear due to Torsion	5	5	5	5

Shear Stress Summary	4	4	7	7
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Outside	Inside	Outside	Inside

Two * Max Shear Stress	125	112	186	180
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Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 457.2 + 2 \cdot 1.65 \cdot \sqrt{410.6(18.0 - 3.2)}$$

$$= 714.449 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	821.200 mm
Nozzle Outside Diameter	(d) =	657.200 mm
Vessel Thickness used	(T) =	14.800 mm
Nozzle Thickness used	(t) =	120.600 mm
T / t	=	0.123
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 5.961$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.044
Mr/P	=	0.070
M0/P	=	0.020
N0/P	=	0.034
MrD/Mc	=	0.241



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 177 of 190

Nozzle Calcs.: N4-Shell outlet

Nozl: 12 10:31am

Sep 9, 2026

NrDL/Mc = 0.091
MØd/Mc = 0.070
NØDL/Mc = 0.061
MrD/Ml = 0.046
NrDL/Ml = 0.039
MØD/Ml = 0.015
NØDL/Ml = 0.030

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stresses Normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	18	-18	18	-18
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-5	-5	5	5
Outplane Bending (ML)	-17	17	17	-17
Normal Pressure Stress	32	33	32	33
Outplane Stress Summary	33	31	79	8

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	6	6	6	6
Inplane Bending (P)	64	-64	64	-64
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-7	-7	7	7
Inplane Bending (ML)	-54	54	54	-54
Inplane Pressure Stress	16	16	16	16
Inplane Stress Summary	25	5	149	-88

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	4	4	4	4
Shear Stress Summary	5	5	2	2

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress	35	32	149	97

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	18	-18	18	-18
Outplane Membrane (Mc)	-12	-12	12	12
Outplane Bending (Mc)	-82	82	82	-82



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 178 of 190

Nozzle Calcs.: N4-Shell outlet Nozl: 12 10:31am Sep 9,2026

Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	16	16	16	16

Outplane Stress Summary	-54	73	134	-67
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	6	6	6	6
Inplane Bending (P)	64	-64	64	-64
Inplane Membrane (Mc)	-17	-17	17	17
Inplane Bending (Mc)	-284	284	284	-284
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	32	33	32	33

Inplane Stress Summary	-199	242	406	-291
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	4	4	4	4

Shear Stress Summary	2	2	5	5
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside

Two * Max Shear Stress	199	242	406	291
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WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	15	16	15	16	15	16	15	16
Circ. Pl (SUS)	-8	-8	16	16	-5	-5	9	9
Circ. Q (SUS)	-14	14	30	-30	-66	66	102	-102
Long. Pm (SUS)	7	7	7	7	7	7	7	7
Long. Pl (SUS)	-1	-1	5	5	-4	-4	12	12
Long. Q (SUS)	-25	25	61	-61	-34	34	50	-50
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	15.0	16.0	15.0	16.0	15.0	16.0	15.0	16.0
Pm+Pl (SUS)	7.6	8.4	31.1	32.0	10.1	11.1	24.2	25.2
Pm+Pl+Q (Total)	20.2	32.5	73.3	51.2	56.2	77.1	126.3	77.3



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 179 of 190

Nozzle Calcs.: N4-Shell outlet

Nozl: 12 10:31am

Sep 9,2026

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	16.00	138.00	Passed
Pm+Pl (SUS)	32.05	207.00	Passed
Pm+Pl+Q (TOTAL)	126.28	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	32	33	32	33	32	33	32	33
Circ. Pl (SUS)	0	0	10	10	-11	-11	23	23
Circ. Q (SUS)	1	-1	35	-35	-220	220	348	-348
Long. Pm (SUS)	16	16	16	16	16	16	16	16
Long. Pl (SUS)	-1	-1	13	13	-7	-7	17	17
Long. Q (SUS)	10	-10	118	-118	-64	64	100	-100
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	32.0	33.0	32.0	33.0	32.0	33.0	32.0	33.0
Pm+Pl (SUS)	32.1	33.1	42.1	43.1	21.1	22.1	55.0	56.0
Pm+Pl+Q (Total)	35.4	32.9	147.1	97.2	199.1	242.1	403.1	292.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	33.00	138.00	Passed
Pm+Pl (SUS)	56.04	207.00	Passed
Pm+Pl+Q (TOTAL)	403.09	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	13	15	13	15	13	15	13	15
Circ. Pl (SUS)	-8	-8	16	16	-4	-4	12	12
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	6	6	6	6	6	6	6	6
Long. Pl (SUS)	-8	-8	10	10	-8	-8	10	10
Long. Q (SUS)	-45	45	99	-99	-114	114	168	-168



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 180 of 190

Nozzle Calcs.: N4-Shell outlet

Nozl: 12 10:31am

Sep 9, 2026

Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	5	5	5	5	5	5	5	5
Pm (SUS)	13.0	15.0	13.0	15.0	13.0	15.0	13.0	15.0
Pm+Pl (SUS)	7.3	9.2	29.1	31.1	11.2	13.2	25.1	27.1
Pm+Pl+Q (Total)	53.4	44.0	115.2	114.3	125.3	112.2	184.2	179.4

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	15.00	118.00	Passed
Pm+Pl (SUS)	31.07	177.00	Passed
Pm+Pl+Q (TOTAL)	184.23	353.99	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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FileName : E-0102

Page 181 of 190

ASME TS Calc:

Case: 1 10:31a Sep 9,2026

Input Echo, Tubesheet Number 1, Description: U-Tube

Shell Data:

Main Shell Description: Main Shell

Shell Maximum Design Pressure	Psd,max	1200.00 KPa.
Shell Maximum Operating Pressure	Psox,max	1200.00 KPa.
Shell Minimum Operating Pressure	Psox,min	101.35 KPa.
Shell Thickness	ts	18.0000 mm
Shell Internal Corrosion Allowance	cas	3.2000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	800.000 mm
Shell Temperature for Internal Pressure	Ts	82.00 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	138.00 N./mm ²
Shell Allowable Stress at Ambient		138.00 N./mm ²

Channel Data:

Channel Description: Left Ch shell

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	1050.00 KPa.
Channel Maximum Operating Pressure	Ptox,max	1050.00 KPa.
Channel Minimum Operating Pressure	Ptox,min	0.00 KPa.
Channel Thickness	tc	14.0000 mm
Channel Internal Corrosion Allowance	cac	3.2000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	800.000 mm
Channel Design Temperature	TEMPC	230.00 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	138.00 N./mm ²
Channel Allowable Stress at Ambient		138.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	170
Tube Wall Thickness	et	2.3200 mm
Tube Outside Diameter	D	25.4000 mm
Total Straight Tube Length	Lt	2438.00 mm
Straight Tube Length (bet. inner tubsht faces) L		2373.00 mm
Design Temperature of the Tubes		230.00 °C
Tube Material		SA-334 1
Tube Material UNS Number		K03008
Is this a Welded Tube		No
Tube Material Specification used	Smls. & wld. tube	
Tube Allowable Stress at Temperature		108.00 N./mm ²
Tube Allowable Stress At Ambient		108.00 N./mm ²
Tube Yield Stress At design Temperature	Syt	173.20 N./mm ²
Tube Pitch (Center to Center Spacing)	P	38.1000 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	2.3200 mm
Groove Weld Leg	ag	2.3200 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	370.687 mm
Largest Center-to-Center Tube Distance	U1	0.0000 mm



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 182 of 190

ASME TS Calc:

Case: 1 10:31a Sep 9,2026

Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	5.0000 mm
Area of Tube Layout	Ap	2517.1 cm ²
Perimeter of Tube Layout	Cp	2062.1 mm

Tubesheet Data:

Tubesheet TYPE: U-tube, Gasketed both Sides, Conf. d

Tubesheet Design Metal Temperature	T	230.00 °C
Tubesheet Material		SA-765 II

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number		K03047
Tubesheet Allowable Stress at Temperature	S	136.80 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	138.00 N./mm ²
Thickness of Tubesheet	h	65.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	3.2000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	3.2000 mm
Tubesheet Outside Diameter	A	980.000 mm

Additional Data for Stepped Tubesheets:

Is the Tubesheet Stepped?		Yes
Is the Tubesheet Flat on Tubeside?		No
Step 1 Diameter on the Tubeside	dt1	832.00 mm
Step 1 Depth on the Tubeside	ht1	8.00 mm
Step 2 Diameter on the Tubeside	dt2	797.00 mm
Step 2 Depth on the Tubeside	ht2	8.00 mm
Is the Tubesheet Flat on Shellside?		No
Calculated Tubesheet Diameter per Div 2, 4.18.4		832.00 mm

Note: Tubesheet diameter is now: 832.000 mm Div 2, 4.18.4.

Dimension G for the Channel Side	Gc	819.250 mm
Area of the Untubed Lanes	AL	0.0 cm ²

Junction Stress Reduction option		Perform Plastic Calculation
Perform Differential Pressure Design		No
Run Multiple Load Cases		Yes
Channel Side Min. Design Pressure	Ptd,min	103.4195 KPa.

Additional Data for Gasketed Tubesheets:

Tubesheet Gasket on which Side		Both
Flange Outside Diameter	A	980.000 mm
Flange Inside Diameter	B	800.000 mm
Flange Face Outside Diameter	Fod	835.000 mm
Flange Face Inside Diameter	Fid	800.000 mm
Gasket Outside Diameter	Go	832.000 mm
Gasket Inside Diameter	Gi	806.000 mm
Small end Hub thk.	g0	12.0000 mm
Large end Hub thk.	g1	18.0000 mm
Gasket Factor,	m	4.25
Gasket Design Seating Stress	y	69.64 N./mm ²
Flange Facing Sketch		Code Sketch 1a
Gasket Thickness	tg	3.0000 mm
Full face Gasket Flange Option		Program Selects
Length of Partition Gasket	lp	732.000 mm
Width of Partition Gasket	wp	10.0000 mm
Partition Gasket Factor,	mPart	2.7500
Partition Gasket Design Seating Stress	yPart	25.51 N./mm ²
Partition Gasket Facing Sketch		Code Sketch 1a
Partition Gasket Column for Gasket Seating		Code Column II

Bolting Information:



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 183 of 190

ASME TS Calc:

Case: 1 10:31a Sep 9,2026

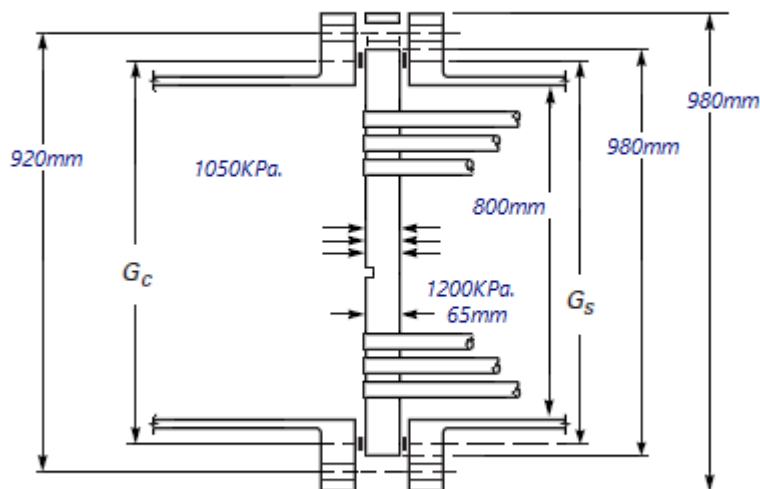
Diameter of Bolt Circle	C	920.000 mm
Nominal Bolt Diameter	dB	24.0000 mm
Thread Series	TEMA Metric Thread	
Number of Bolts	n	36
Bolt Material	SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00 N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00 N./mm ²
Weld between Flange and Shell/Channel		0.0000 mm

Alternate Flange Operating Bolt Load, Wm1	84516.24 Kg
Alternate Flange Seating Bolt Load, Wm2	126883.20 Kg
Alternate Flange Design Bolt Load, W	197464.48 Kg

Note: The full bolt load $Sa \cdot Ab$, was used for all load cases.

Tubesheet Integral with	None
Tubesheet Extended as Flange	Yes
Thickness of Extended Portion of Tubesheet	Tf 49.0000 mm
Is Bolt Load Transferred to the Tubesheet	Yes

Tubesheet Gasketed w/Shell and Channel



Configuration d:

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties :

Shell - TM-1 Carbon Steels with C> 0.3%

Elastic Mod. at Design Temperature	82.0 °C	0.19784E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20132E+09 KPa.

Channel - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	230.0 °C	0.19046E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tubes - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Tubsht. Design Temp.	230.0 °C	0.19046E+09 KPa.
--------------------------------------	----------	------------------



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FileName : E-0102

Page 184 of 190

ASME TS Calc:

Case: 1 10:31a Sep 9, 2026

Elastic Mod. at Ambient Temperature 21.1 °C 0.20270E+09 KPa.

TubeSheet - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature 230.0 °C 0.19046E+09 KPa.

Elastic Mod. at Ambient Temperature 21.1 °C 0.20270E+09 KPa.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:

= $(P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)

= $(1050.0 \cdot (25.4/2 - 0)) / (108.0 \cdot 1.0 + 0.4 \cdot 1050.0)$

= $0.1230 + 0.0000 = 0.1230$ mm

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart CS-2 at 230.00 °C
Elastic Modulus for Material 189879744.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.3200	25.40	2743.69	10.95	50.0000	0.0091770	107.35

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 107.346) / (3 \cdot 10.9483) = 13072.3555$ KPa.

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.5353	25.40	2743.69	47.45	50.0000	0.0004886	46.39

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 46.3881) / (3 \cdot 47.4493) = 1303.4366$ KPa.

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.5353 mm
Allowable Internal Pressure at Corroded thickness	21282.78 KPa.
Required Internal Design Pressure	1050.00 KPa.
Allowable External Pressure at Corroded thickness	13072.36 KPa.
Required External Design Pressure	1303.42 KPa.
Required Thickness due to Shell Side pressure	0.5353 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Intermediate Calculations For Tubesheets Extended As Flanges:

ASME Code, Section VIII Division 1, 2025

Gasket Contact Width,	$N = (Goc - Gic) / 2$	13.000 mm
Basic Gasket Width,	$b0 = N / 2$	6.500 mm
Effective Gasket Width,	$b = \sqrt{b0} \cdot 2.5$	6.422 mm
Gasket Reaction Diameter,	$G = Go - 2.0 \cdot b$	819.156 mm

Results for ASME U-tube Tubesheet Calculations for Configuration d:

Per Edition 2025, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:

= $2 \cdot ro + dt$
 = $2 \cdot 370.687 + 25.4$
 = 766.774 mm

Basic ligament efficiency for shear [μ]:

= $(p - dt) / p$
 = $(38.1 - 25.4) / 38.1$



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FileName : E-0102

Page 185 of 190

ASME TS Calc: Case: 1 10:31a Sep 9, 2026

= 0.333

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ps]:

= G_s / D_o (Configurations d, e, f)

= 819.1555/766.774

= 1.0683

Ratio [pc]:

= G_c / D_o (Configurations d)

= 819.2501/766.774

= 1.0684

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [p]:

= l_{tx} / h

= 0/65.0

= 0.000 (must be $0 \leq p \leq 1$)

Effective tube hole Diameter [d*]:

= $\max(d_t - 2t_t(E_t T / E) \cdot (S_t T / S) \cdot (p); d_t - 2t_t)$

= $\max(25.4 - 2 \cdot 2.32 (0.19046E+09 / 0.19046E+09) \cdot$

$(107/136) \cdot (0); 25.4 - 2 \cdot 2.32)$

= 25.4000 mm

Effective tube pitch [p*]:

= $p / \sqrt{ (1 - 4 \cdot \min(A_l; 4 \cdot D_o \cdot p) / (\pi \cdot D_o^2)) }$

= $38.1 / \sqrt{ (1 - 4 \cdot \min(0; 4 \cdot 766.774 \cdot 38.1) / (\pi \cdot 766.774^2)) }$

= 38.1000 mm

Effective ligament efficiency for bending [μ^*]:

= $(p^* - d^*) / p^*$

= $(38.1 - 25.4) / 38.1$

= 0.3333

$h/p = 1.706037$; $\mu^* = 0.333333$; $E^* = 64044960$ KPa.

$E^*/E = 0.336268$; $v^* = 0.324996$

Skip Step 4 for Configuration d :

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

= A / D_o

= 832.0/766.774

= 1.0851

Coefficient [F]:

= $(1 - v^*) / E^* \cdot (E \cdot \ln(K))$

= $(1 - 0.32) / 64044960 \cdot (0.19046E+09 \cdot \ln(1.09))$

= 0.1639

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

= $D_o^2/16 \cdot [(p_s - 1) \cdot (p_s^2 + 1) \cdot P_s - (p_c - 1) \cdot (p_c^2 + 1) \cdot P_t]$

= $766.774^2/16 \cdot [(1.068 - 1) \cdot (1.068^2 + 1) \cdot 1200.0 -$

$(1.068 - 1) \cdot (1.068^2 + 1) \cdot 1050.0]$

= 795402.5000 KPa·mm²

Moment acting on unperforated tubesheet rim [M*]

= $M_{ts} + W^* \cdot (G_c - G_s) / (2 \cdot \pi \cdot D_o)$

= $795402 + 197464.5 \cdot (819.25 - 819.25) / (2 \cdot \pi \cdot 766.774)$



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FileName : E-0102

Page 186 of 190

ASME TS Calc:

Case: 1 10:31a Sep 9, 2026

$$= 795402.5000 \text{ KPa} \cdot \text{mm}^2$$

Note: W^ is the maximum of the bolt loads between the shell and channel sides.*

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$\begin{aligned} &= ((M^*) - D_o^2/32 \cdot F (P_s - P_t)) / (1 + F) \\ &= ((795402) - 766.774^2/32 \cdot 0.164 (1200.0 - 1050.0)) / (1 + 0.16) \\ &= 295351.1250 \text{ KPa} \cdot \text{mm}^2 \end{aligned}$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$\begin{aligned} &= M_p + D_o^2/64 \cdot (3 + \nu^*) \cdot (P_s - P_t) \\ &= 295351.12 + 766.774^2/64 \cdot (3 + 0.325) \cdot (1200.0 - 1050.0) \\ &= 4877160.0000 \text{ KPa} \cdot \text{mm}^2 \end{aligned}$$

Maximum bending moment acting on tubesheet [M]:

$$\begin{aligned} &= \max(| M_p | ; | M_o |) \\ &= \max(| 295351.125 | ; | 4877160 |) \\ &= 4877160.0000 \text{ KPa} \cdot \text{mm}^2 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$\begin{aligned} &= 6 \cdot M / (\mu^* (h - h_g')^2) \\ &= 6 \cdot 4877160 / (0.3333 (65.0 - 5.0)^2) \\ &= 24.3872 \text{ N./mm}^2 \end{aligned}$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$\begin{aligned} &= 2 \cdot S \\ &= 2 \cdot 136.8 \\ &= 273.60 \text{ N./mm}^2 \end{aligned}$$

Tubesheet bending stress at final thickness [σ]:

$$\begin{aligned} &= 6 \cdot M / (\mu^* (h - h_g')^2) \\ &= 6 \cdot 4792351 / (0.3333 (22.7626 - 5.0)^2) \\ &= 273.4225 \text{ N./mm}^2 \end{aligned}$$

Required tubesheet thickness for Bending Stress [HreqB]:

$$\begin{aligned} &= h + C_{ats} + C_{atc} \\ &= 22.7626 + 0 + 0 \\ &= 22.7626 \text{ mm} \end{aligned}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$\begin{aligned} &= \max(H_{reqB} ; H_{reqS}) \\ &= \max(22.7626 ; 0.5019) \\ &= 22.7626 \text{ mm} \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [$| P_s - P_t |$]:

$$\begin{aligned} &= | P_s - P_t | \\ &= | 1200.0 - 1050.0 | \\ &= 150.000 \text{ KPa} \end{aligned}$$

Shear Stress check [τ_{limit}]:

$$\begin{aligned} &= 4 \cdot \mu \cdot h / D_o \cdot \min(0.8 \cdot S ; 0.533 \cdot S_y) \\ &= 4 \cdot 0.333 \cdot 65.0 / 766.77 \cdot \min(0.8 \cdot 136.798 ; 0.533 \cdot 208.0) \\ &= 12.370 \text{ N./mm}^2 \end{aligned}$$

$| P_s - P_t |$ is $\leq$ the Shear Stress limit (τ_{limit}). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [HreqS]:

$$= 1 / (4 \cdot \mu \cdot \min(0.8 \cdot S ; 0.533 \cdot S_y)) \cdot (4 \cdot A_p / C_p) \cdot | P_s - P_t | + C_{ats} + C_{atc}$$



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FileName : E-0102

Page 187 of 190

ASME TS Calc:

Case: 1 10:31a Sep 9, 2026

$$= 1 / (4 \cdot 0.333 \cdot 109.439) \cdot (4 \cdot 2517.1 / 2062.09) \cdot | 1200.0 - 1050.0 | + 0$$

$$= 0.5019 \text{ mm}$$

Note: Analysis Completed for Tubesheet Configuration d.

Tube Metal Area [At]:

$$= \pi (d_o - t) t$$

$$= \pi (25.4 - 2.32) 2.32$$

$$= 1.682 \text{ cm}^2$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$= \pi \cdot t (d_o - t) S$$

$$= \pi \cdot 2.32 (25.4 - 2.32) 107.998$$

$$= 1852.511 \text{ Kg}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t)$$

$$= 0.85 \cdot \pi \cdot 2.32 (25.4 + 0.67 \cdot 2.32) 108.0$$

$$= 1838.9659 \text{ Kg}$$

Weld Strength Factor [fw]:

$$= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$$

$$= 107.998 / 107.998$$

$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t)$$

$$= 0.55 \cdot \pi \cdot 2.32 (25.4 + 0.67 \cdot 2.32) 108.0$$

$$= 1189.9191 \text{ Kg}$$

Minimum Length of Weld [ar]:

$$= 2 (\sqrt{ ((0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d) } - 0.75 \cdot d_o)$$

$$= 2 (\sqrt{ ((0.75 \cdot 25.4)^2 + 1.07 \cdot 2.32 \cdot (25.4 - 2.32) 1.0 \cdot 1.0) } - 0.75 \cdot 25.4)$$

$$= 2.897 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.4487 mm

Minimum Required Groove Weld Leg [agr]: 1.4487 mm

Tube-Tubesheet Jt allowable, 1852.51 is $\geq$ tube strength 1852.51 Kg

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	24.4 $\leq$	273.6 N./mm ²	Ok
Tubesheet shear stress	0.8 $\leq$	109.4 N./mm ²	Ok

Thickness results for loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	22.763	65.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.449	2.320	Ok
Tube-Tubesheet Groove Weld Leg :	1.449	2.320	Ok

U-Tube Tubesheet results per ASME Div. 2 Part 4.18.7 2025



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FileName : E-0102

Page 188 of 190

ASME TS Calc:

Case: 1 10:31a Sep 9,2026

Results for 8 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Bend Allwd	Stresses Shear Allwd	Case Type	Pass/ Fail
D1uc	52.530 ...	171	274	6 109	Ps+Pt D1	Ok
D2uc	57.895 40.509	212	274	7 109	Ps+Pt D2	Ok
D3uc	22.763 ...	24	274	1 109	Ps+Pt D3	Ok
D4uc	19.756 ...	17	274	1 109	Ps+Pt D4	Ok
D1c	55.625 ...	191	274	7 109	Ps+Pt-c D1	Ok
D2c	61.038 40.509	237	274	8 109	Ps+Pt-c D2	Ok
D3c	25.929 ...	27	274	1 109	Ps+Pt-c D3	Ok
D4c	22.904 ...	19	274	1 109	Ps+Pt-c D4	Ok
Max:	61.0377 40.509 mm	0.866	0.074	(Str. Ratio)		

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the Load Case Tables

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

Summary of Thickness Comparisons for 8 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	61.038	65.000	Ok
Tubesheet thickness flanged extension :	40.509	49.000	Ok
Tube thickness :	0.535	2.320	Ok
Tube-tubesheet fillet weld leg :	1.449	2.320	Ok
Tube-tubesheet groove weld leg :	1.449	2.320	Ok

Note: This is a full strength Tube to Tubesheet Joint.

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	0 shellside Stress Rat.	Shellside MAWP	0 tubeside Stress Rat.
Tubesheet Bending Stress	1505.56	1.000	1505.97	1.000
Tubesheet Shear Stress	17511.86	1.000	17511.86	1.000
Tube Pressure Stress	21282.77	1.000	13072.34	1.000
Tubesheet Extension Stress	No Calc	No Calc	2816.04	...
Minimum MAWP	1505.56		1505.97	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	0 shellside Stress Rat.	Shellside MAPnc	0 tubeside Stress Rat.
Tubesheet Bending Stress	1693.86	1.000	1693.86	1.000



PV Elite 28 Licensee: SL User

FileName : E-0102

Page 189 of 190

ASME TS Calc:

Case: 1 10:31a Sep 9, 2026

Tubesheet Shear Stress	19594.70	1.000	19594.69	1.000
Tube Pressure Stress	21282.77	1.000	14945.89	1.000
Tubesheet Extension Stress	No Calc	No Calc	2816.04	...
Minimum MAPnc	1693.86		1693.86	

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

Tubesheet MDMT Calculations:

Determine the governing MDMT considering the governing condition:

Governing thickness on the shell side per figure UCS-66.3 (c):

= tubesheet thickness/4

= 65.0/4

= 16.250 mm

Thickness Ratio = 0.708, Temperature Reduction per Fig. UCS 66.1 = 16 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C

Min Metal Temp. at Required thickness (UCS 66.1) -30 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

where the MDMT reduction ratio per UCS 66 (b)(1)(b) is:

= max(pt/Tubeside MAPnc, ps/Shellside MAPnc), must be ≤ 1

= max(1050.0/1693.86, 1200.0/1693.86)

= 0.708

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FileName : E-0102

Page 190 of 190

Problems/Failures Summary:

Step: 33 10:31am Sep 9, 2026

Noted issues for the current model:

Listed below are the known problem areas for the current design. If one or more of the design flags are turned on, please re-run the analysis. Some of these issues may be resolved when using updated input values.

**** Warning: A standard flange seems to be governing the Channelside MAWP!**

Please review all reports carefully!

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PV Elite® Pressure Vessel Design Notes

Project Information	
Project Number	
Client	
Date	
Designer	
Design Code	
Additional Standards	

Design Conditions		
Internal Design Pressure:		
External Design Pressure:		
Design Int. Temperature:		
Design Ext. Temperature:		
Corrosion Allowance:		

Vessel Geometry		
Vessel Type		
Diameter		
Tangent-to-Tangent Length		
Head Type		
Number of Segments		

Material Specifications	
Shell:	
Head:	
Nozzle:	

Nozzle/Opening Details	ID/OD	Size	Rating	Matl.	Reinf.

Miscellaneous		
Support Type		
Number of Supports		
Support Material		
Seismic Code		
Wind Code		
Insulation		
Pressure Test		
Radiography		
Fatigue Analysis		

Analysis Notes: